

KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine



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Lesson content for each week

Ch0_Introduction to the Class

Ch6_Piston Fittings and Crankshaft

Ch1_Overview of the Internal Combustion Engine

Ch7_Lube Oil System

Ch2_Thermodynamics and Compressed Air System

Ch8_Cooling Water System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch9_Introduction to the Main Engine

Ch4_Intake and Exhaust System

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch5_Turbocharger

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

Midterm

Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

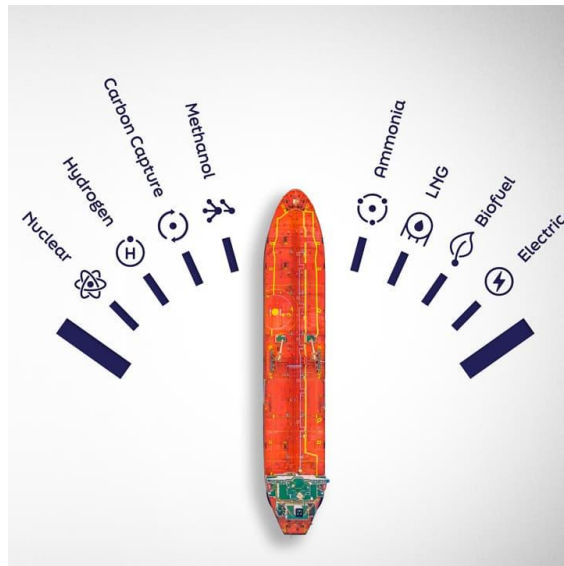
What are alternative fuels in maritime industry?

The use of conventional fuels (especially HFO) is being restricted due to strengthening environmental regulations.

However, since current technology cannot completely replace conventional fuels, alternative fuels are used along with conventional fuels.

Alternative fuels in the maritime field include:

- 1) Liquefied Natural Gas (LNG)
- 2) Liquefied Petroleum Gas (LPG)
- 3) Biofuel
- 4) Methanol
- 5) Ammonia
- 6) Electricity
- 7) Hydrogen
- 8) Nuclear



Alternative fuels

Properties and practicalities

Energy Storage Type	Specific Energy MJ/kg	Energy Density MJ/L	Required Tank Volume m ³ . ¹	Supply Pressure Bar	Injection Pressure Bar	Emmision Reduction Compared To HFO Tier II			
						SO _x	NO _x	CO ₂	PM
HFO	40,5	35	1000	7-8	950				
Liquefied natural gas (LNG -162 °C)	50	22	1590	300 METHANE	300 METHANE	90-99%	20-30%	24%	90%
				380 ETHANE	380 ETHANE	90-97%	30-50%	15%	90%
LPG (including Propane / Butane)	42	26	1346	50	600-700	90-100%	10-15%	13-18%	90%
Methanol	18	15	2333	10	500	90-97%	30-50%	5%	90%
Ethanol	26	21	1750	10	500				
Ammonia* (liquid -33 °C)	18,6	12,5	2800	50	600-700				
Hydrogen (liquid -253 °C)	142	10	3500						
Marine battery market leader Corvus battery rack	0,29	0,33	106.060						
Tesla model 3 battery Cell 2170*. ²	0,8	2.5	14000						

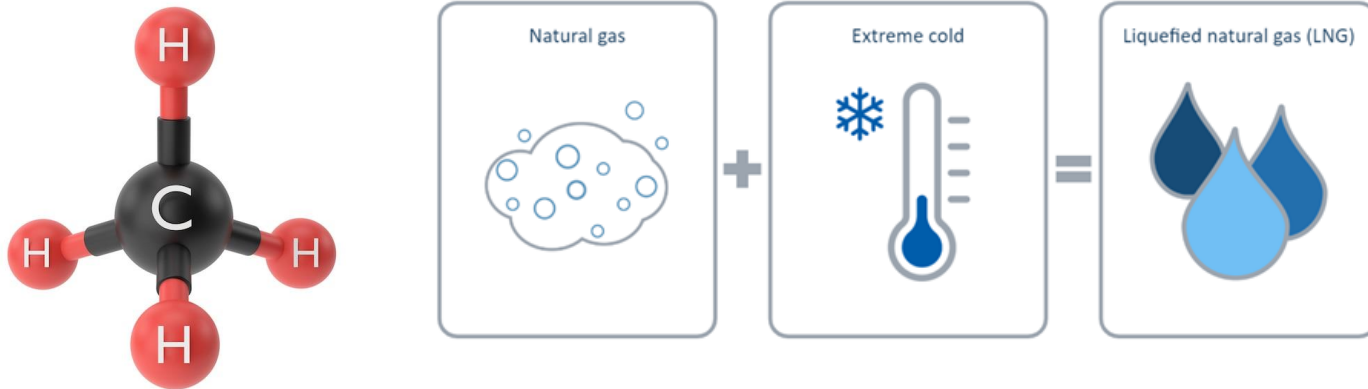
• ¹: Given a 1000 m³ tank for HFO. Additional space for insulation is not calculated for in above diagram. All pressure values given a high pressure Diesel injection principle.

• ²: Values for Tesla battery doesn't contain energy/mass obtained for cooling/safety/classification . <https://insideevs.com/tesla-model-3-2170-energy-density-compared-bolt-p100d/>

LNG (Liquefied Natural Gas)

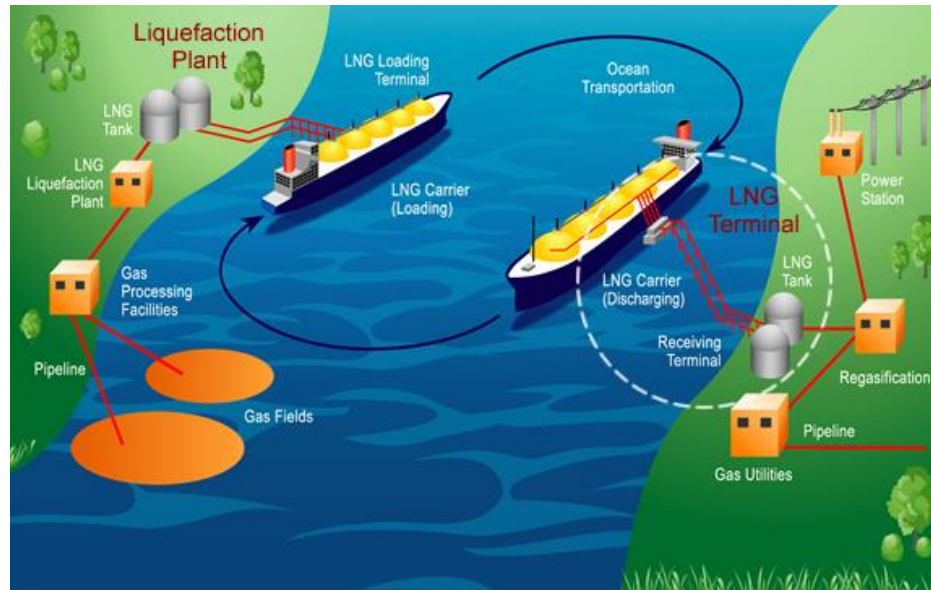
The chemical formula of LNG follows the main component, methane (CH_4), but it is a mixture containing other hydrocarbons such as ethane (C_2H_6), propane (C_3H_8), and other inert gases, so it can be represented by CH_4 .

LNG is natural gas that has been cooled to a liquid state, at about -162°C , for shipping and storage. The volume of natural gas in its liquid state is about 600 times smaller than its volume in its gaseous state. This process makes it possible to transport natural gas to places pipelines do not reach.



LNG

Liquefying natural gas is a way to move natural gas long distances when pipeline transport is not feasible. Markets that are too far away from producing regions to be connected directly to pipelines have access to natural gas because of LNG. In its compact liquid form, natural gas can be shipped in special tankers to terminals around the world. At these terminals, the LNG is returned to its gaseous state and transported by pipeline to distribution companies, industrial consumers, and power plants.

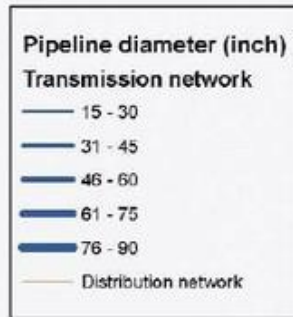


NG



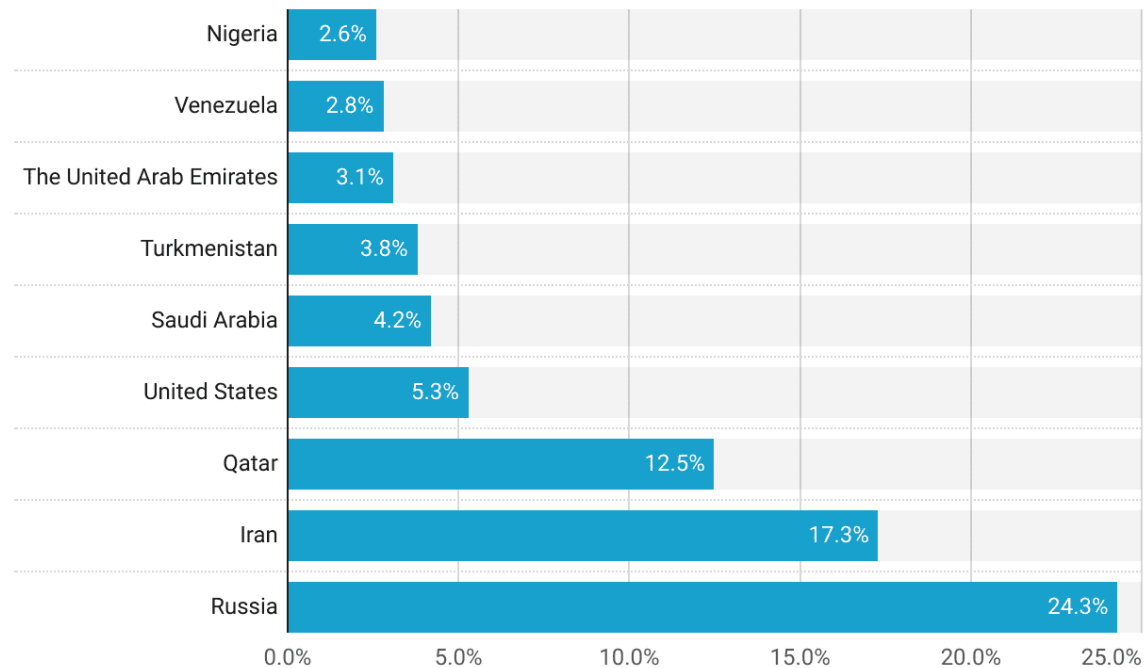
Source: S&P Global Platts

NG



Natural Gas Reserves Top Countries by Share

Share By Percentage

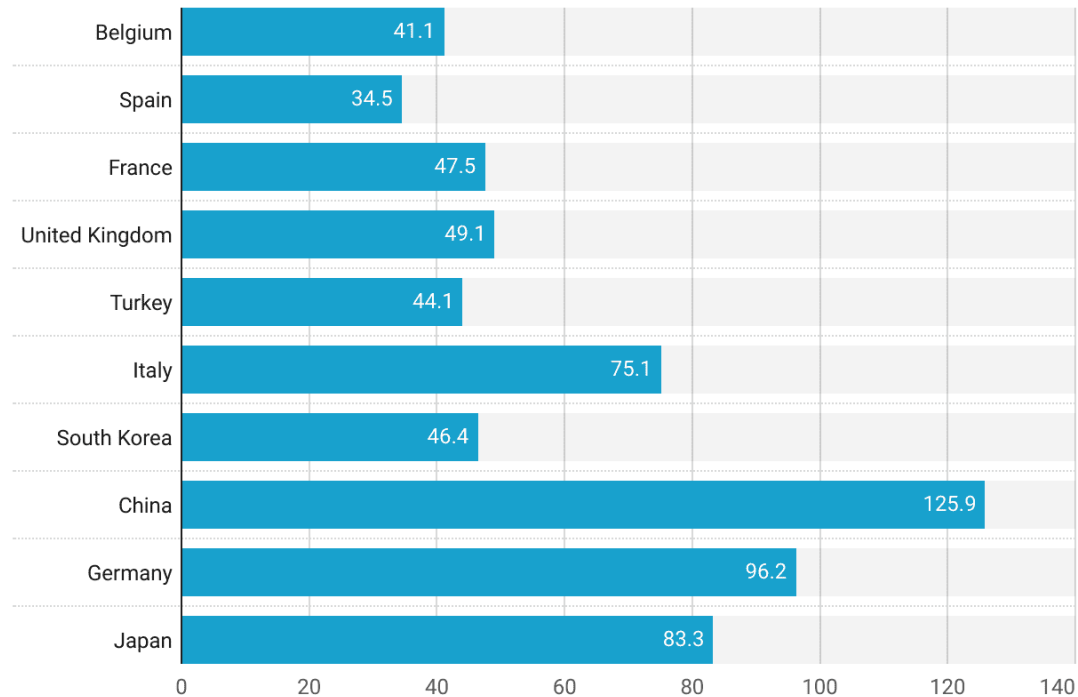


(Share By %)

Source: Market.us Media

Top Natural Gas Importing Countries

■ Natural Gas Imports (bcm/year)

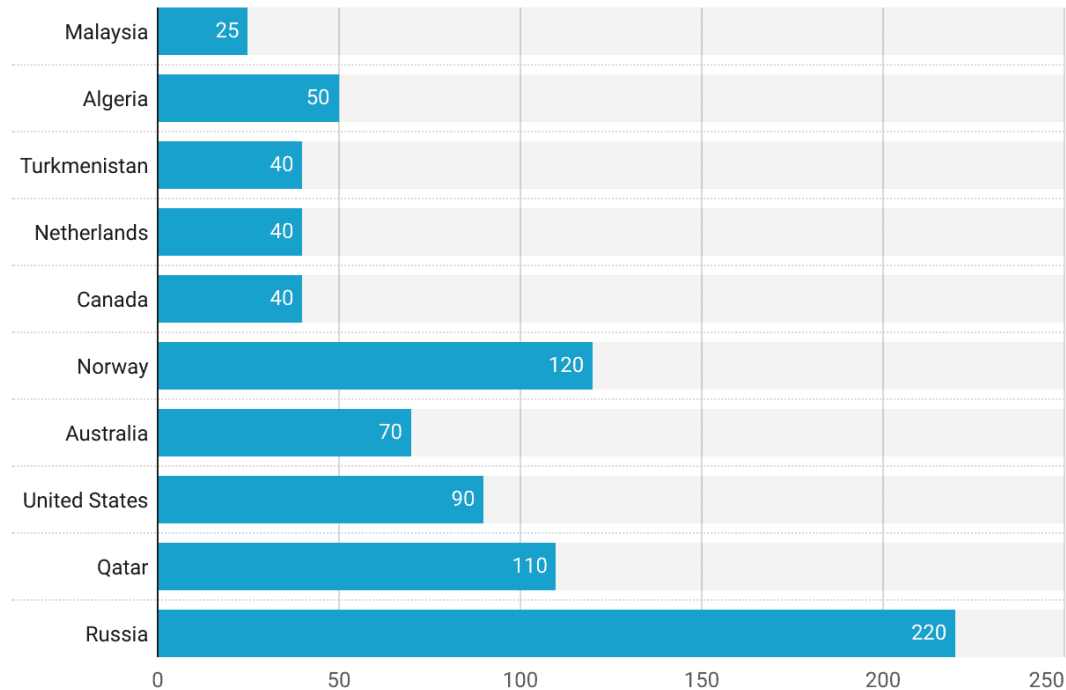


(bcm/year)

Source: Market.us Media

Top Natural Gas Exporting Countries

■ Natural Gas Exports (bcm/year)

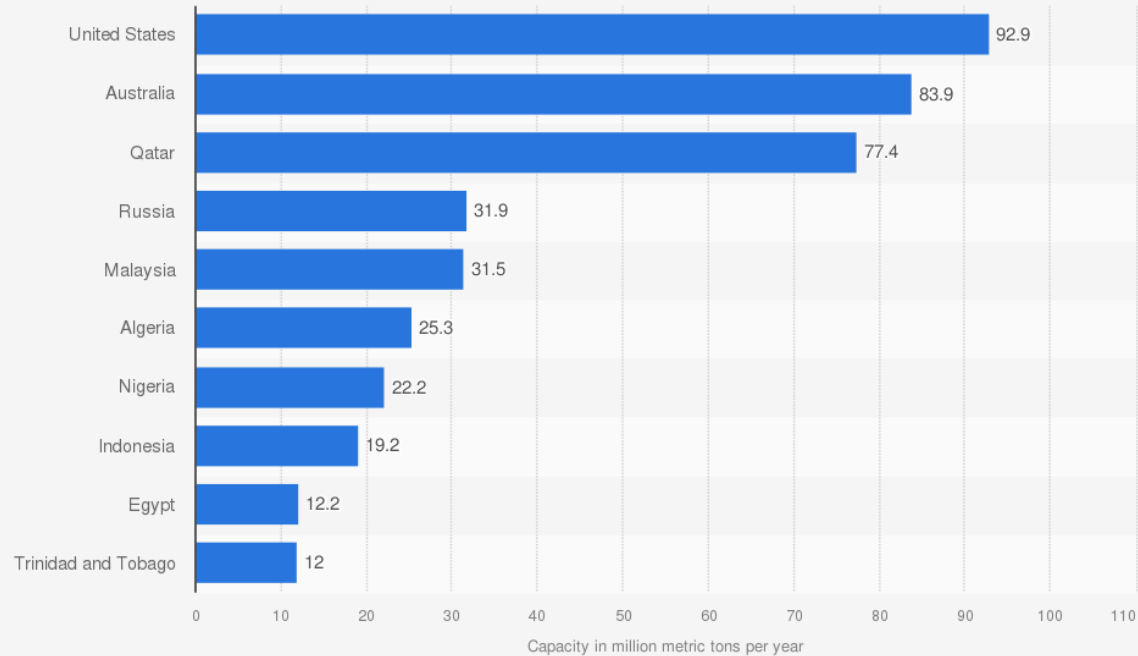


(bcm/year)

Source: Market.us Media

LNG

Countries with largest liquefied natural gas (LNG) export capacity in operation worldwide as of September 2024 (in million metric tons per year)



Source
Global Energy Monitor
© Statista 2024

Additional Information:
Worldwide; September 2024



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LNG

TV로 보는
골라둔다큐

130,000,000ℓ

도시가스 배달 왔습니다

영하 **167°C**

액화천연가스 어떻게 하역할까?



LNG

LNG ships that transport LNG as cargo are divided into MOSS and membrane types.



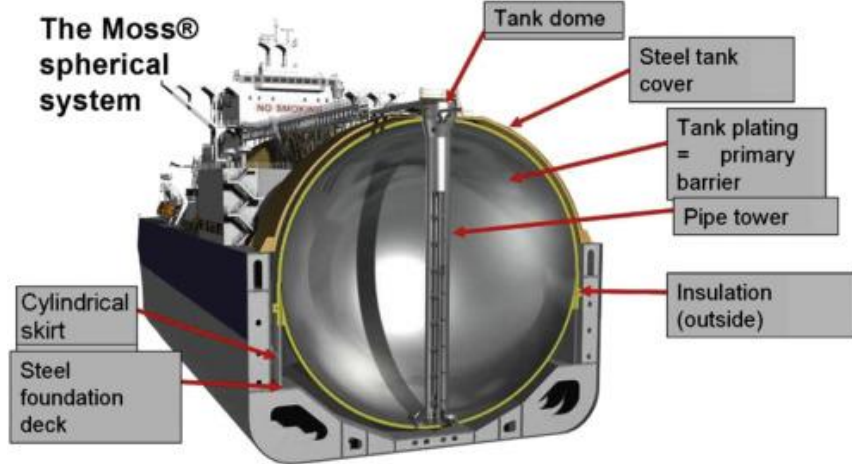
MOSS type



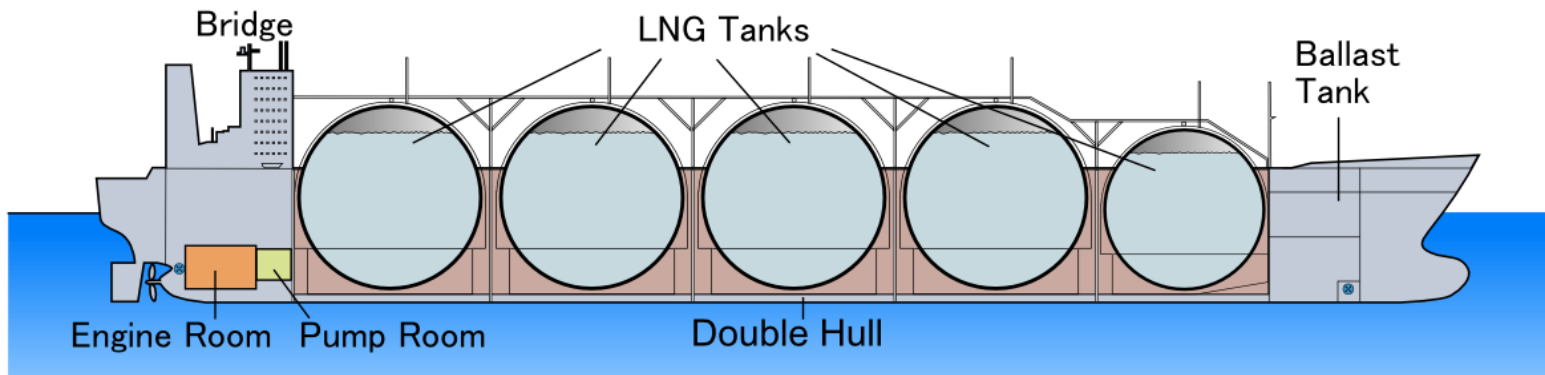
Membrane type

LNG

MOSS type

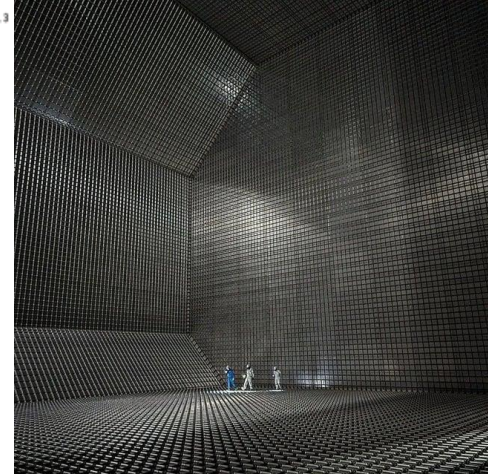
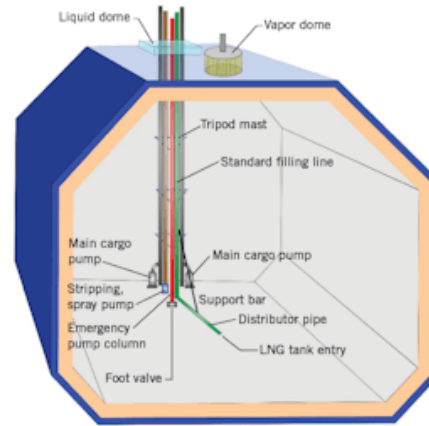


LNG tanker (side view)

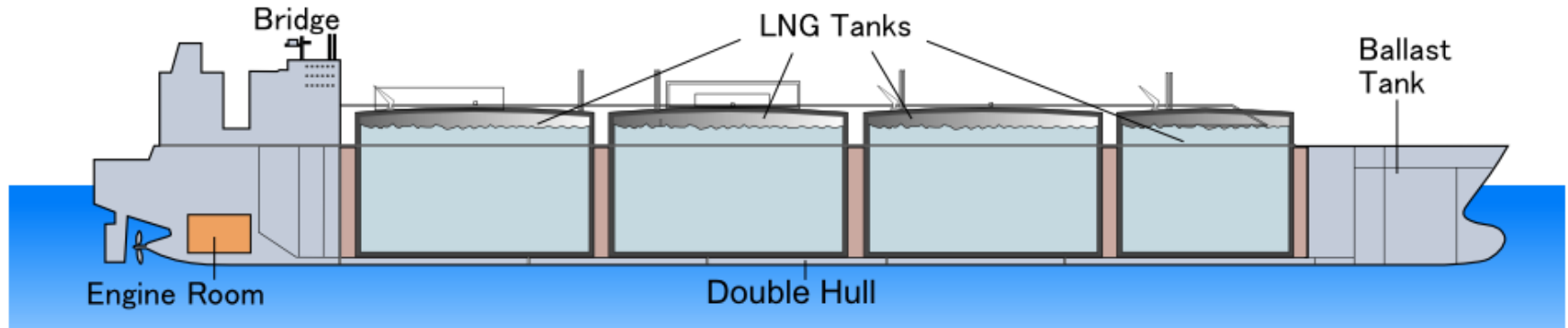


LNG

Membrane type

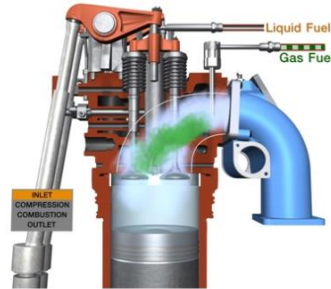
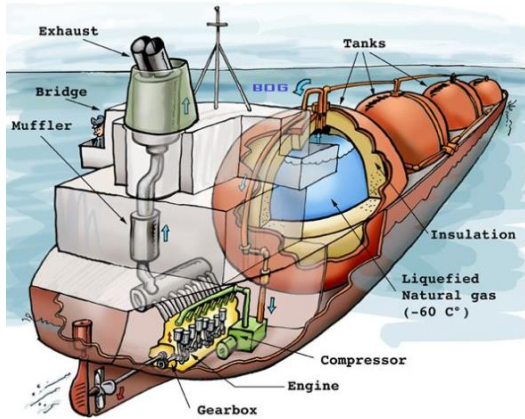


LNG tanker GTT Mk.III(side view)



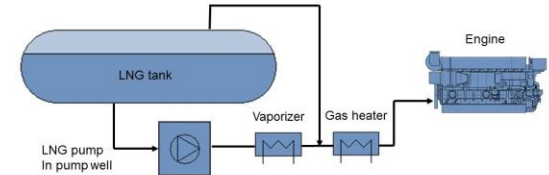
LNG

- 1) LNG ships that carry LNG as cargo use LNG as fuel.
- 2) General commercial ships are equipped with LNG tanks and use LNG as fuel.

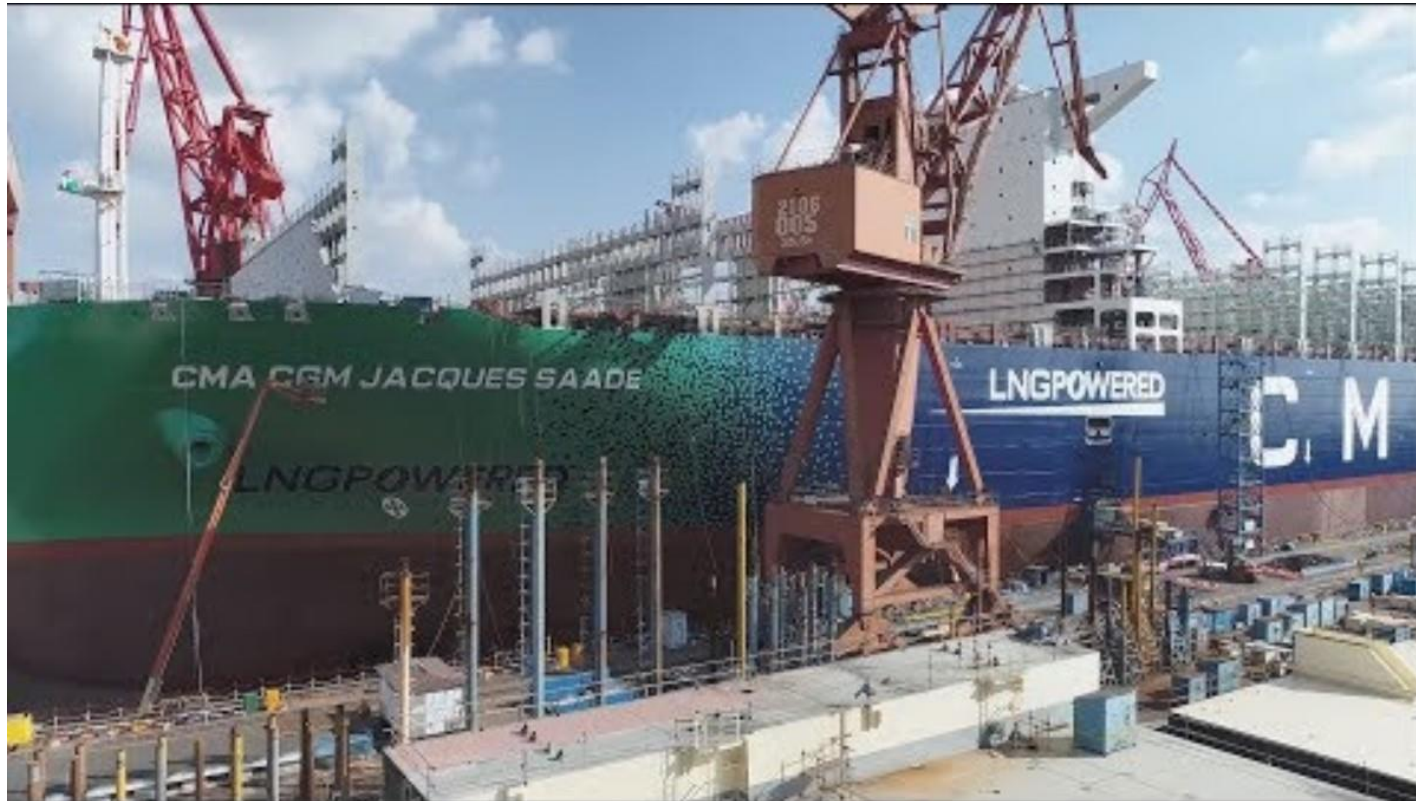


LNG fuel system with pump

Above picture is only for understanding of LNG used in LNG carrier



LNG



LNG



LNG



LNG (ME-GI)

ME-GI stands for Everllence B&W (MAN B&W) LNG fuelled engine.

ME → MAN B&W's electronically controlled engine series

GI → Gas Injection (high-pressure gas injection)

ME-GI = Electronically Controlled Gas Injection Engine

ME-GI is possible to change between two different fuel modes: Dual-fuel and fuel oil mode.

In case of a problem in the gas injection system, the use of gas is stopped and the engine is changed-over to fuel oil operation.

Nearly

1,000

ME-GI engine orders since 2014

More than

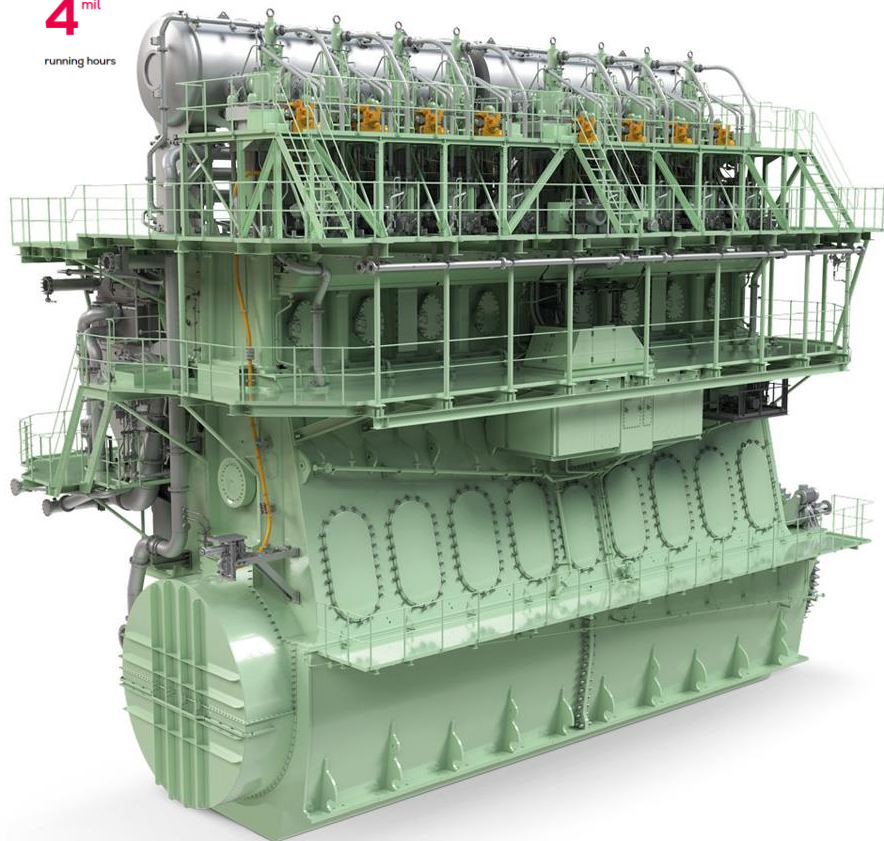
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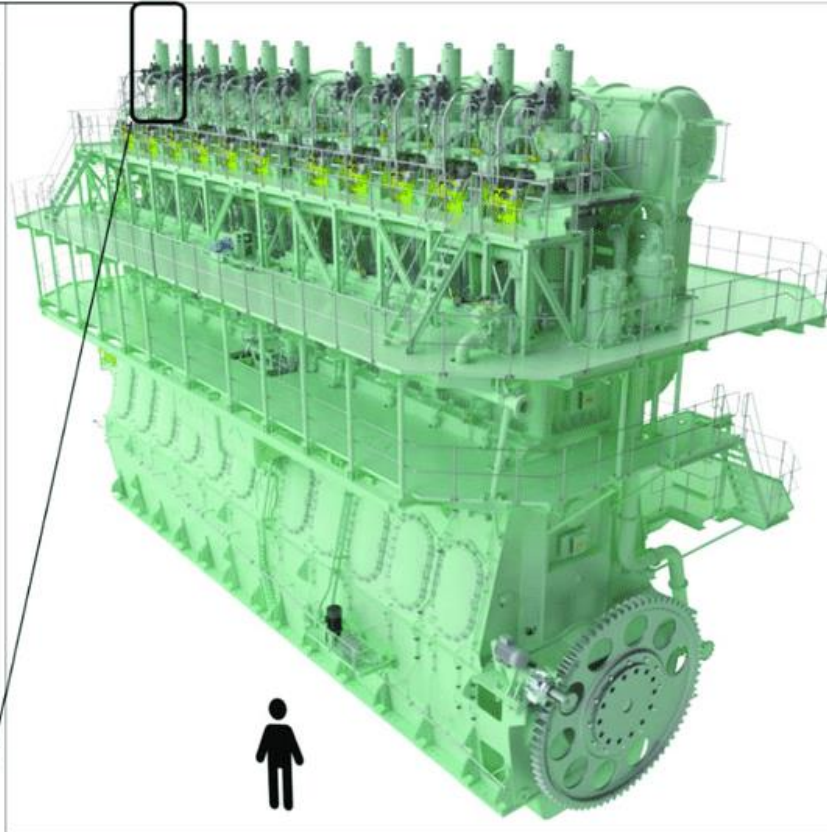
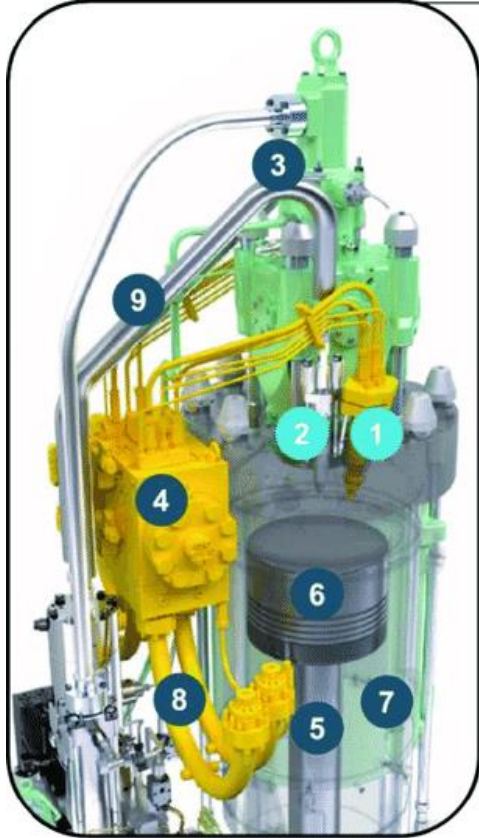
running hours

0.2 g/kWh

methane slip with a tolerance of ± 0.2 g/kWh



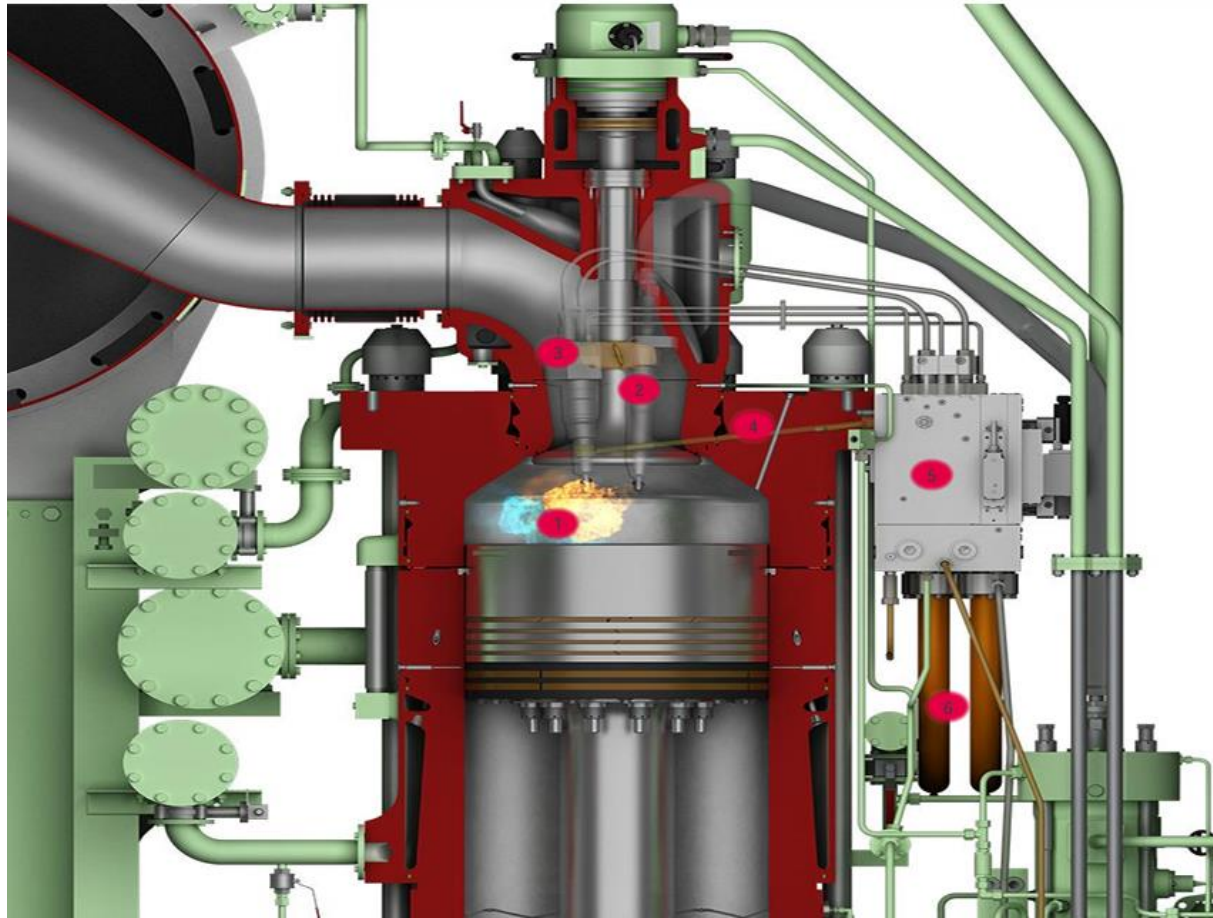
LNG (ME-GI)



- (1) gas injection valve
- (2) fuel oil injection valve
- (3) exhaust gas valve
- (4) gas control block
- (5) piston rod
- (6) piston
- (7) cylinder liner
- (8) gas chain pipes
- (9) fuel oil pipe

ME-GI Concept

Combustion Concept



Pilot fuel oil to create the initial flame and ammonia injected into flame.

① Direct injection of pilot oil and gas

Yellow = pilot oil
Blue = gas

② Conventional slide fuel valve

③ Gas fuel valve

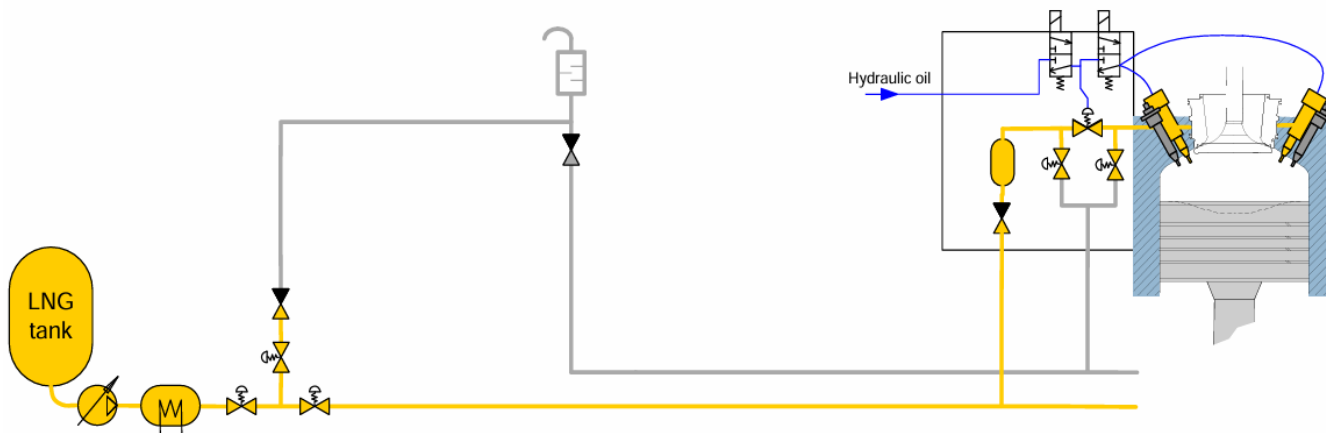
④ Gas distribution channel

⑤ Gas block

⑥ Gas chain link double-walled pipes

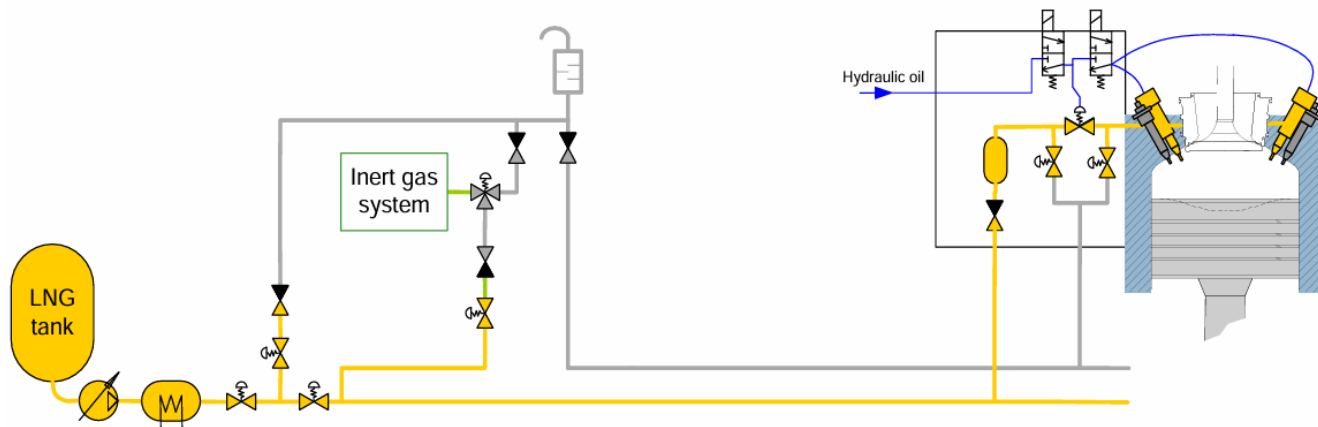
LNG (ME-GI)

ME-GI gas injection



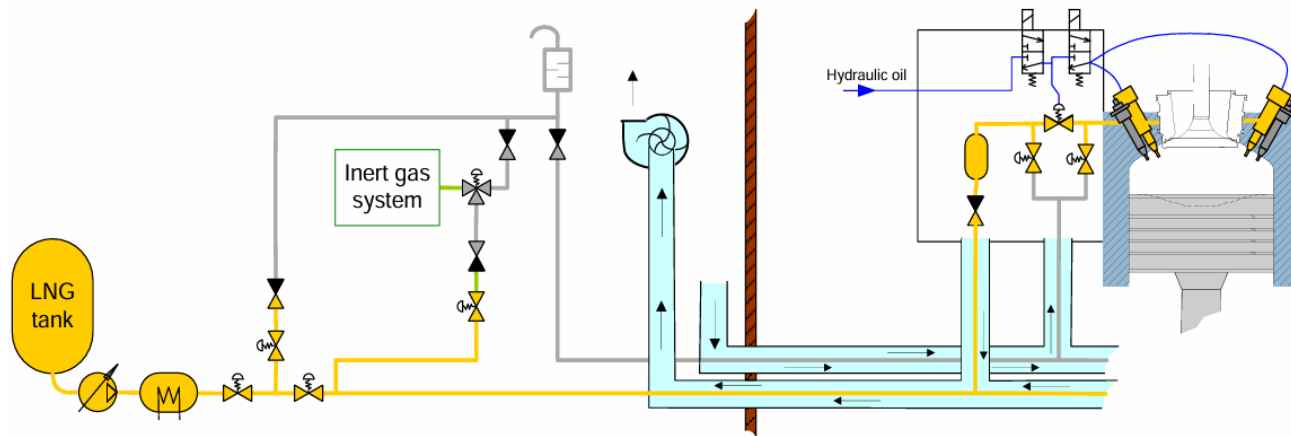
LNG (ME-GI)

Inert gas system
- for purging of gas pipes



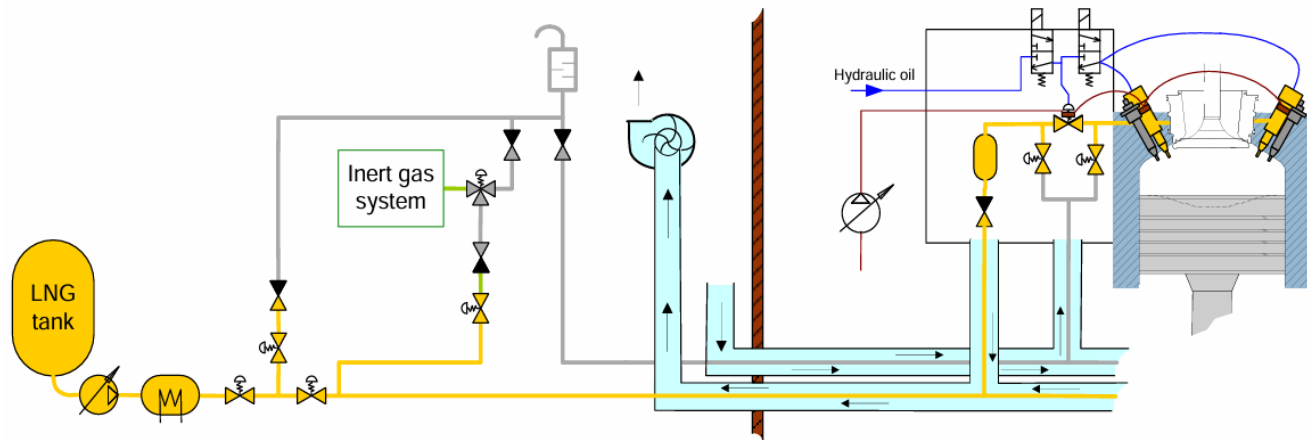
LNG (ME-GI)

Double pipes and ventilation system



LNG (ME-GI)

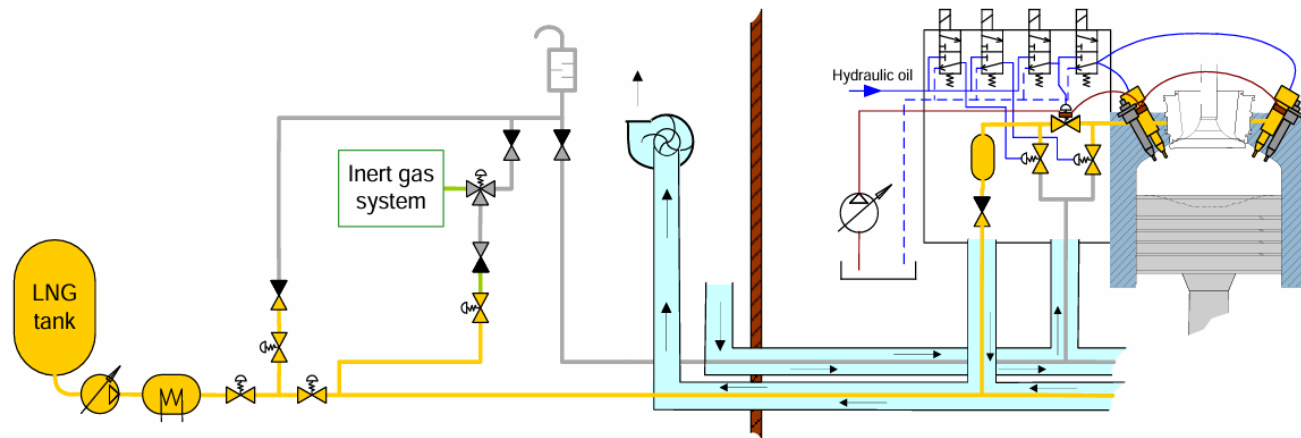
Sealing oil system



LNG (ME-GI)

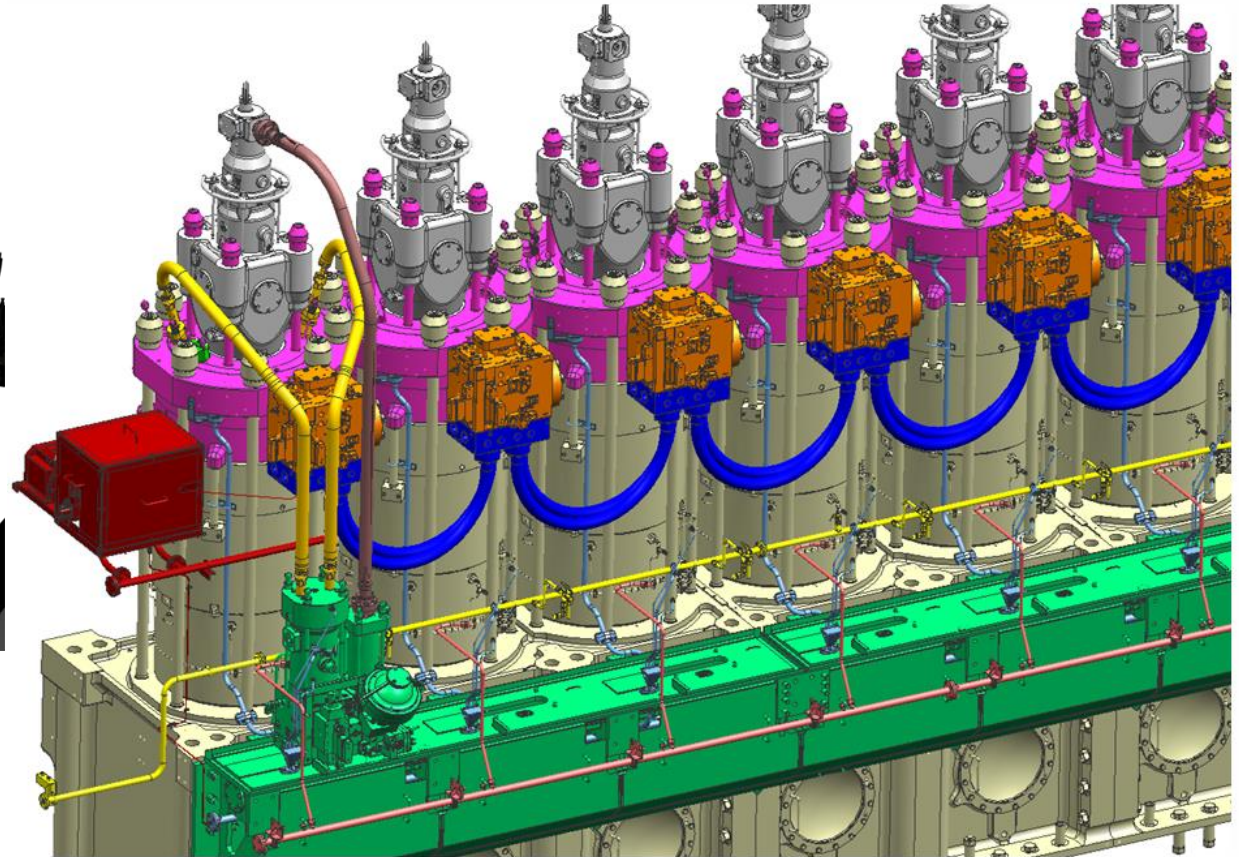
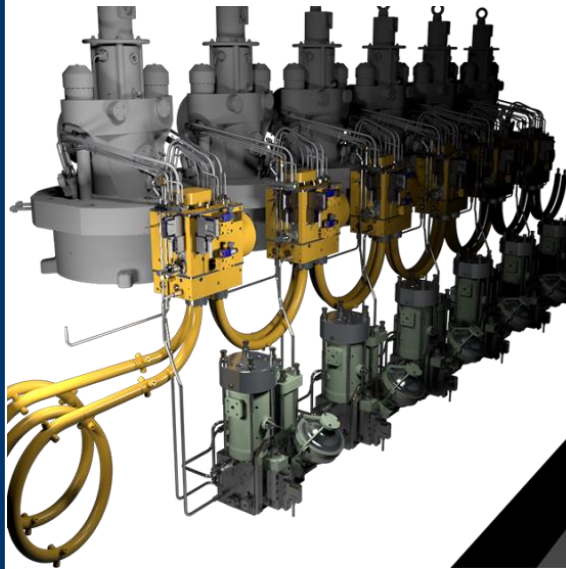
blow-off

Hydraulic activation of blow-off and
purge valves



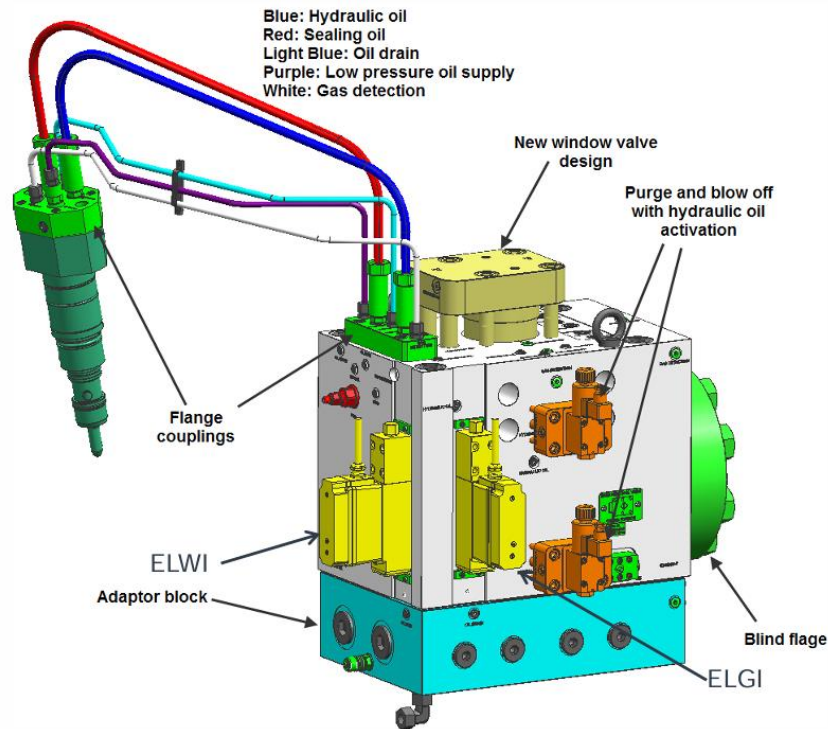
LNG (ME-GI)

ME-GI overview



LNG (ME-GI)

ME-GI Gas control block

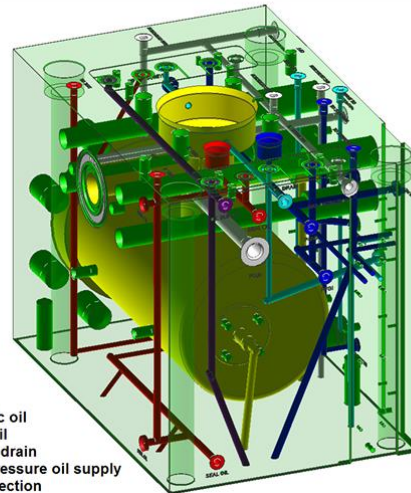
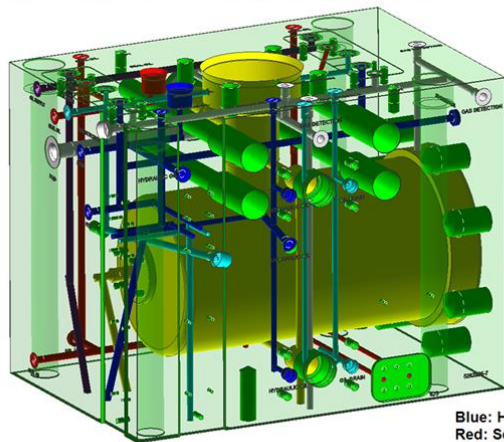


LNG (ME-GI)

ME-GI Gas control block



- Material S45R (393 kg)
- H=396.5mm ;W=505mm; D=340
- Acc. volume for 20 gas injections.

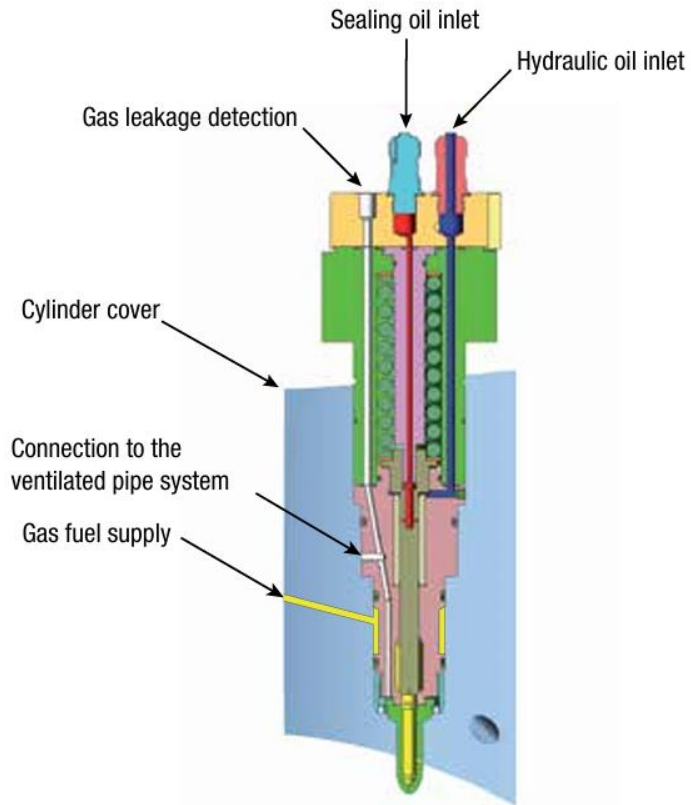
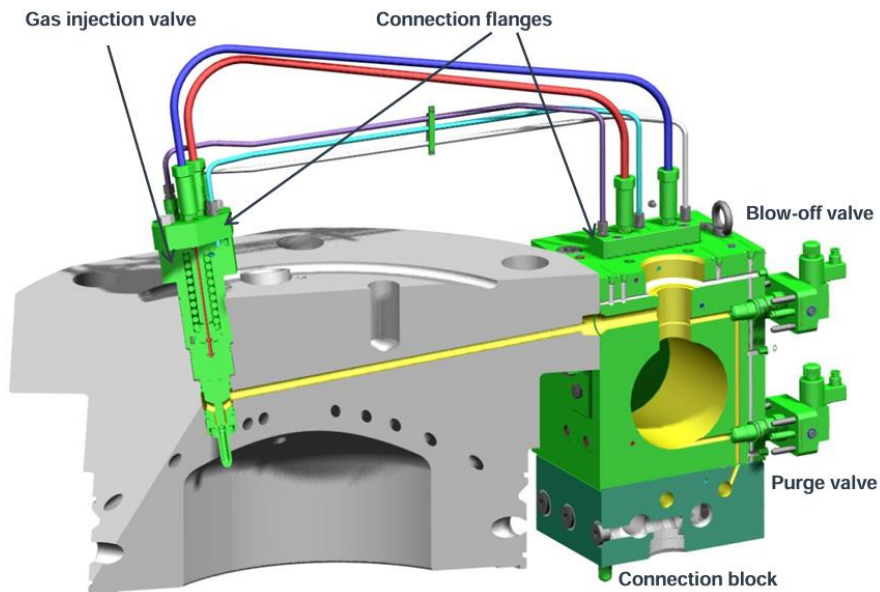


Blue: Hydraulic oil
Red: Sealing oil
Light blue: Oil drain
Purple: Low pressure oil supply
White: Gas detection



LNG (ME-GI)

ME-GI Gas Injection System

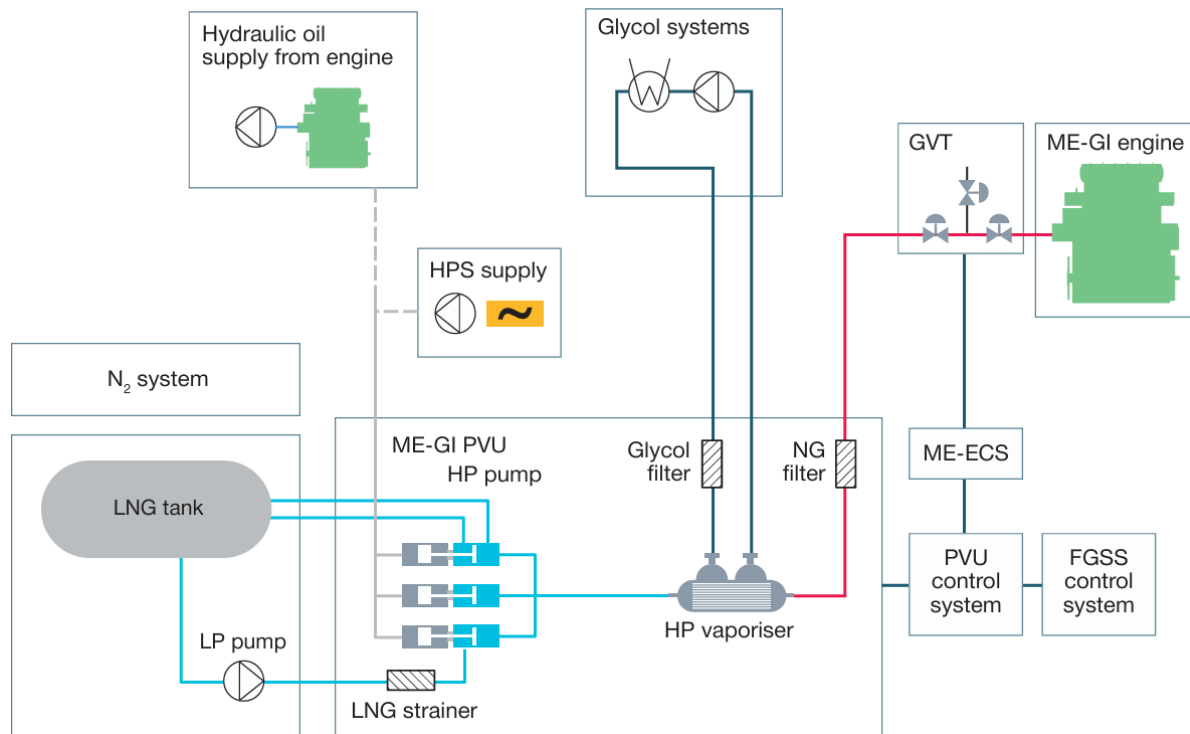


LNG (ME-GI)



LNG (ME-GI)

Fuel gas supply system (FGSS) design



LP : low-pressure

HP : high-pressure

HPS : hydraulic power supply

PVU : pump vaporiser unit

GVT : gas valve train

ECS : engine control system

Fig. 3: Schematic representation of the fuel gas supply system for G90ME-GI, G/S95ME-GI and DF gensets

LNG (ME-GI)

Fuel gas supply system (FGSS) design

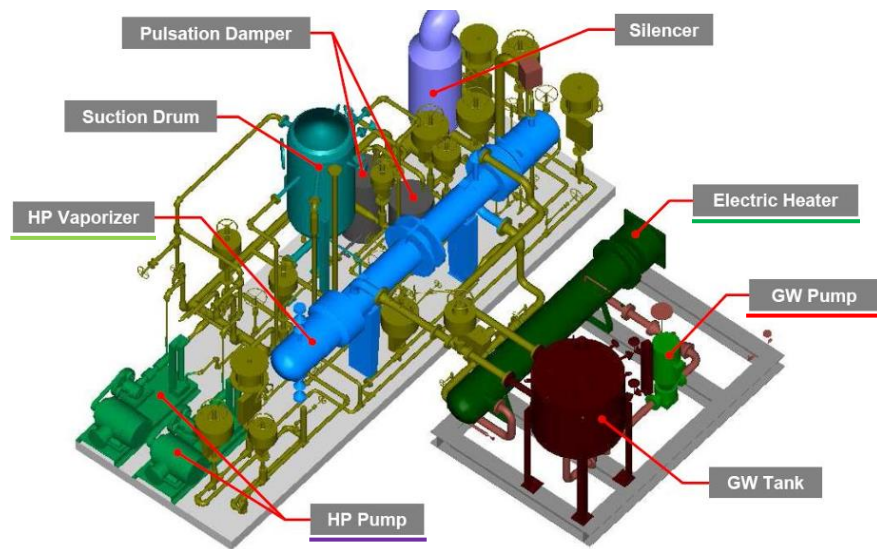
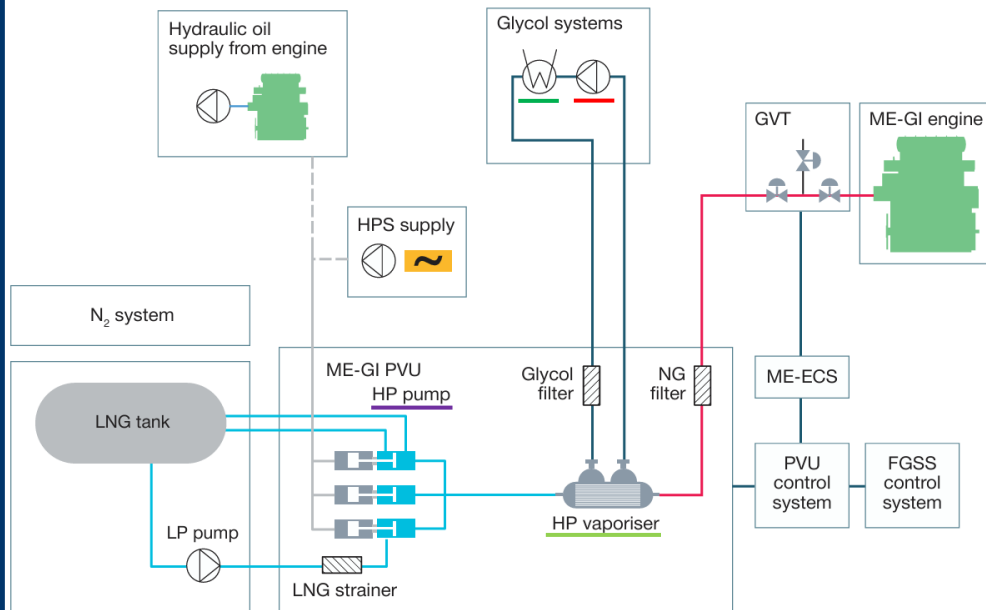


Fig. 3: Schematic representation of the fuel gas supply system for G90ME-GI, G/S95ME-GI and DF gensets

LNG (ME-GI)

Fuel gas supply system (FGSS) design

The requirements to gas supply-pressure and temperature are:

- for the ME-GI engine: high-pressure gas at 300 bar, 45°C
- for the dual-fuel gensets: low-pressure gas at 6 bar, 0-60°C

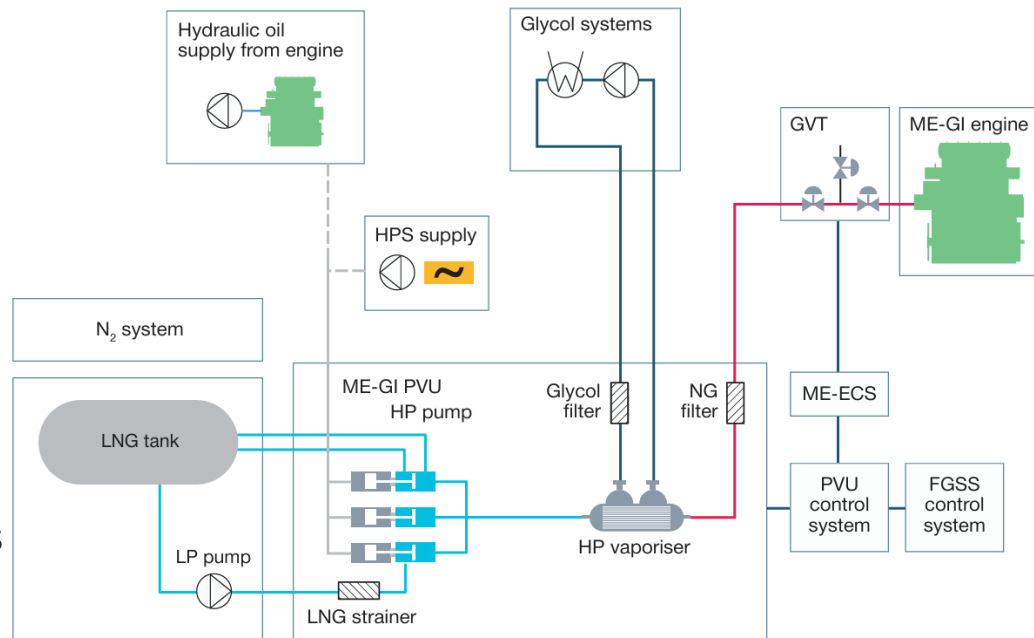


Fig. 3: Schematic representation of the fuel gas supply system for G90ME-GI, G/S95ME-GI and DF gensets

LNG (ME-GI)

Fuel gas supply system (FGSS) design

The engine control system (ME-ECS) receives information from the FGSS control system about the available amount of fuel gas and it calculates the required amount of fuel oil.

Today, the ME-GI engine operates on 3% fuel oil and 97% gas fuel.

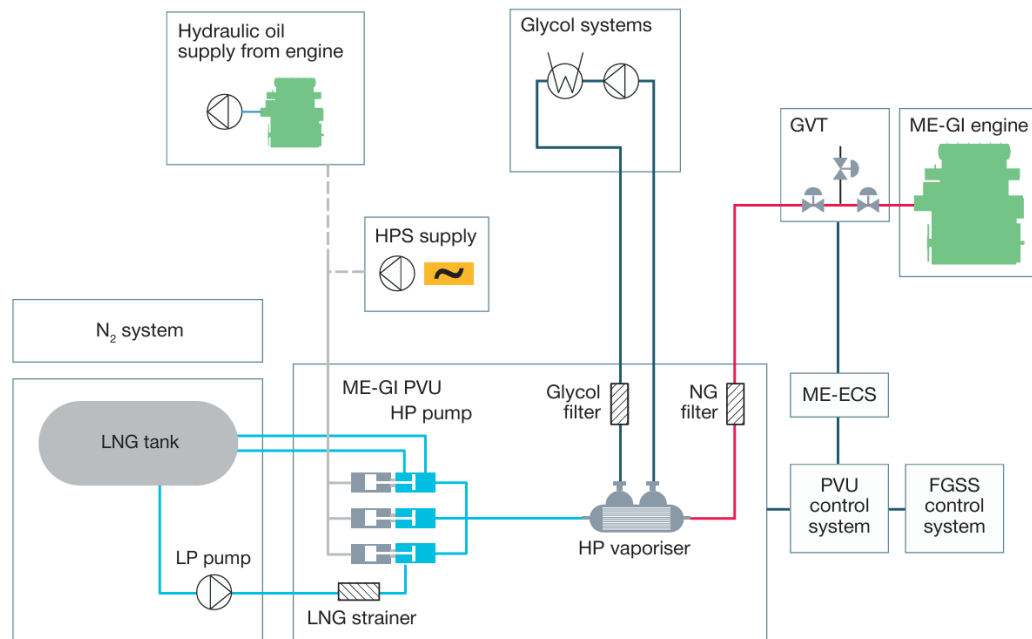
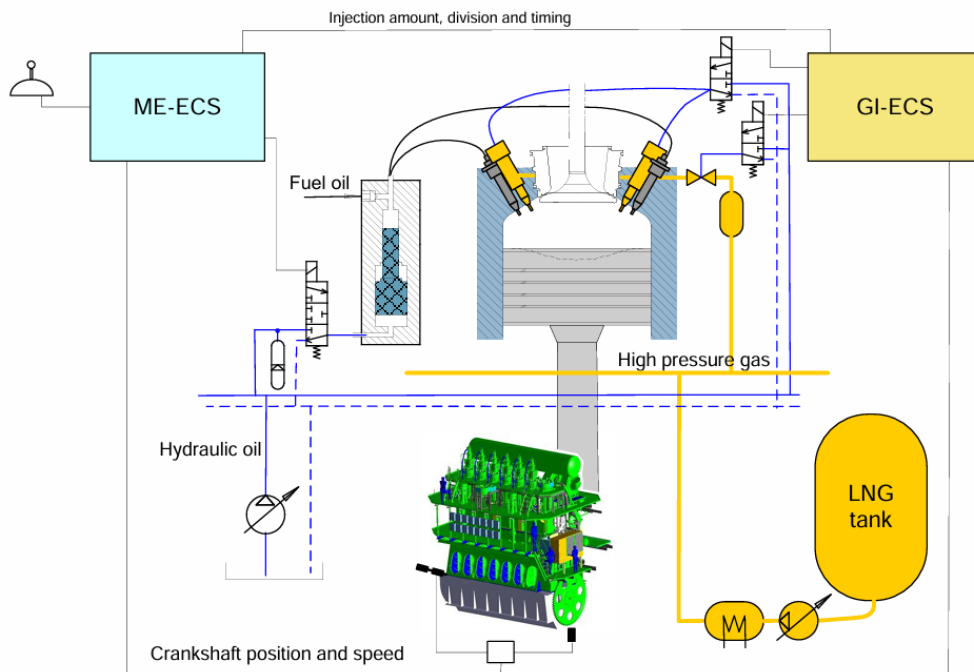


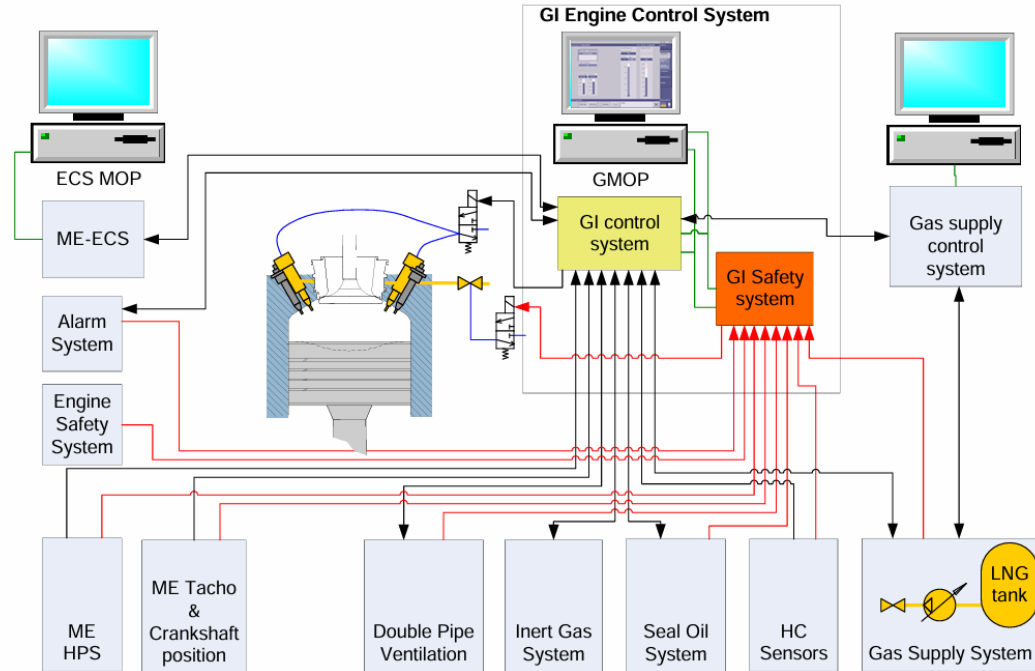
Fig. 3: Schematic representation of the fuel gas supply system for G90ME-GI, G/S95ME-GI and DF gensets

LNG (ME-GI)

GI engine, injection control



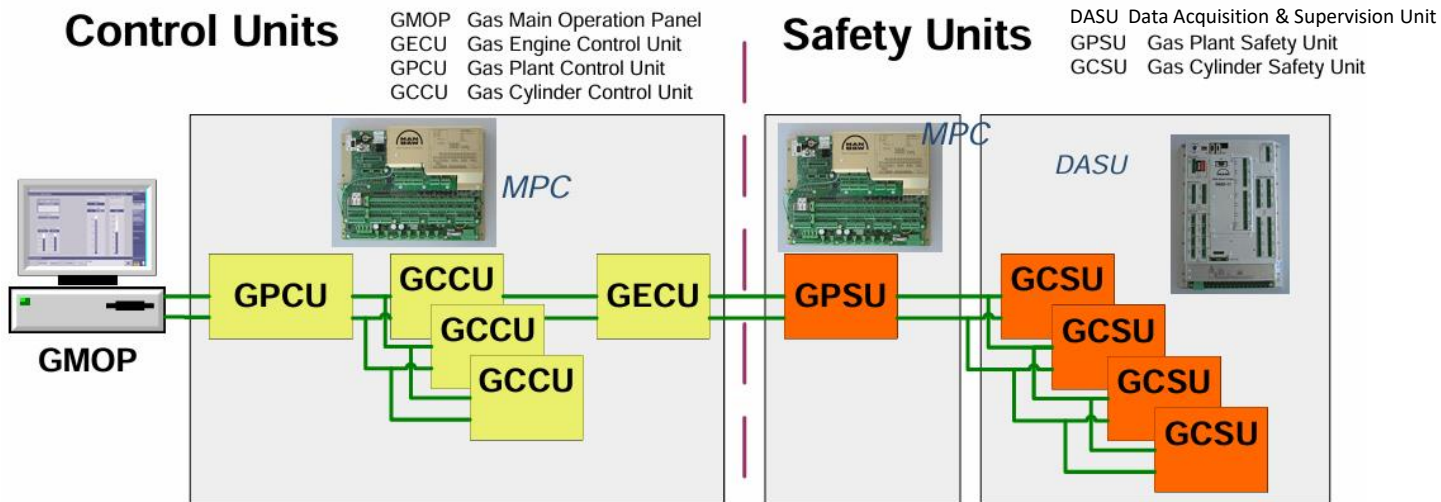
ME-GI Engine Control System Environment



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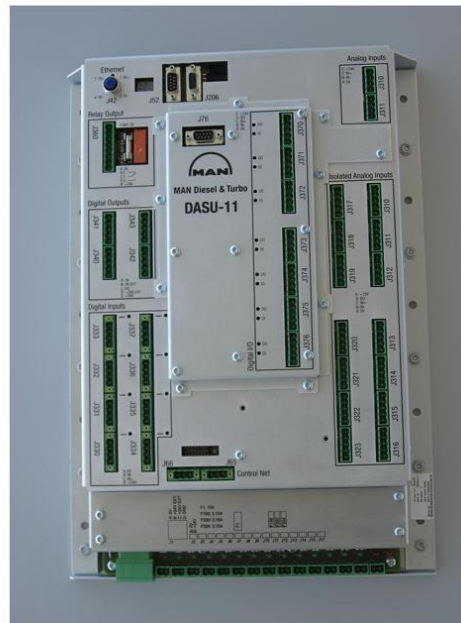
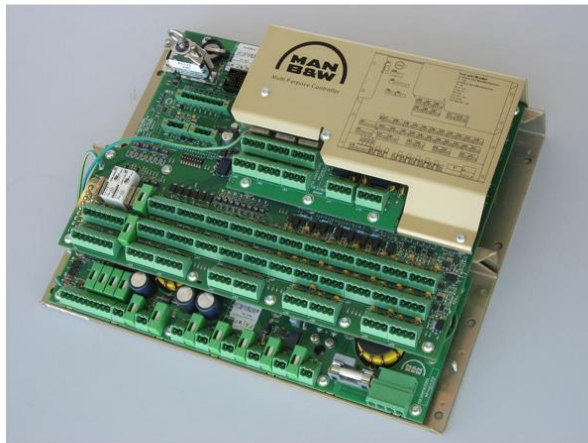
LNG (ME-GI)

GI-ECS configuration



LNG (ME-GI)

Controllers of the system

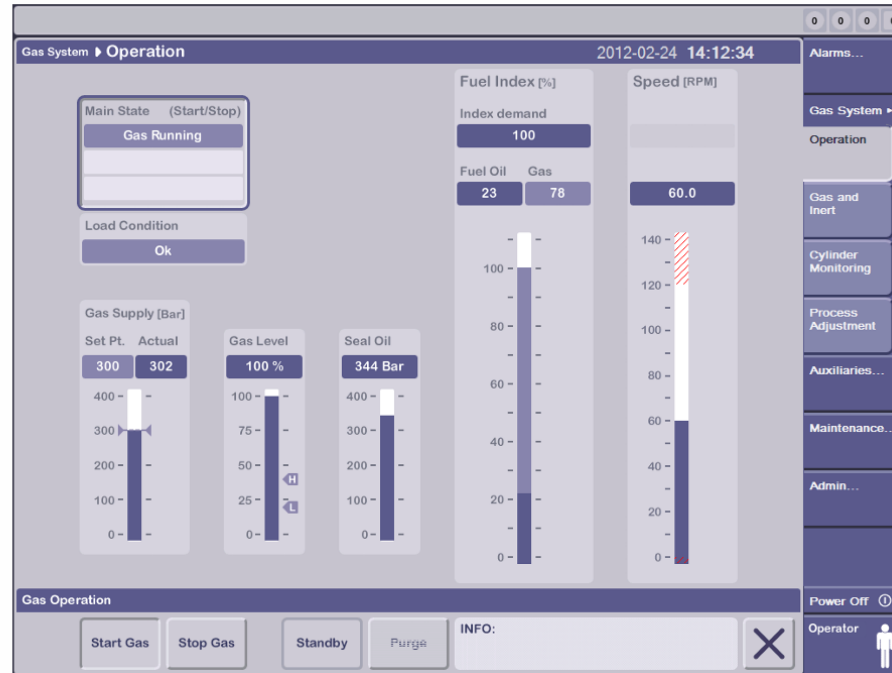


LNG (ME-GI)

ME-GI GMOP Main dashboard display

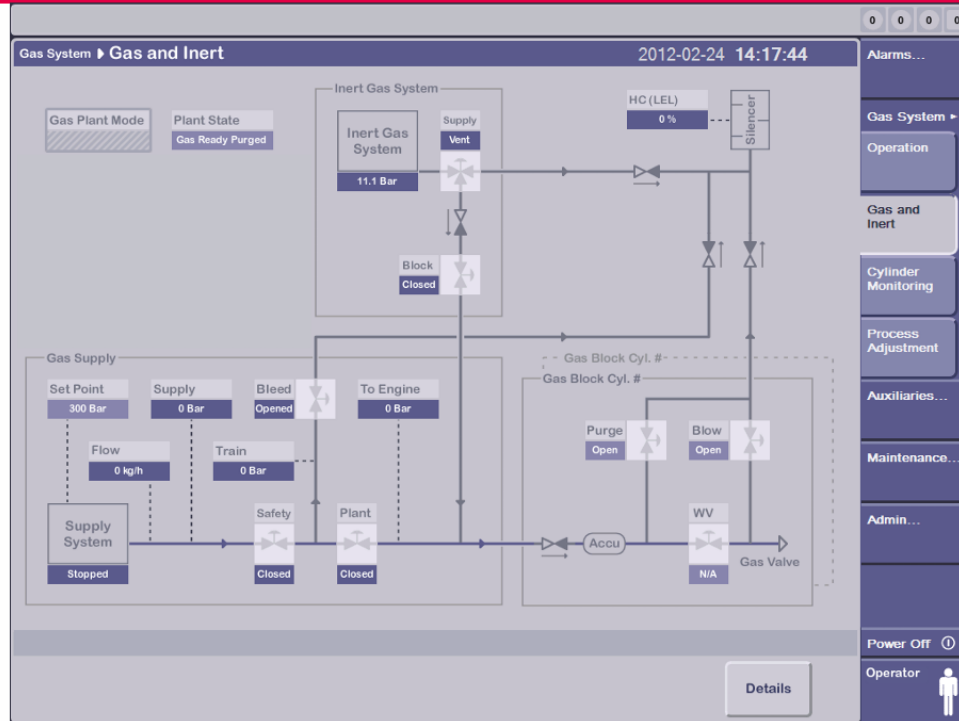


GMOP



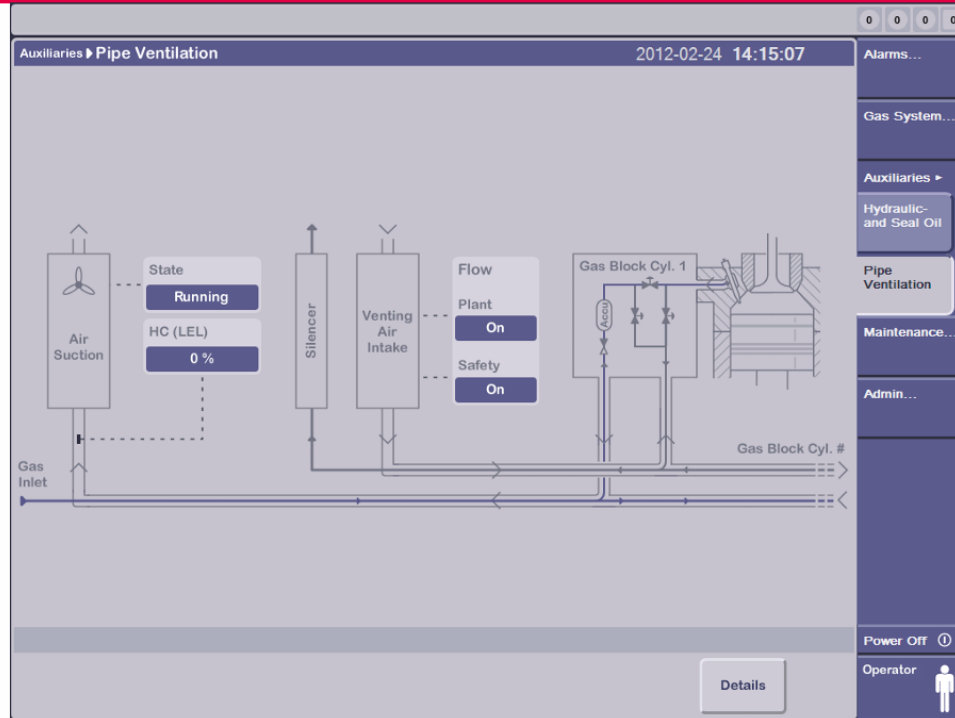
LNG (ME-GI)

ME-GI GMOP Gas supply overview



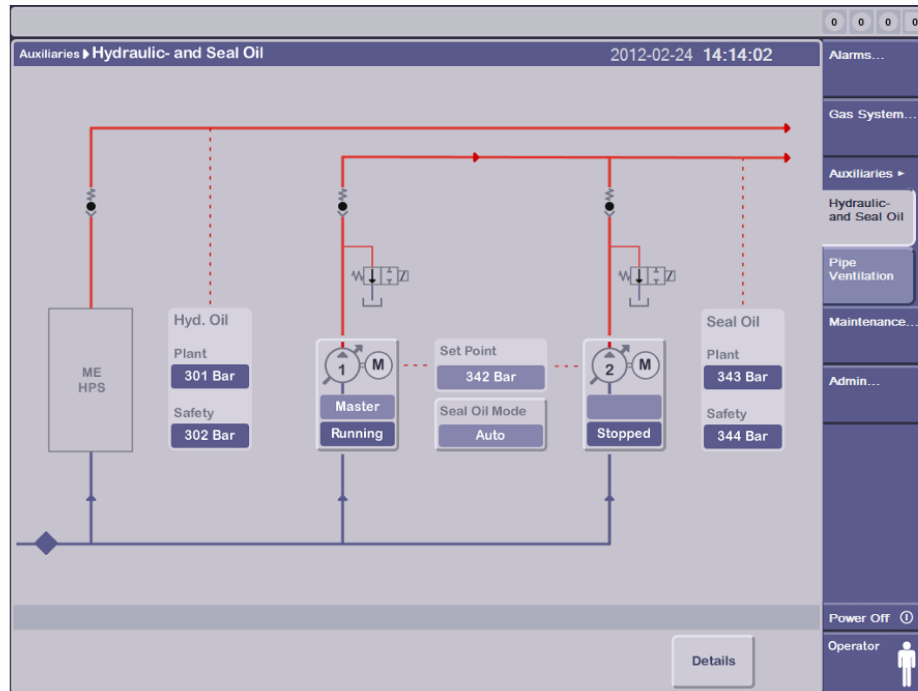
LNG (ME-GI)

ME-GI GMOP Double pipe ventilation system



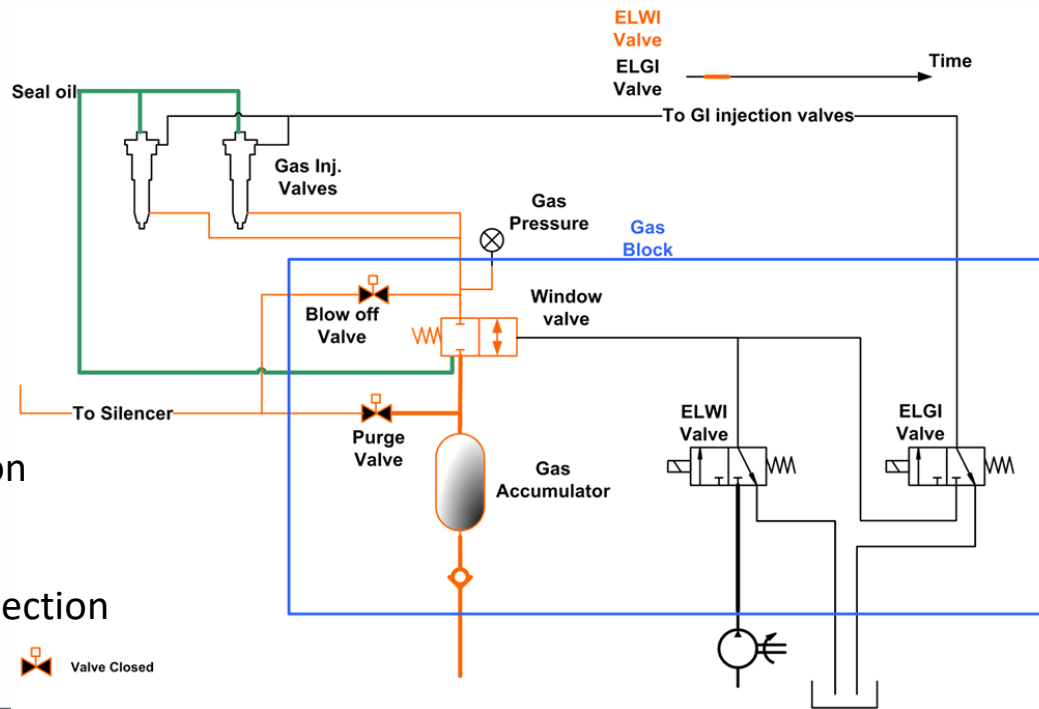
LNG (ME-GI)

ME-GI GMOP Hydraulic- and Seal oil display



LNG (ME-GI)

ME-GI Gas Injection Control



ELGI:
electronic gas injection

ELWI:
electronic window injection

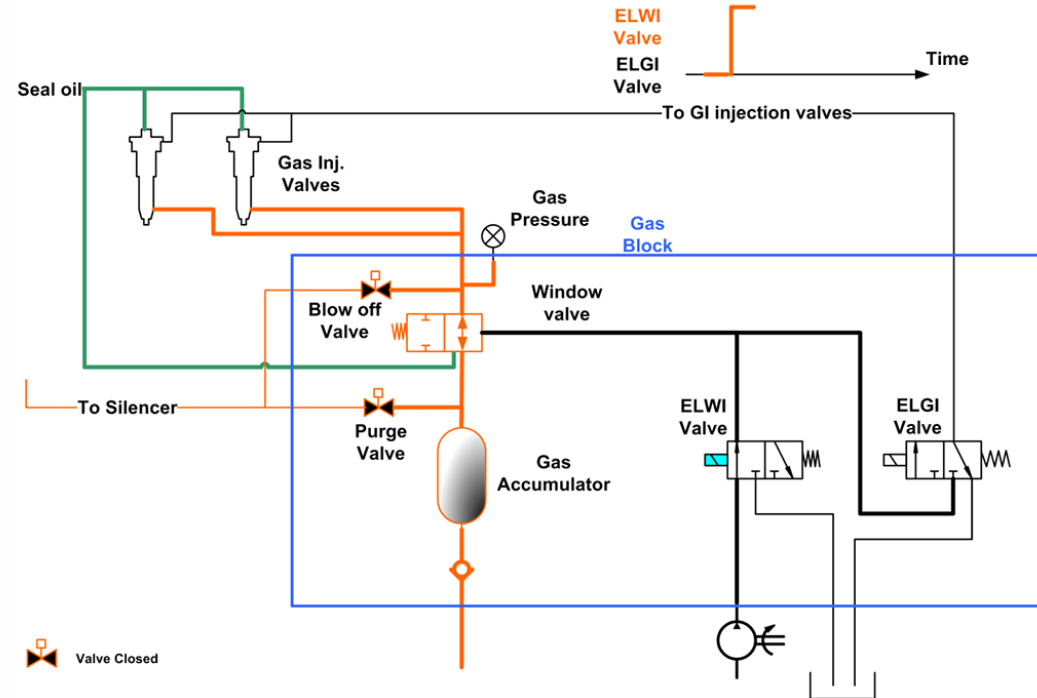


LNG (ME-GI)

ME-GI

Gas Injection Control

Step 1

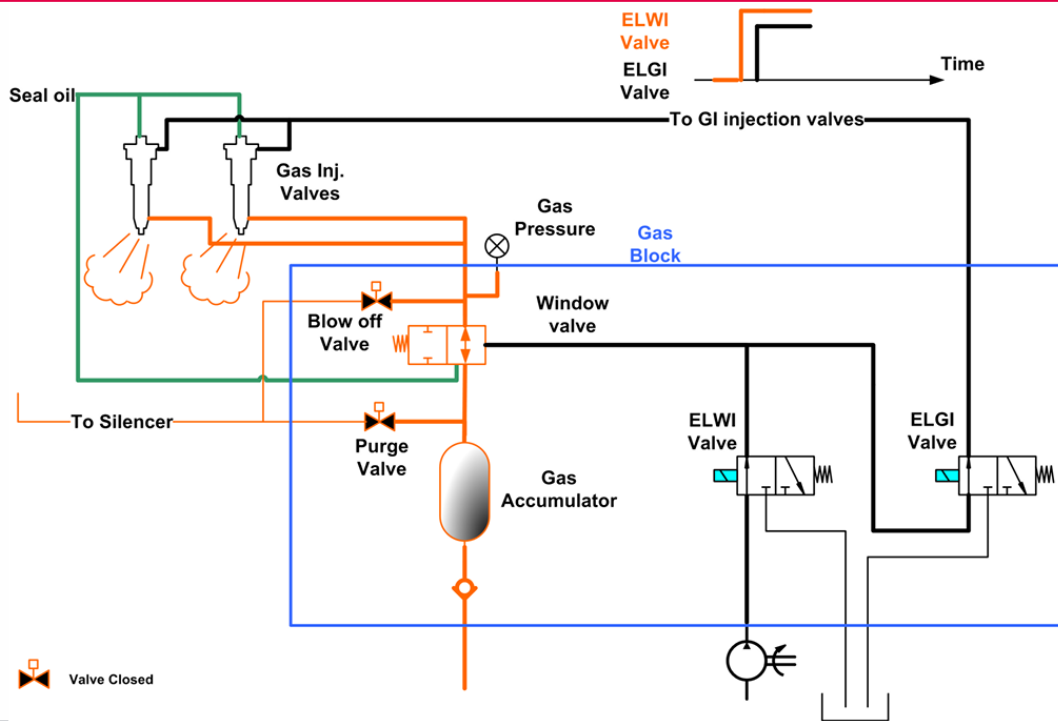


LNG (ME-GI)

ME-GI

Gas Injection Control

Step 2



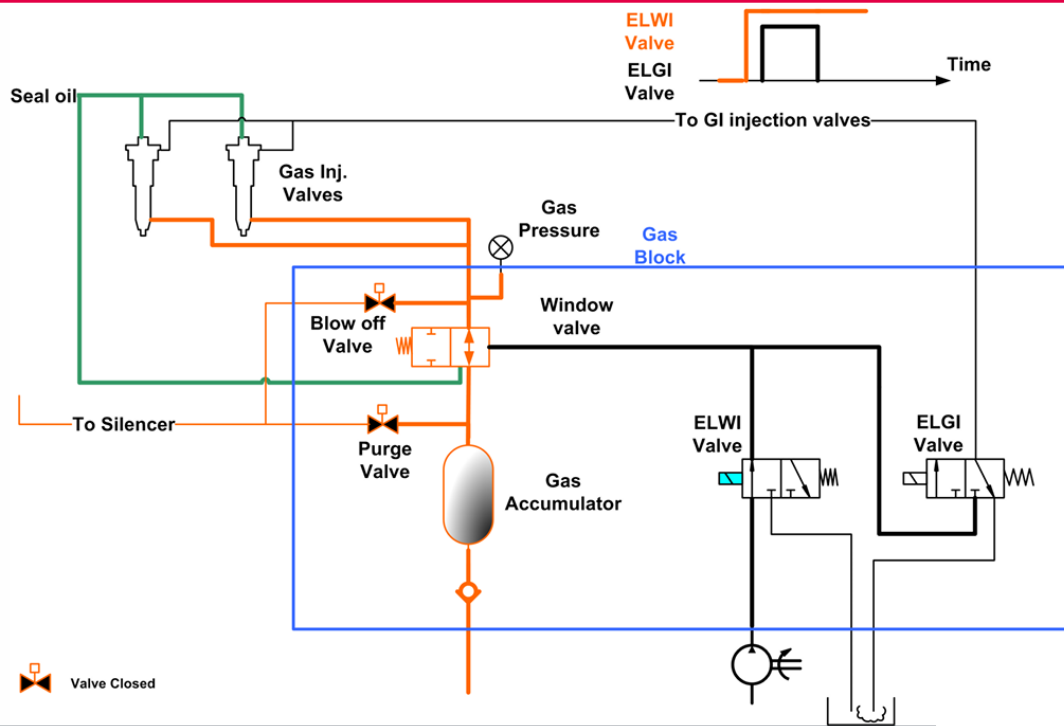
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LNG (ME-GI)

ME-GI

Gas Injection Control

Step 3

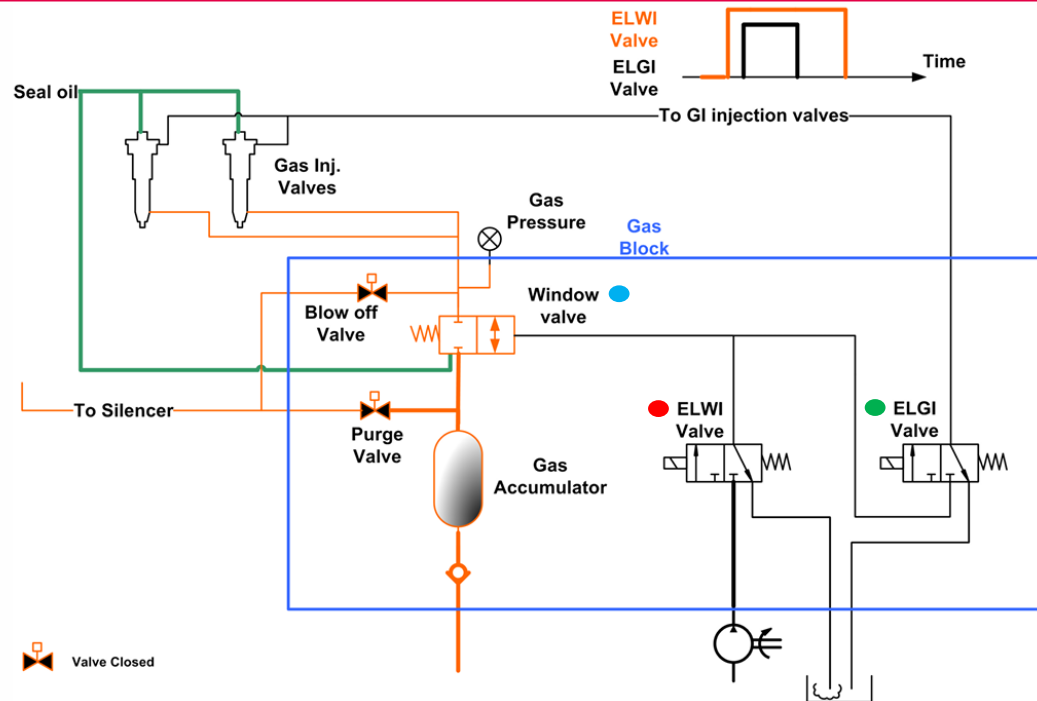


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LNG (ME-GI)

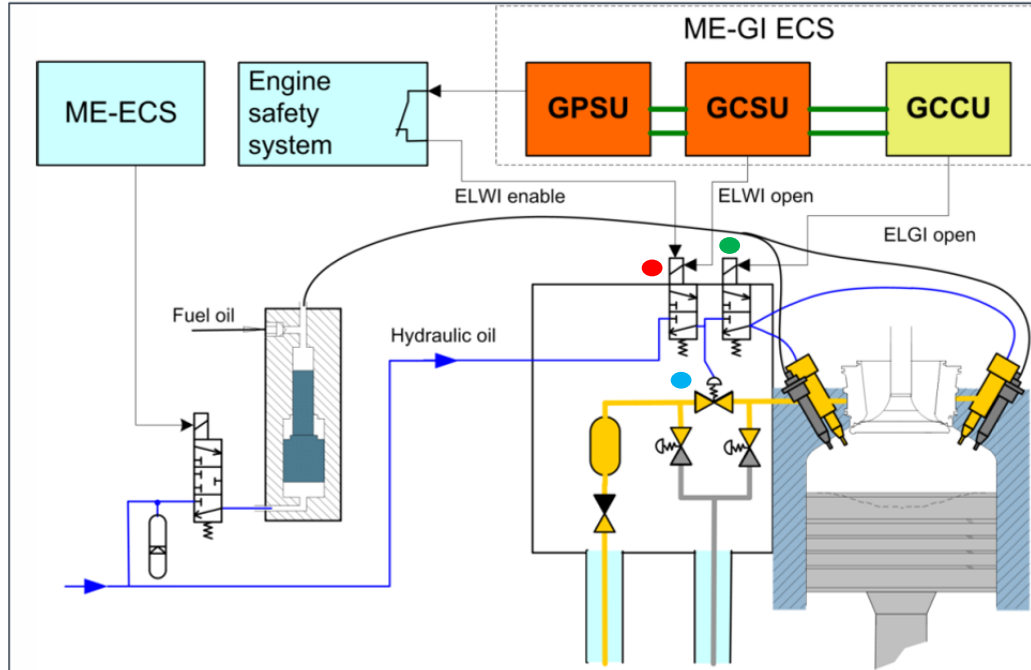
ME-GI Gas Injection Control

Step 4



LNG (ME-GI)

Gas injection - safety



LNG (ME-GI)

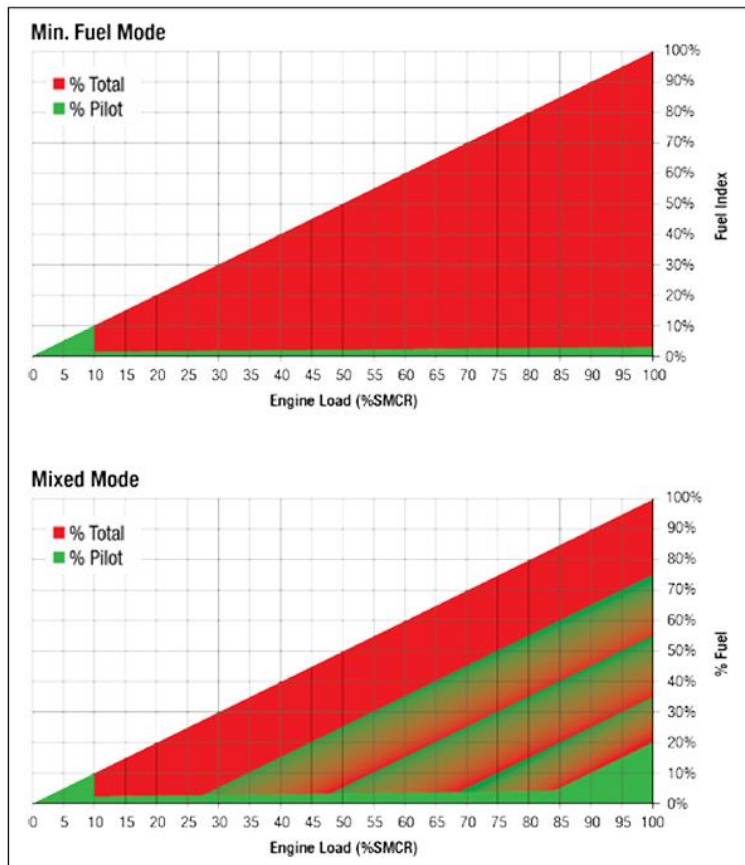


Fig. 12: Fuel type modes for the ME-GI engines for LNG carriers

Comparison of emissions from an HFO burning and a gas burning 70ME type engine

Estimated emissions 6S70ME-C

Load 100% g/kWh

CO₂ 577

O₂ (%) 1359

CO 0.64

NO_x 11.58

HC 0.19

SO_x 10.96

PM (mg/m³) 0.54

Estimated emissions 6S70ME-GI

Load 100% g/kWh

CO₂ 446

O₂ (%) 1340

CO 0.79

NO_x 8.76

HC 0.39

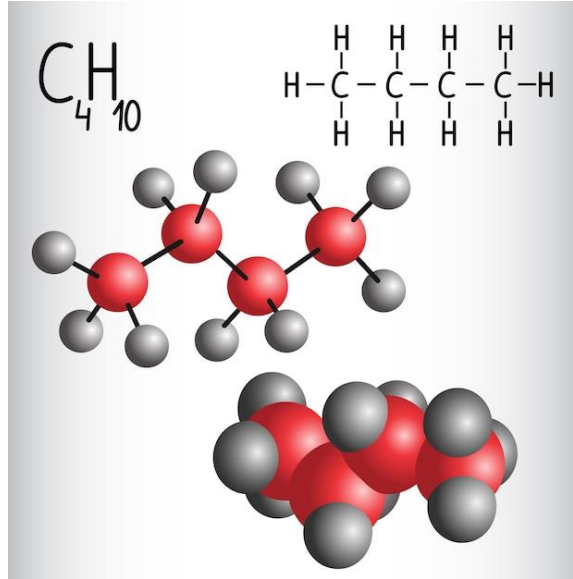
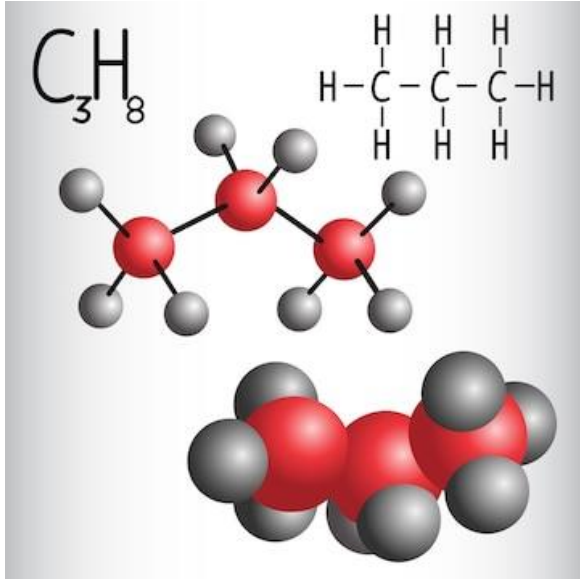
SO_x 0.88

PM (mg/m³) 0.34

Fig. 13

LPG (Liquefied Petroleum Gas)

LPG is a colourless and highly flammable gas. It is a component of natural gas and consists primarily of propane (C_3H_8), butane (C_4H_{10}), or a mixture of both. The gas can be stored, compressed as a liquid in cylinders for later use.



LPG



LPG (ME-LGIP)

ME → MAN B&W's electronically controlled engine series

LGI → Liquid Gas Injection

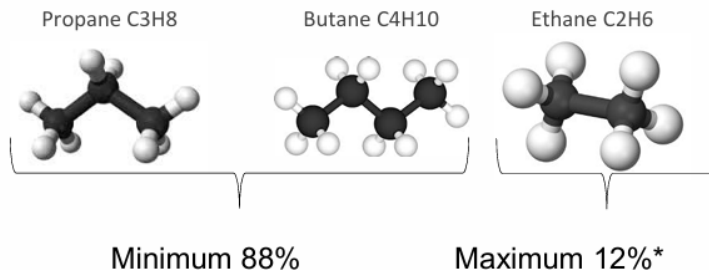
P → Propane (LPG)

ME-LGIP = Electronically Controlled Liquid Gas (LPG) Injection Engine

The new MAN B&W ME-LGIP Engine

MAN B&W ME-LGIP engines are designed to be dual fuelled with LPG as the low-flashpoint fuel.

Engine is flexible with regards to fuel composition.



**Possibility to increase ethane content is under investigation*

MAN B&W

Engine type designation

5G70ME-C10.5-GI-EGRBP

- Tier III technology

(blank)	Tier II only
EGRBP	EGR with bypass matching
EGRTC	EGR with TC cut-out matching
EcoEGR	EGR in Tier II and Tier III mode
HPSCR	High-pressure SCR
LPSCR	Low-pressure SCR

- Fuel injection concept

(blank)	Fuel oil only
GI/GA	Gas injection/gas admission methane
GIE	Gas injection ethane
LGIM	Liquid gas injection methanol
LGIP	Liquid gas injection LPG

- Dot (.) number

- Mark number

- Engine concept

ME-C	Electronically controlled
ME-B	Exhaust valve controlled by camshaft

- Diameter of cylinder bore in cm

- Stroke/bore ratio

G	'Green' ultra long stroke
S	Super long stroke

- Number of cylinders

LPG (ME-LGIP)

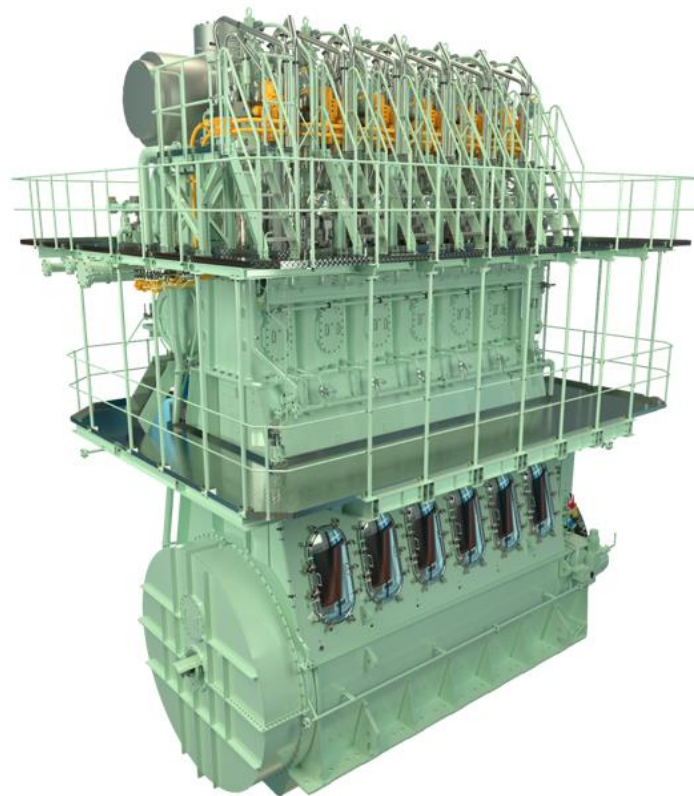
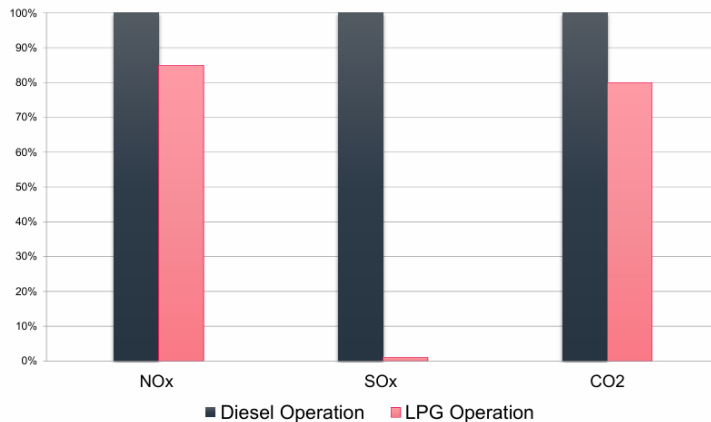
ME-LGIP Conversion

Design



Emission reductions

- **NOx (Nitrogen Oxide)**
 - Reduction ~ 15%
- **SOx (Sulphur Oxide)**
 - Reduction > 90%
- **CO₂ (Carbon Oxide)**
 - Reduction ~ 20%



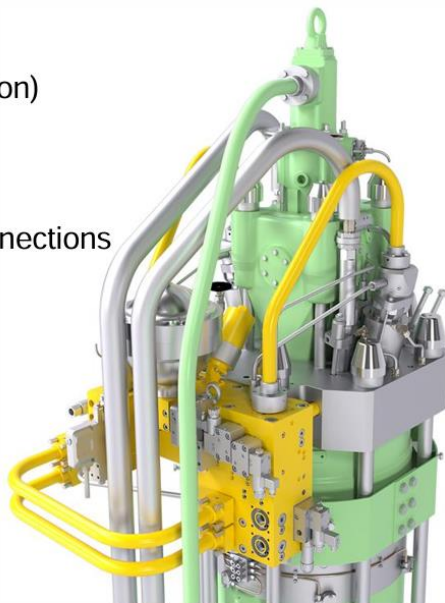
LPG (ME-LGIP)

The new MAN B&W ME-LGIP engine

Same system to be used for NH₃

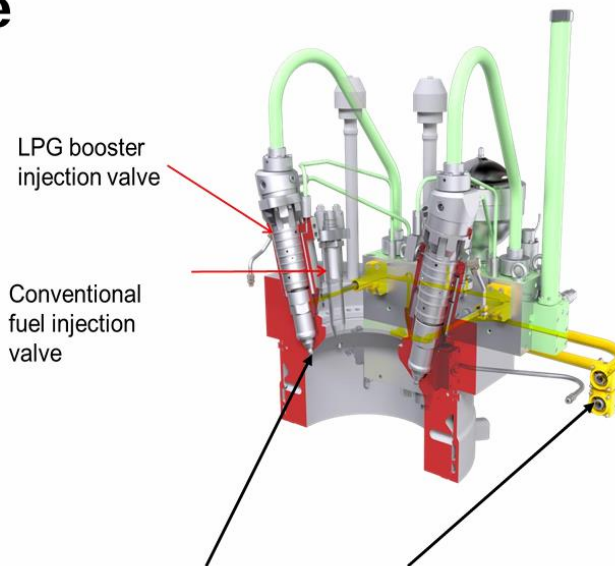
Valve control block:

- ELWI-valve (fuel pressurization)
- ELGI-valve (injection timing)
- Hydraulic accumulator
- Hydraulic and sealing oil connections



Double-wall gas piping:

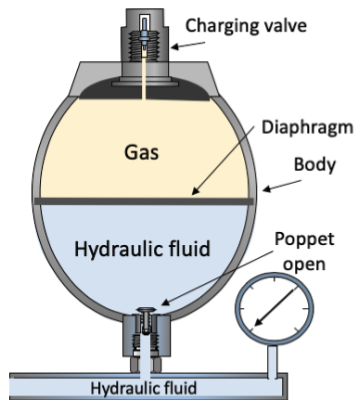
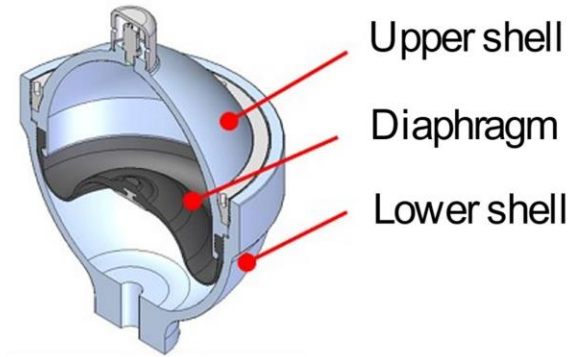
- LPG inlet
- LPG return



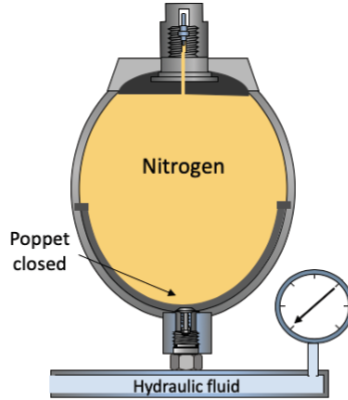
	Injection pressure	Supply pressure	
	bar		
ME-GI	200-380	200-380	Common rail
ME-LGI	600-800	8-50	Pressure booster valve

Hydraulic accumulator

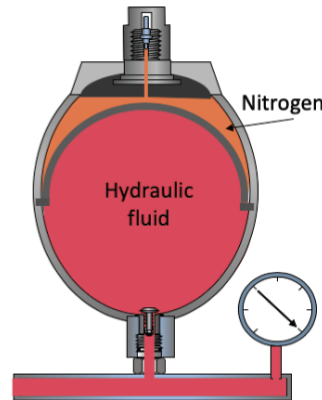
A hydraulic accumulator is a critical component that maintains constant pressure, stores and recaptures energy, reduces pressure peaks, and dampens shock, vibration, and pulsations to make the mechanical system run smoothly.



Diaphragm accumulator
before precharge

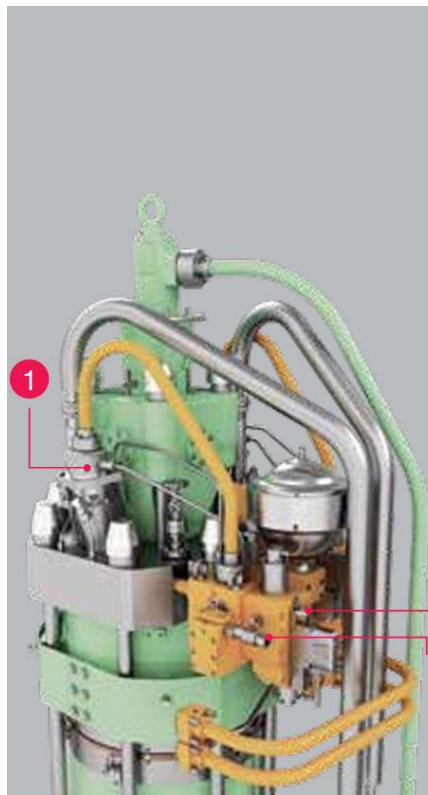


Diaphragm accumulator
precharged

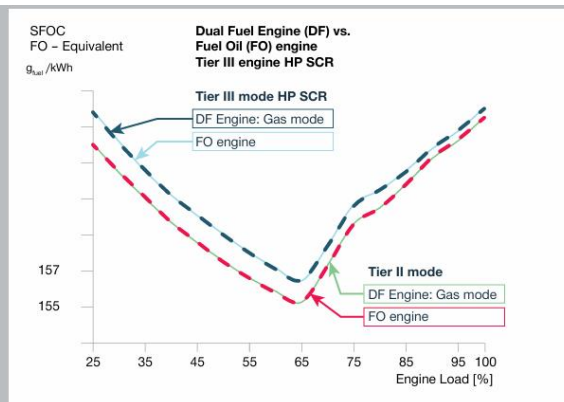


Diaphragm accumulator,
fully charged

LPG (ME-LGIP)



Comparison of
fuel consumption



Cylinder cover
with LPG injection valve
and gas block

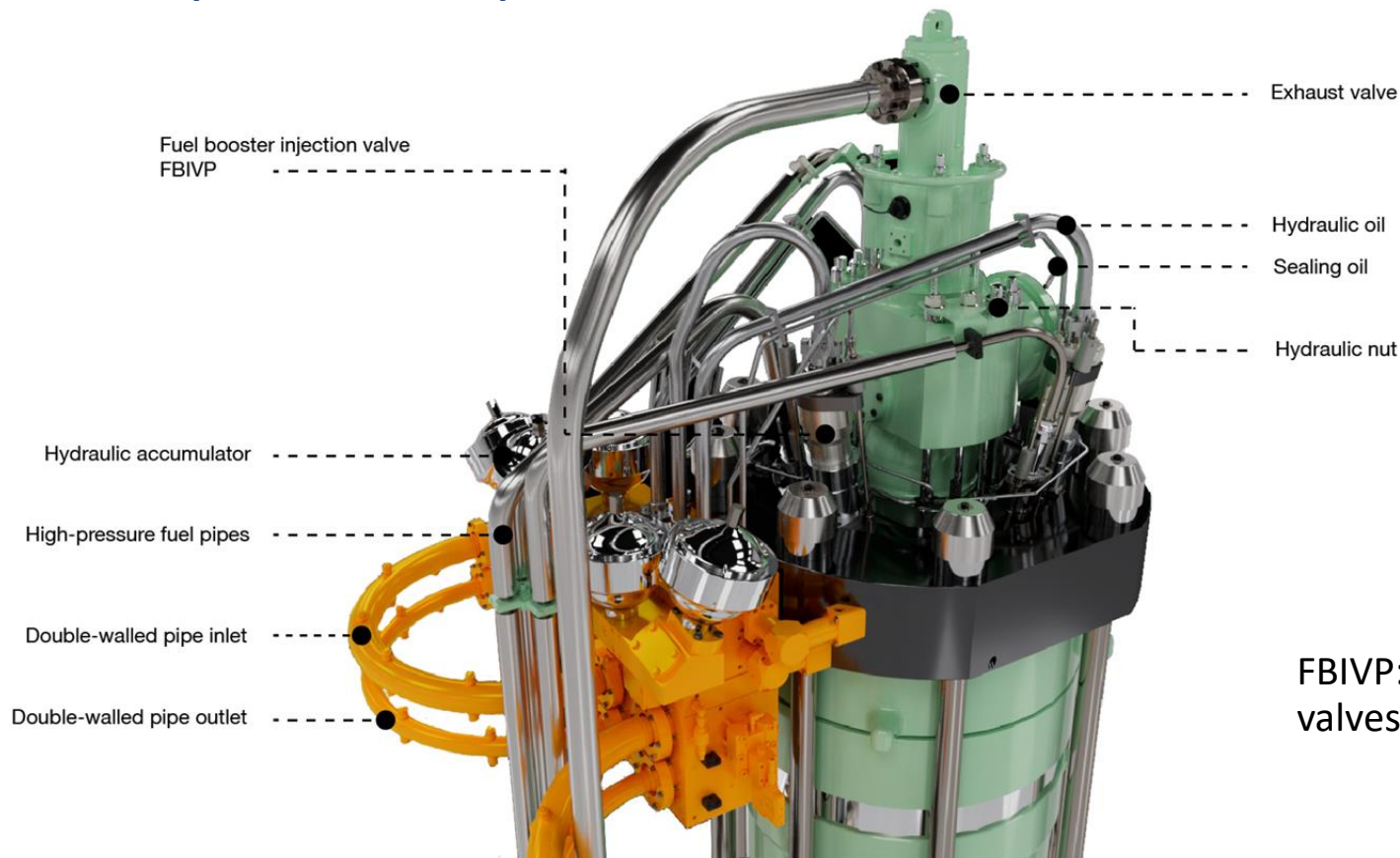
- ① LPG injection valve
- ② ELGI valve (injection timing)
- ③ ELWI valve (fuel pressurization)

ELGI: electronic gas injection
ELWI: electronic window injection



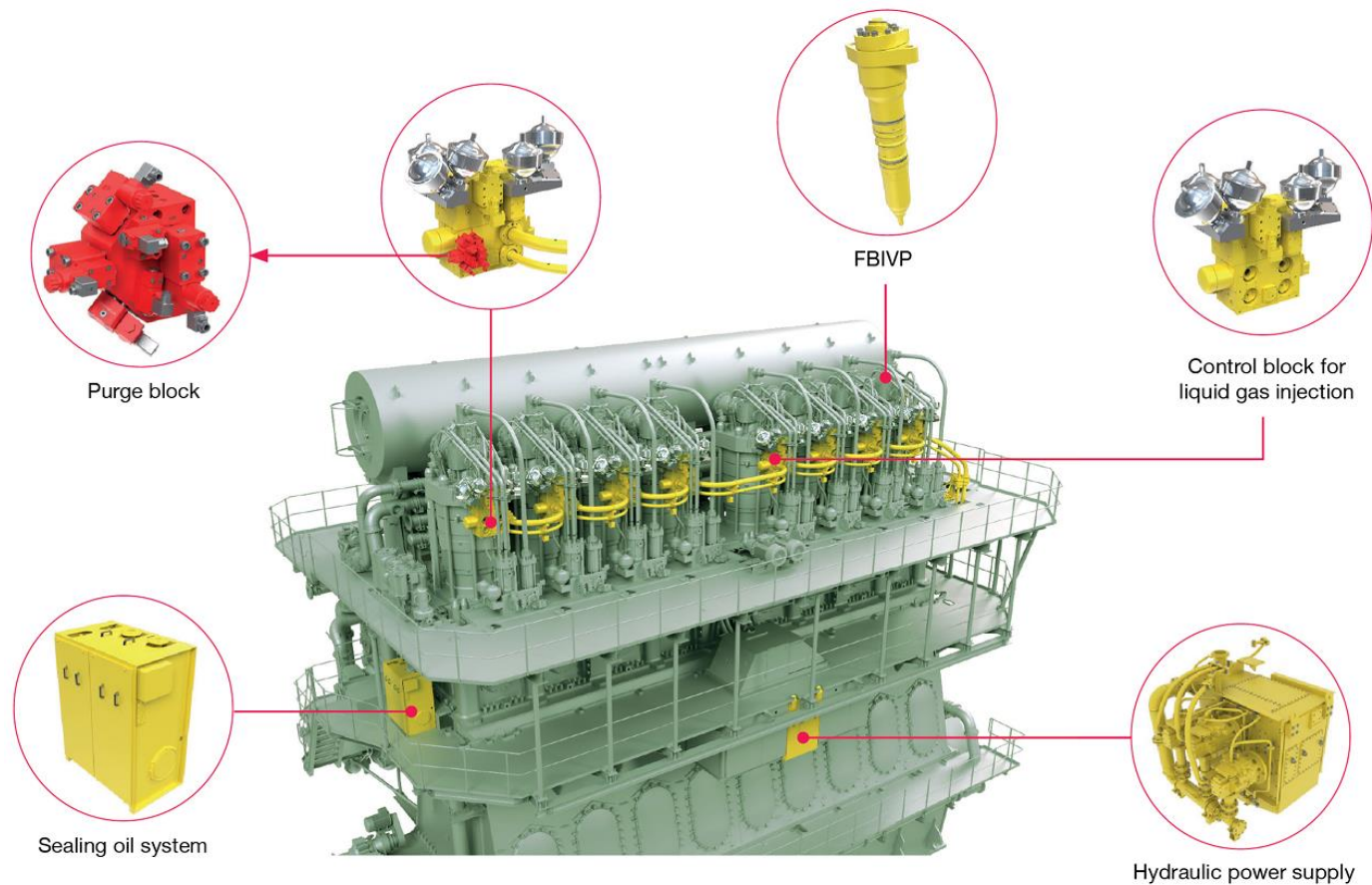
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LPG (ME-LGIP)



FBIVP: fuel booster injection valves for propane

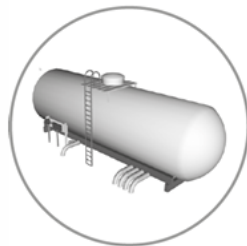
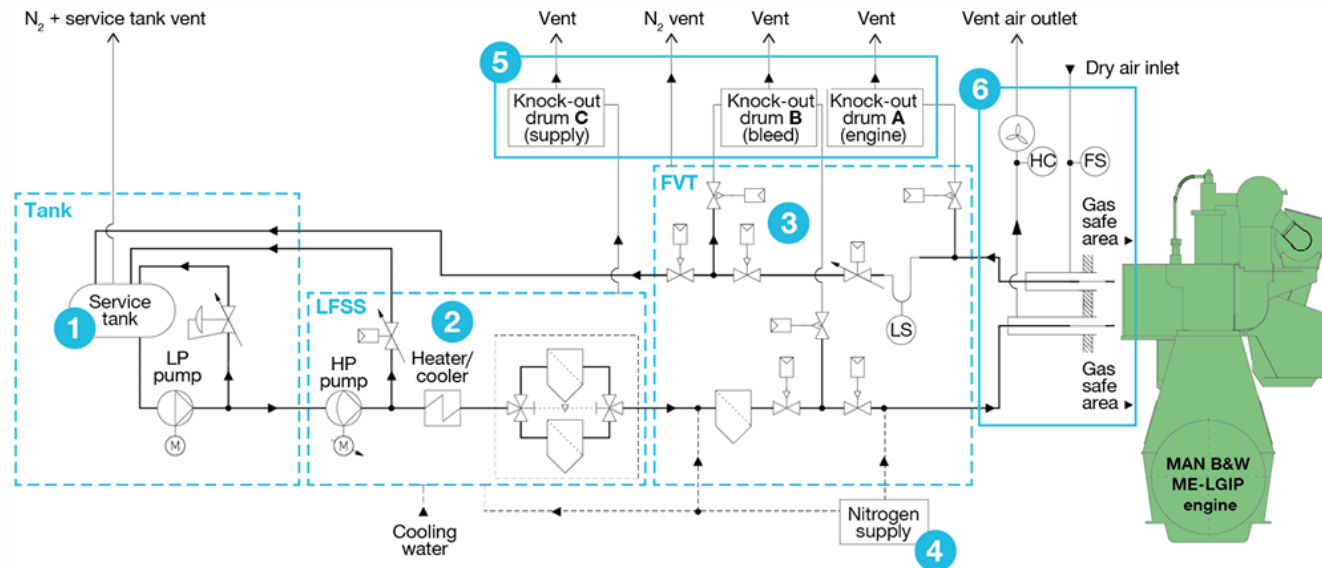
LPG (ME-LGIP)



LPG

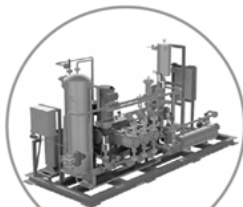
The MAN B&W ME-LGIP engine

This engine type can be modified to burn ammonia as well.



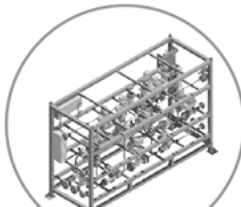
1

LPG service tank



2

Low-flashpoint fuel supply system



3

Fuel valve train



4

Nitrogen storage



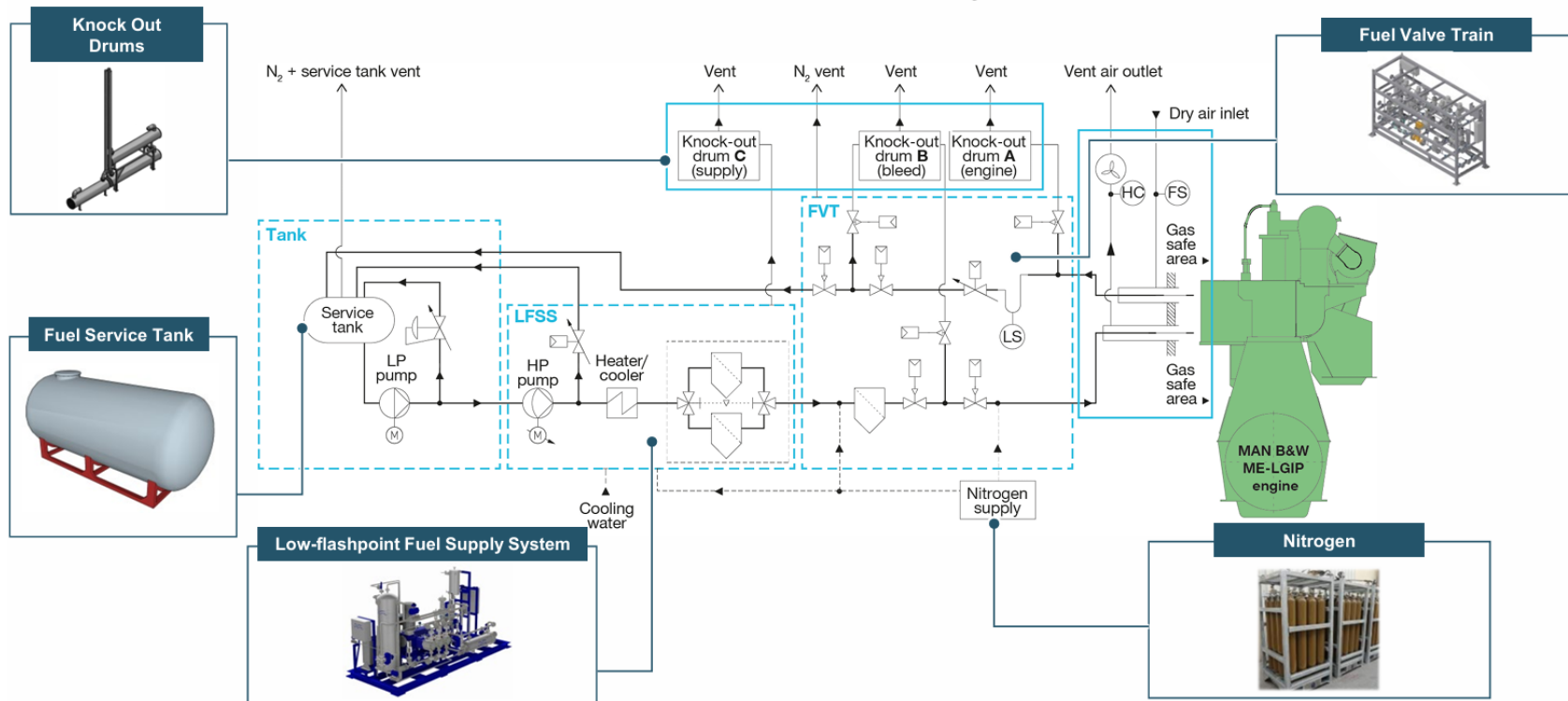
5

Knock-out drums

LPG

The new MAN B&W ME-LGIP Engine

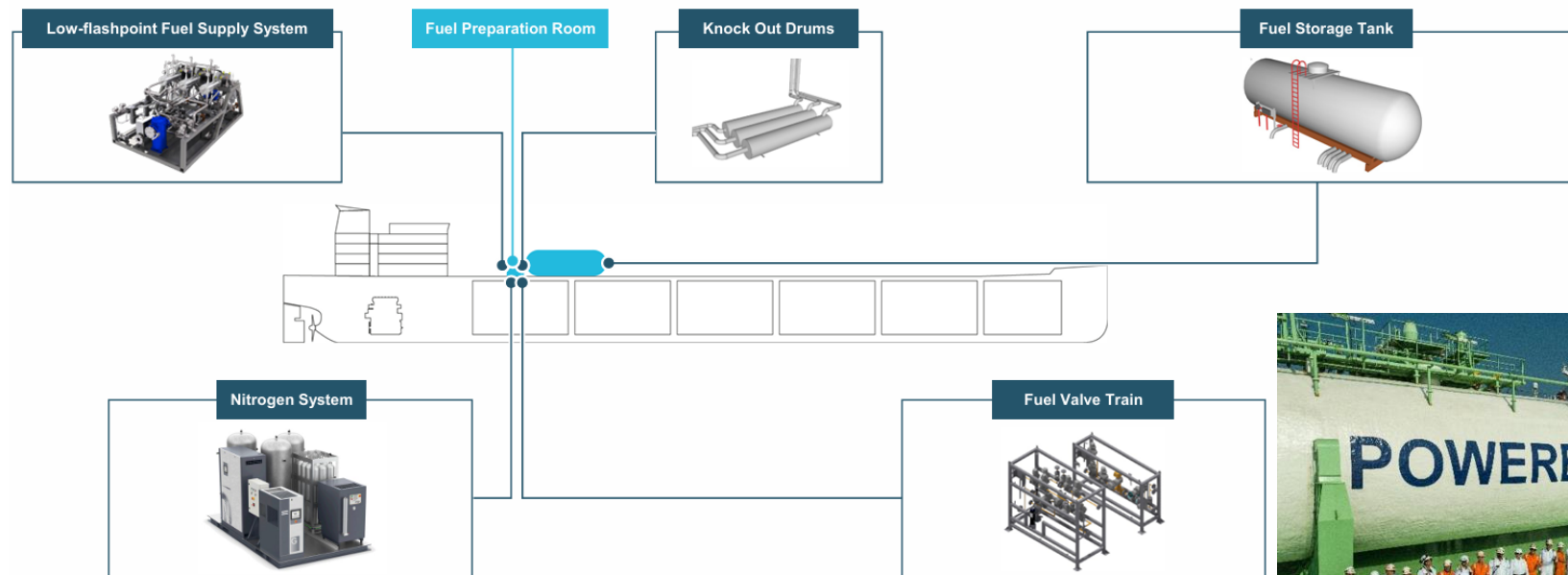
ME-LGIP auxiliaries – Test centre setup – Research Centre Copenhagen



LPG (ME-LGIP)

The new MAN B&W ME-LGIP Engine

ME-LGIP auxiliaries – LR1 Tanker – with large capacity fuel storage tank(s)



LPG (ME-LGIP)

Pilot oil injection amount versus engine load

Dual-fuel operation is possible down to 10% load.

The minimum pilot oil amount in dual-fuel mode is 5% at MCR (in L1), see Fig. 7.00.03.

An engine output with minimum pilot oil amount can be obtained even with an LCV of the LPG as low as 38 MJ/kg. Below 38 MJ/kg, a higher pilot oil amount might be required.

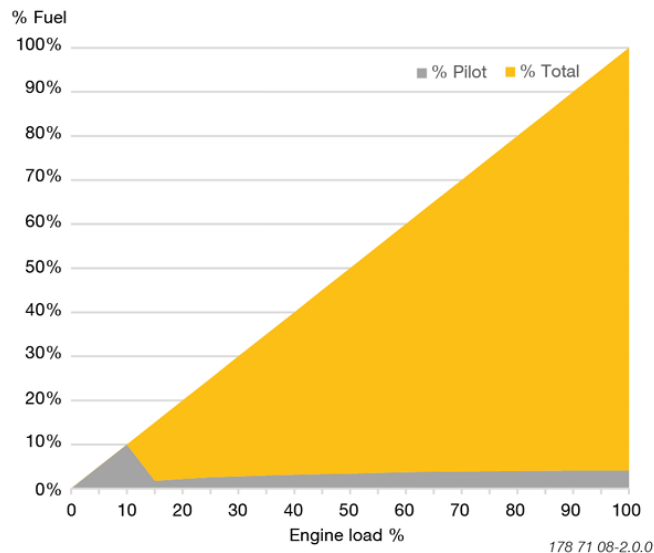


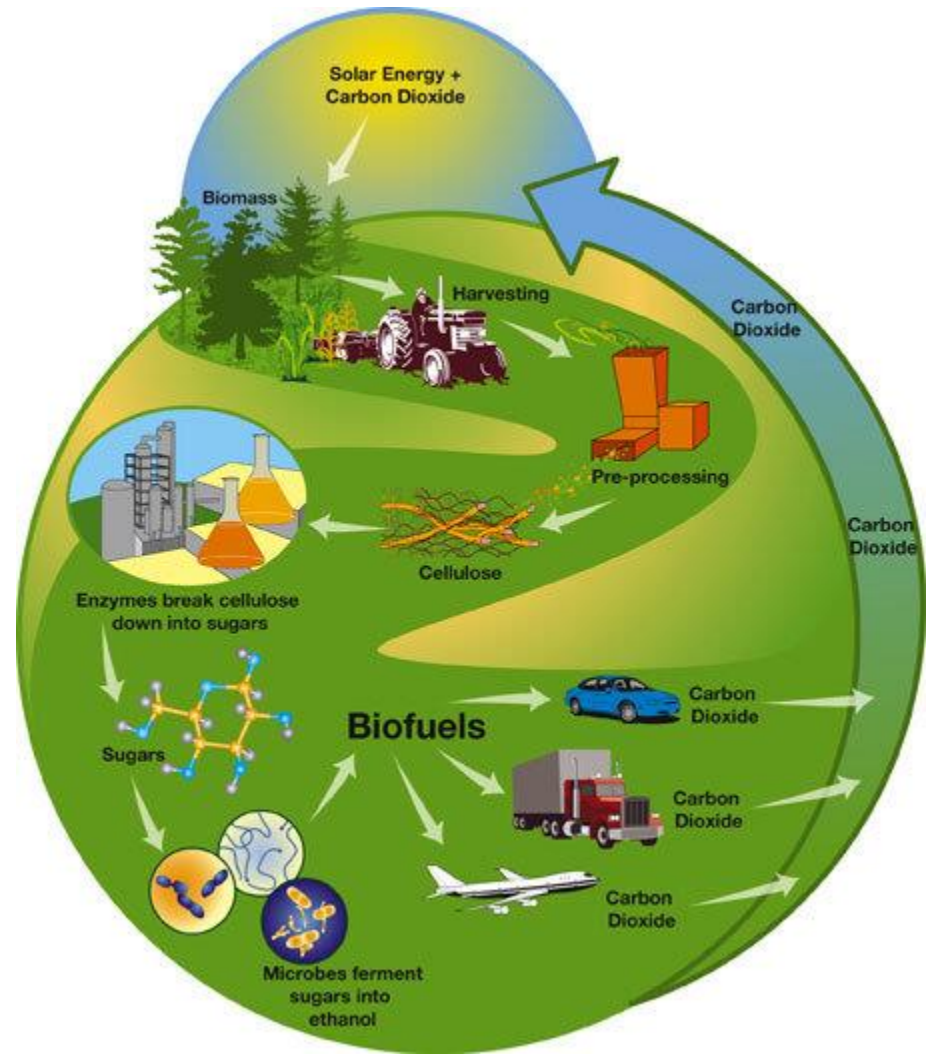
Fig. 7.00.03: Fuel index in dual-fuel mode

Biofuel

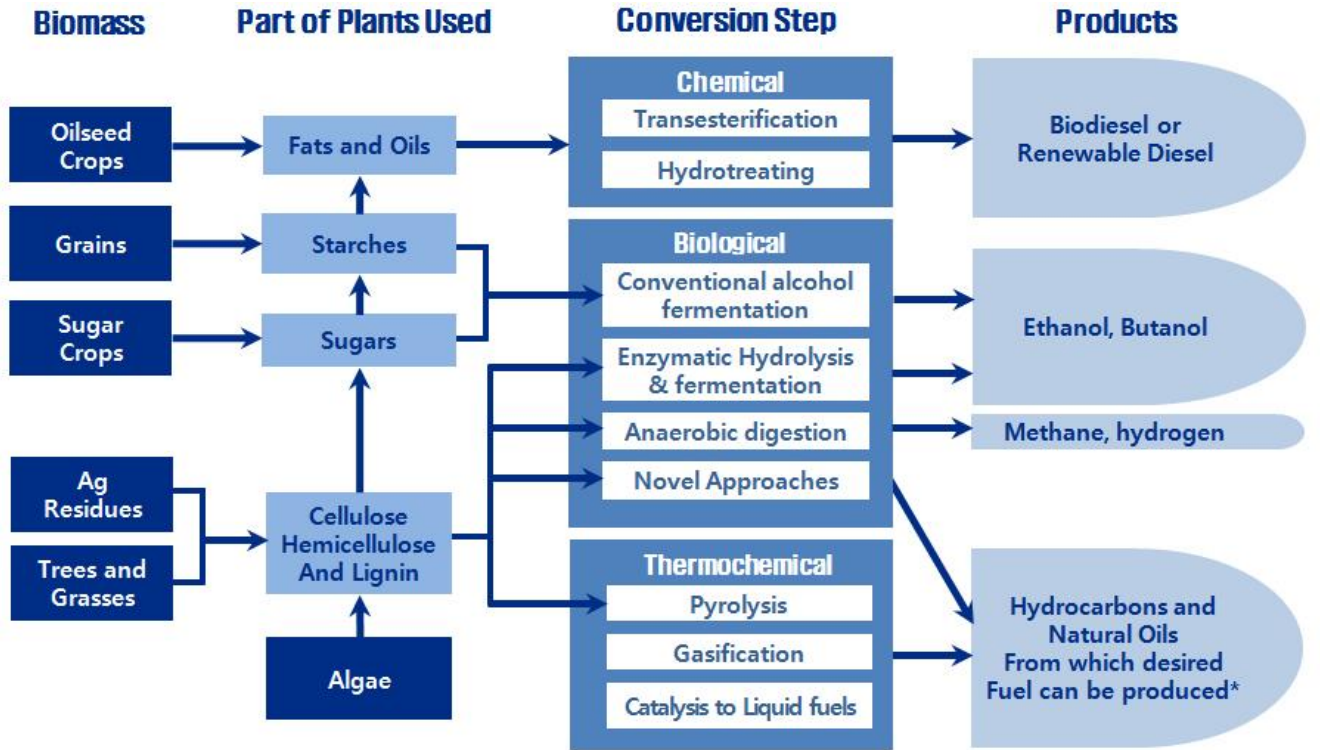
Biofuels produced from biomass are decomposed into water and carbon dioxide during combustion.

The carbon dioxide discharged is re-absorbed by biomass by photosynthesis as the organism grows and is used to synthesize the components of the biomass.

Therefore, biofuels are carbon neutral fuels that are recycled without increasing the concentration of the entire carbon dioxide in the atmosphere caused by bioethanol combustion.



Biofuel



Source: Peria and Sheehan, 2007.

*e.g. gasoline or diesel equivalents, syngas, and hydrogen

Biofuel conversion technology flow from biomass



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Biofuel

HMM accelerates green sailing service with marine biofuel

<Seoul, 15 September 2023> - HMM announced that it conducted a trial use of a sustainable marine biofuel on its 6,400TEU containership HMM TACOMA calling in Busan, South Korea.

HMM TACOMA, operated in FIL service connecting Far East, India and Latin America, was refuelled with the biofuel blend during its bunkering at the Port of Busan.

This marine biofuel, supplied by GS Caltex, consists of 30 percent biodiesel derived from used cooking oil and 70 percent high-sulfur fuel oil(HSFO), reducing greenhouse gas emissions by 24% compared to conventional fuels. Biofuel has the merit of being compatible with current ship engines without technological modification.

HMM plans to expand the use of biofuel gradually, reaching about 5-10% of annual fuel consumption. Despite the higher cost of biofuel, HMM remains committed to heavily investing in the pathway to carbon neutrality, including the introduction of clean alternative energies.

An HMM official said, "We continue to find a way to go green, making meaningful progress toward a carbon-free future. To this end, we will seek to enhance our environmental competence and thereby be positioned as a top-rated carrier in responding to climate issues."

Last April, HMM signed a Memorandum of Understanding (MoU) with GS Caltex to secure marine biofuels.

Meanwhile, HMM has strived to extend its reach to the environmental field.



Biofuel

In 2021, HMM completed the first test voyage using biofuel on its 13,100TEU containership HMM DREAM. HMM used a 20% biofuel blend at the time based on very low-sulfur fuel oil(VLSFO).

In February this year, HMM signed newbuilding contracts for nine 9,000 TEU methanol-powered vessels and, last April, reinforced the partnership with LOTTE Fine Chemical to explore feasible solutions to the ammonia supply chain. The company is also preparing operational tests of an onboard carbon capture system (OCCS) for its containership.



[Image] HMM TACOMA was refueled with the biofuel sourced by GS Caltex at the Port of Busan

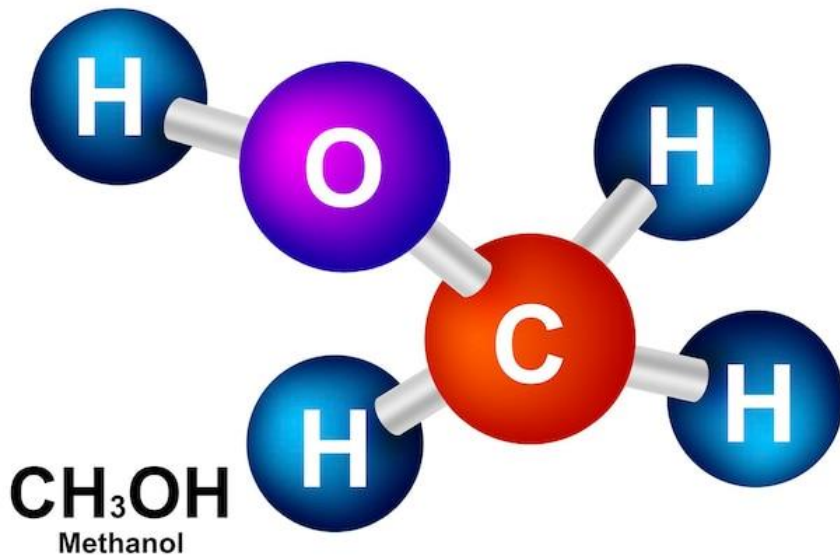


Methanol

Methanol (also called methyl alcohol and wood spirit, amongst other names) is an organic chemical compound and the simplest aliphatic alcohol, with the chemical formula CH_3OH .

It is a light, volatile, colorless and flammable liquid, and is more acutely toxic than the ethanol.

Methanol acquired the name wood alcohol because it was once produced through destructive distillation of wood. Today, methanol is mainly produced industrially by hydrogenation of carbon monoxide.



Methanol

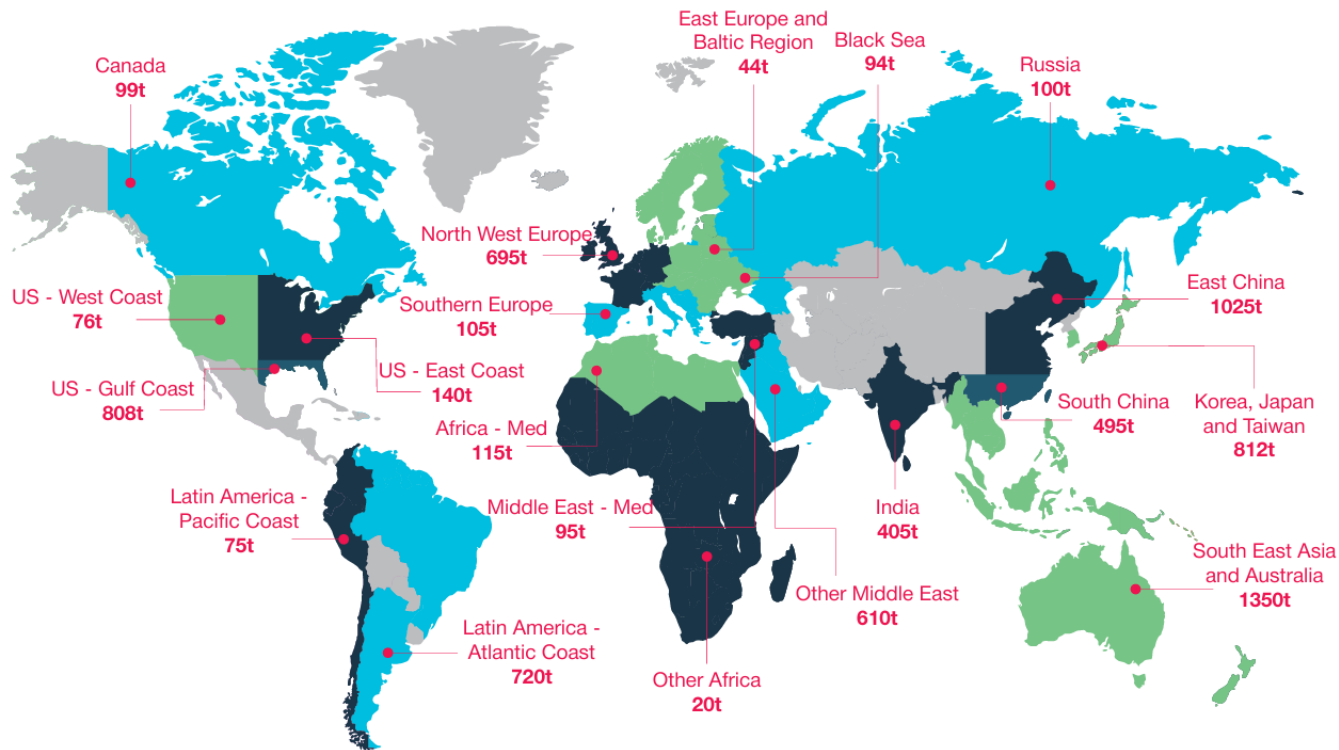


Fig. 2: Methanol storage hubs worldwide (courtesy of the Methanol Institute)

Methanol

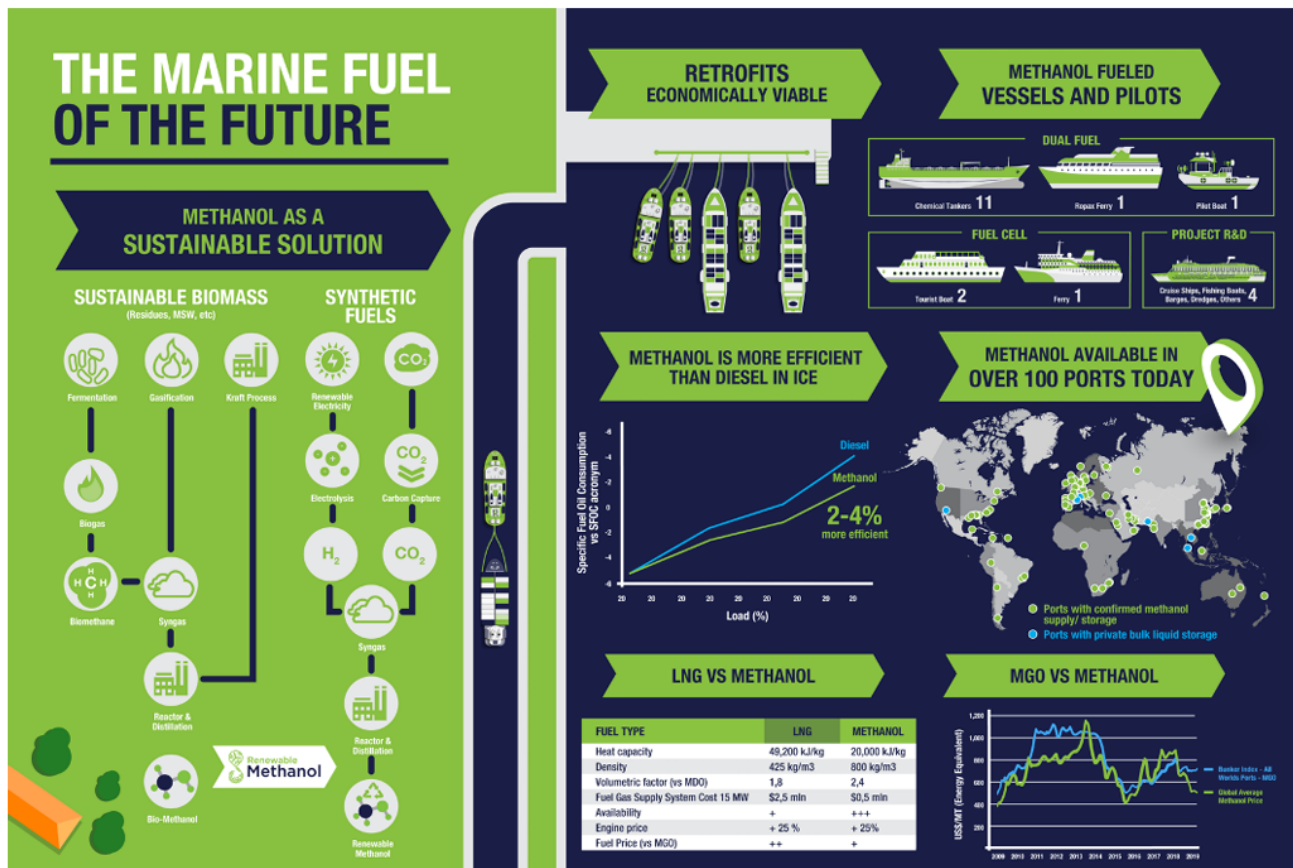


Fig. 3: Methanol as the marine fuel of the future (courtesy of the Methanol Institute)

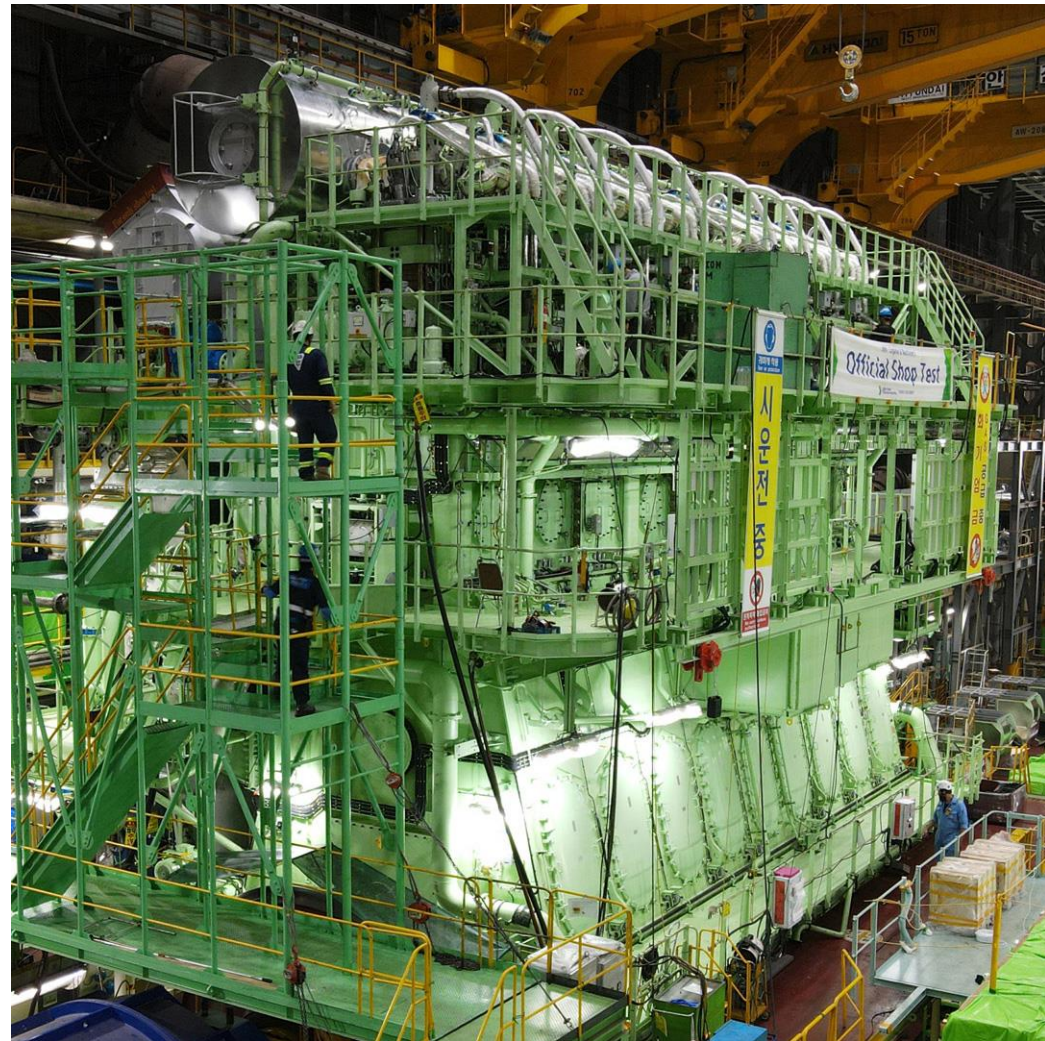
Methanol (ME-LGIM)

ME → MAN B&W's electronically controlled engine series

LGI → Liquid Gas Injection

M → Methanol

ME-LGIM = Electronically Controlled Liquid Gas (Methanol) Injection Engine



Methanol (ME-LGIM)

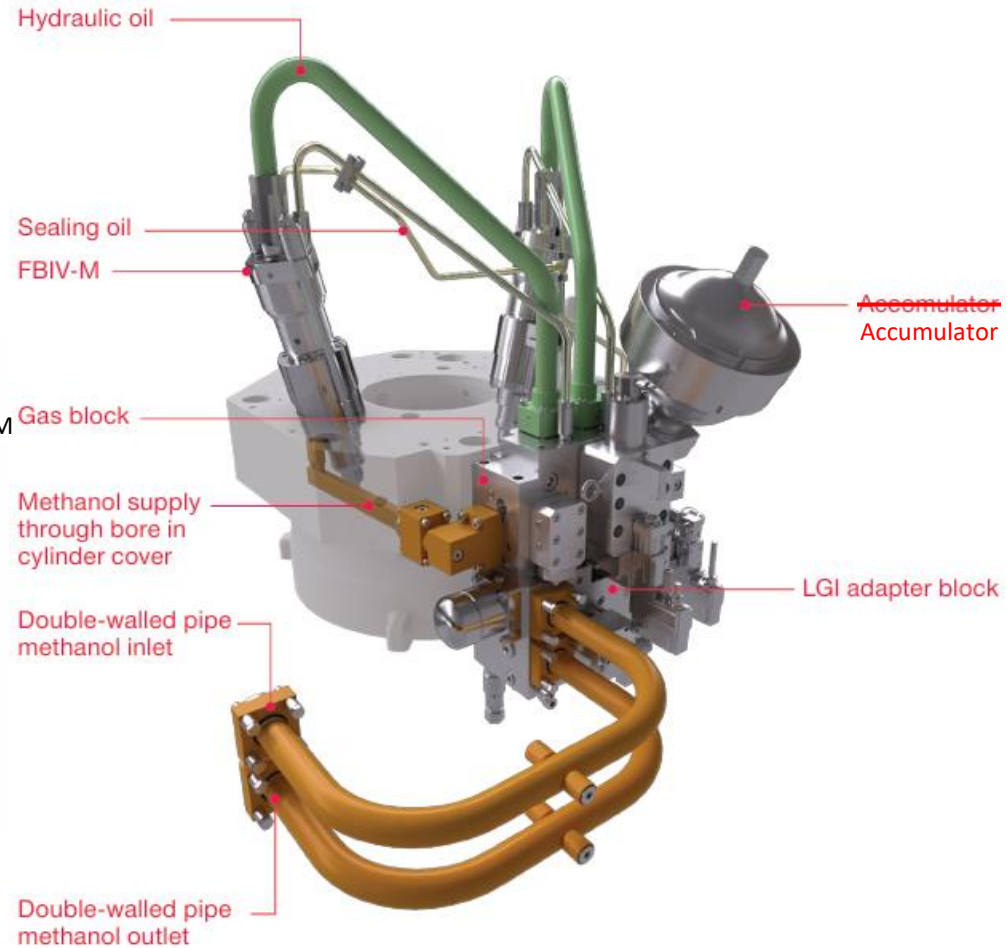
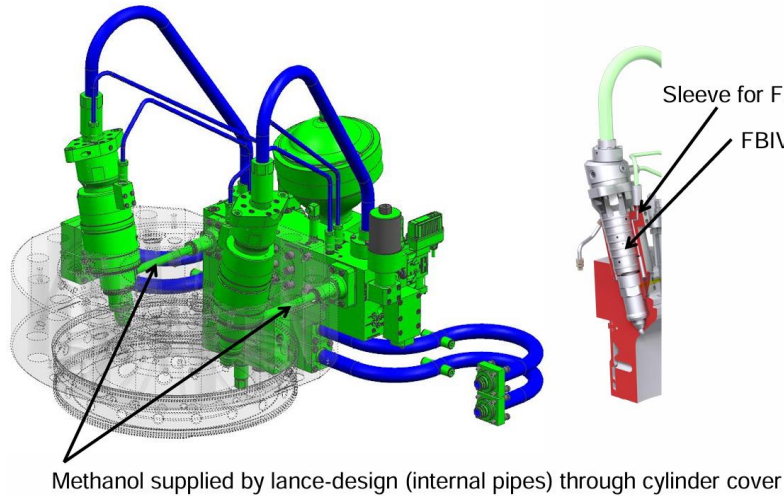


Fig. 6: Methanol components and pipes on each cylinder

Methanol

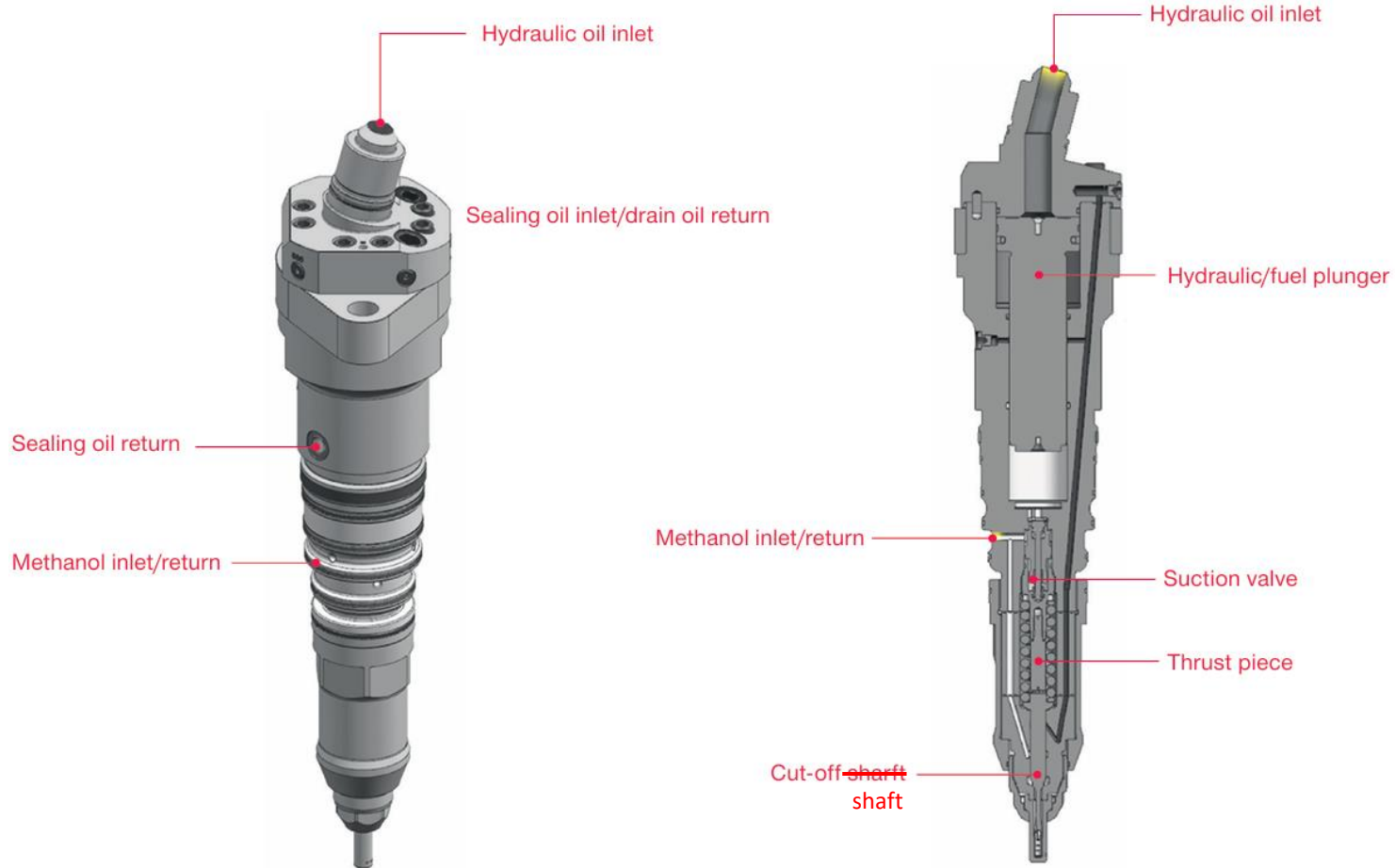
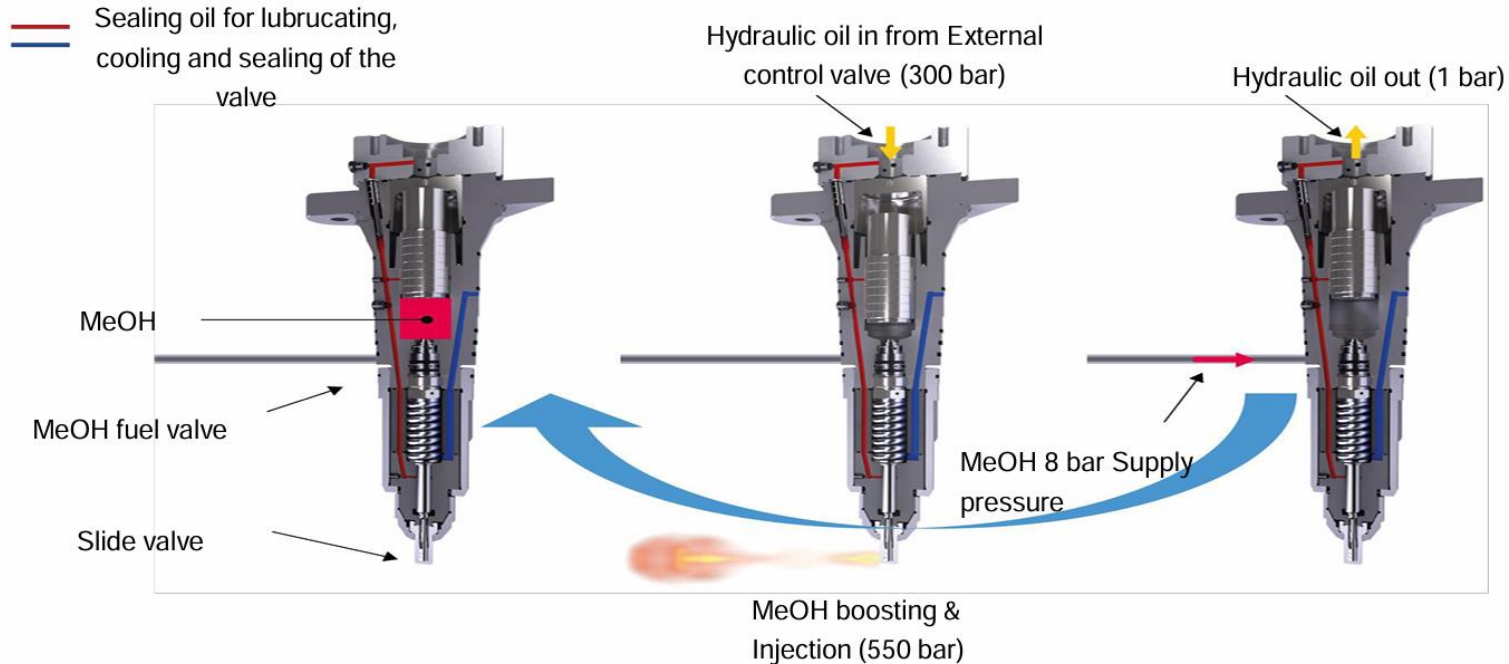


Fig. 7a: Graphic and 7b: Cross-section (right) showing the latest FBIV-M design and points of interest

Methanol (ME-LGIM)

The Fuel Booster Injection Valve

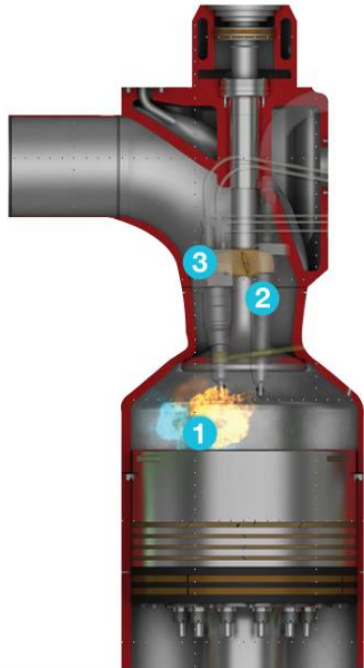
Principle of the FBIV – Fuel Booster Injection Valve



Methanol (ME-LGIM)

The LGIM engine

Concept



1 Combustion illustration

Yellow = pilot oil

Blue = Methanol

2 Conventional slide fuel valve

3 Methanol injection valve (FBIV-M)

The ME-LGIM engine is derived from the industry standard ME engine

- Diesel Cycle operating principle with high efficiency
- High fuel flexibility
- No risk of knocking
- A robust gas consumption – unchanged load response – unaffected by ambient condition

Methanol (ME-LGIM)

Development of LGIM-W (methanol/water and fuel oil/water) Tier III engine

Concept



1 Combustion illustration

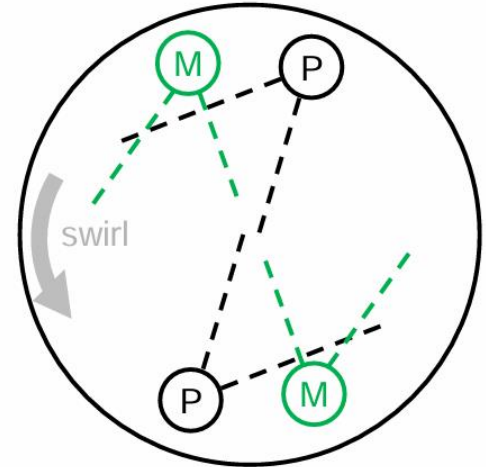
Yellow = pilot oil
Blue = Methanol/Water

2 Conventional slide fuel valve

3 Methanol/water and fuel oil/water injector

Methanol + water

- Water was mixed into the methanol in order to cool the flame, which should reduce the thermal NO_x production



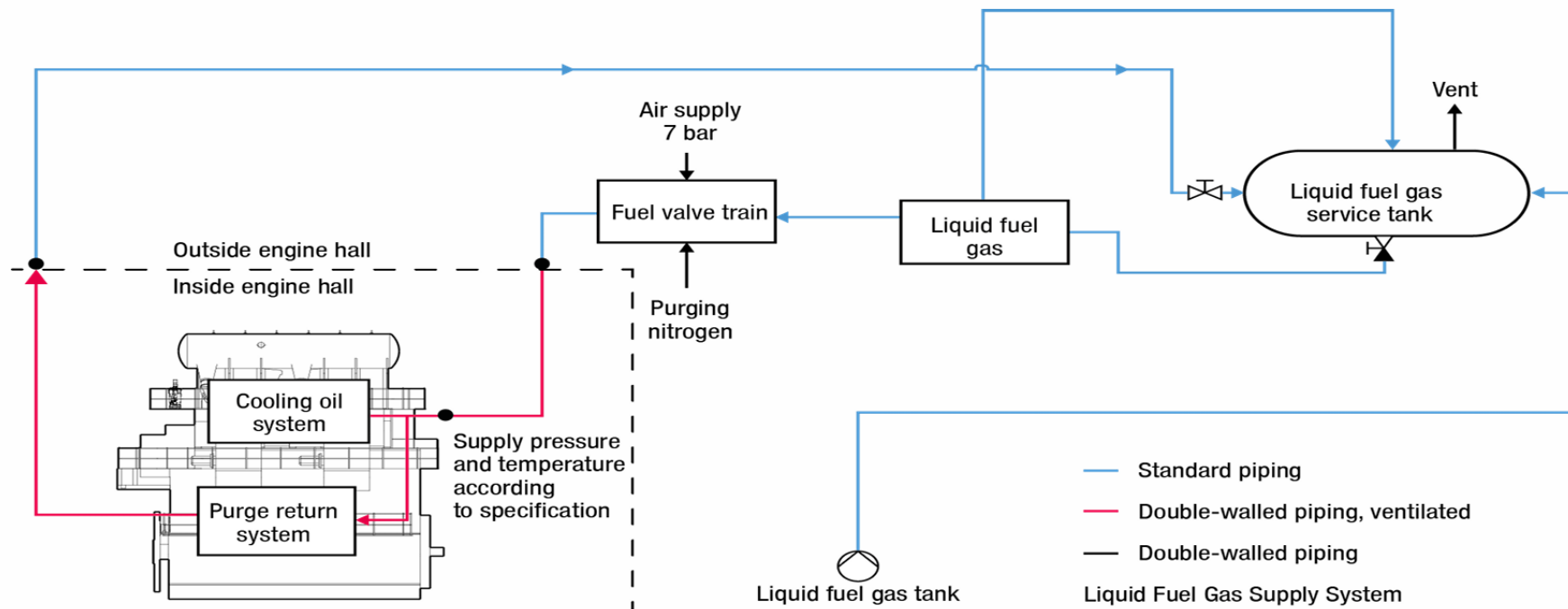
(P) Regular diesel injector for pilot injection

(M) Methanol/water and fuel oil/water injector

Methanol (ME-LGIM)

ME-LGIM System Overview

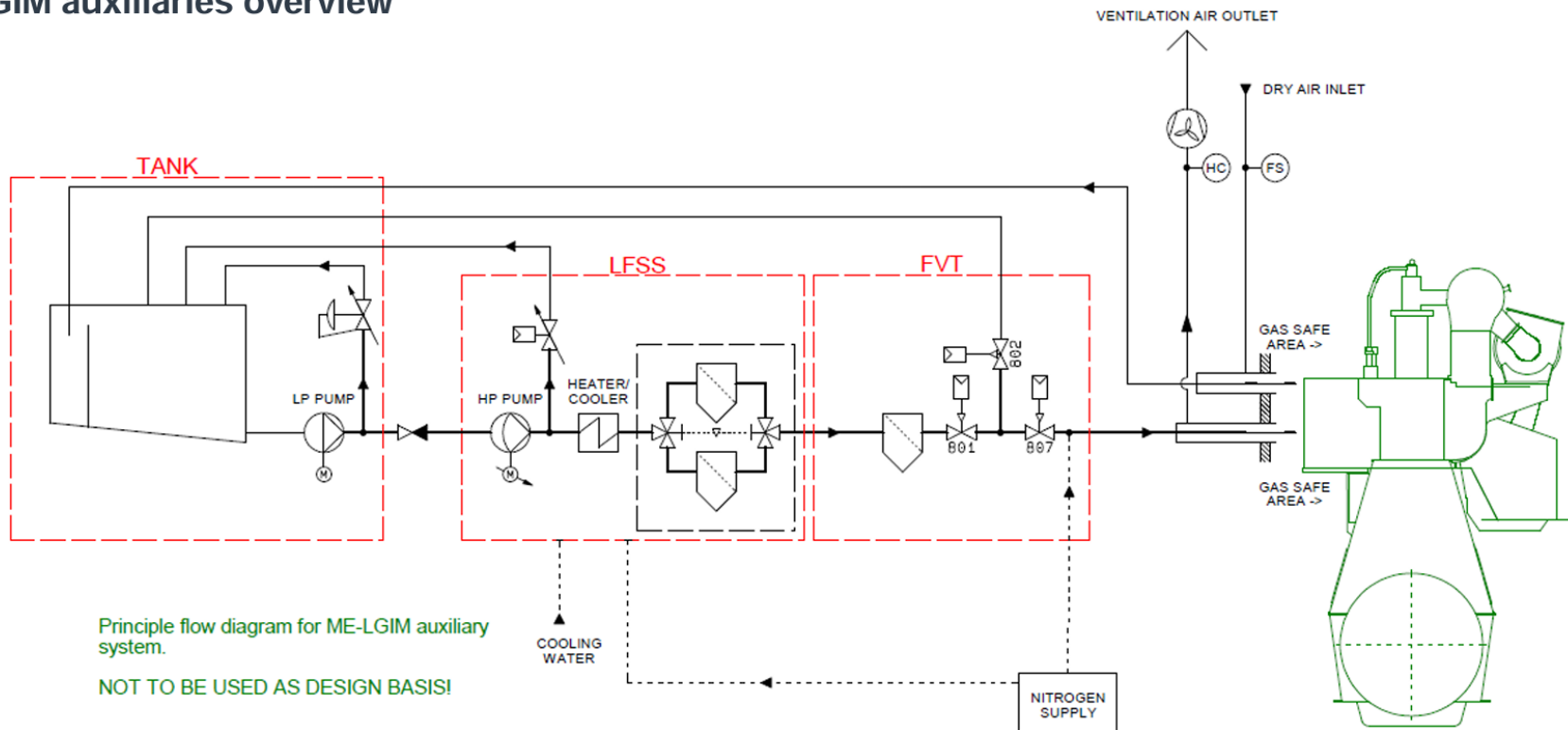
Low pressure supply system



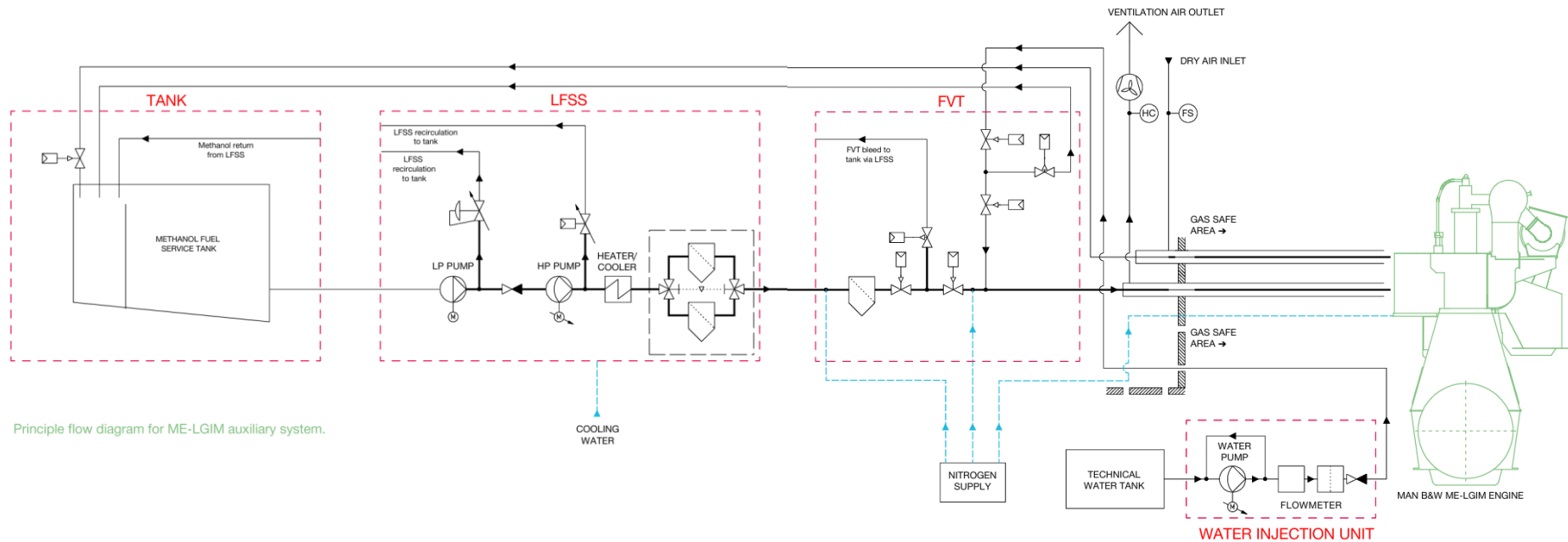
Methanol (ME-LGIM)

Fuel supply system

ME-LGIM auxiliaries overview



Methanol (ME-LGIM)



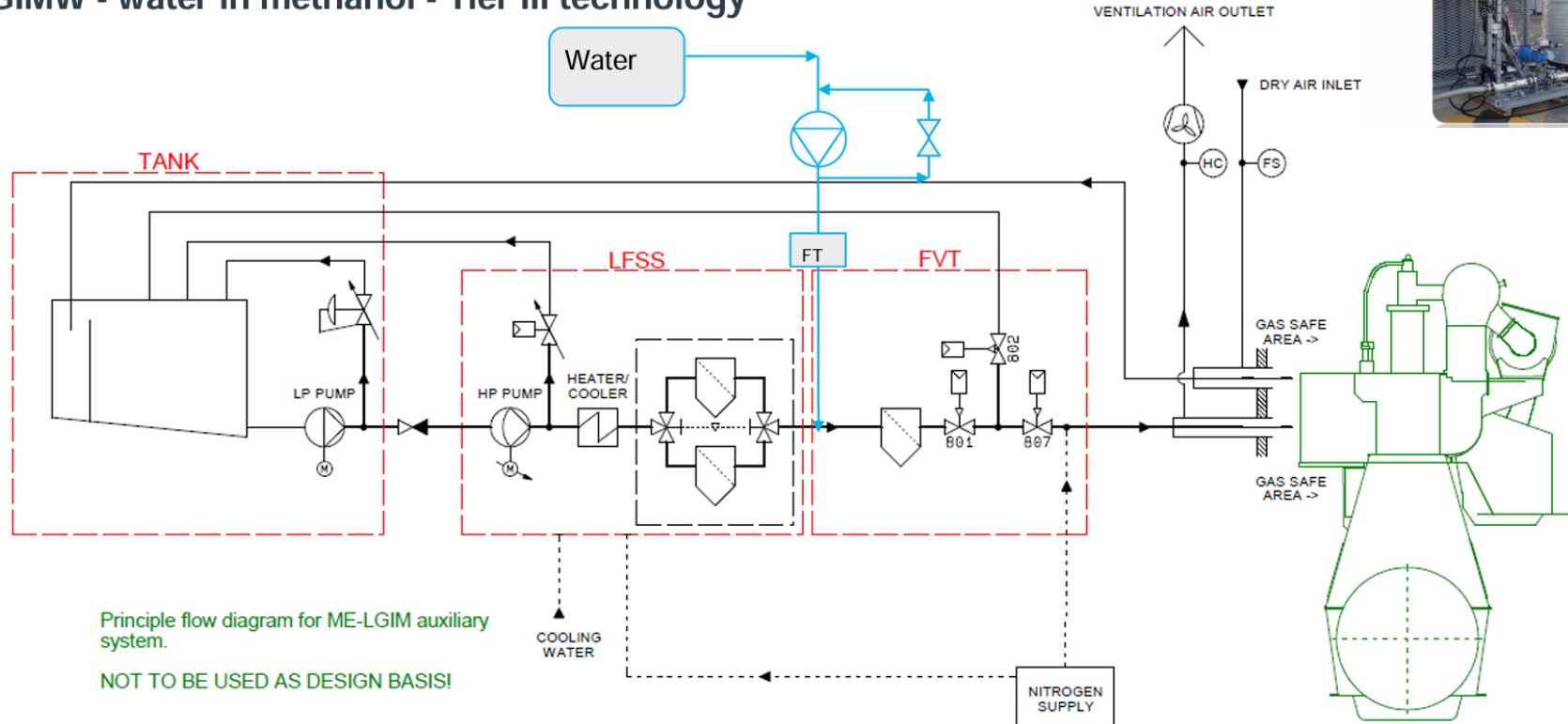
Principle flow diagram for ME-LGIM auxiliary system.

Methanol (ME-LGIM)

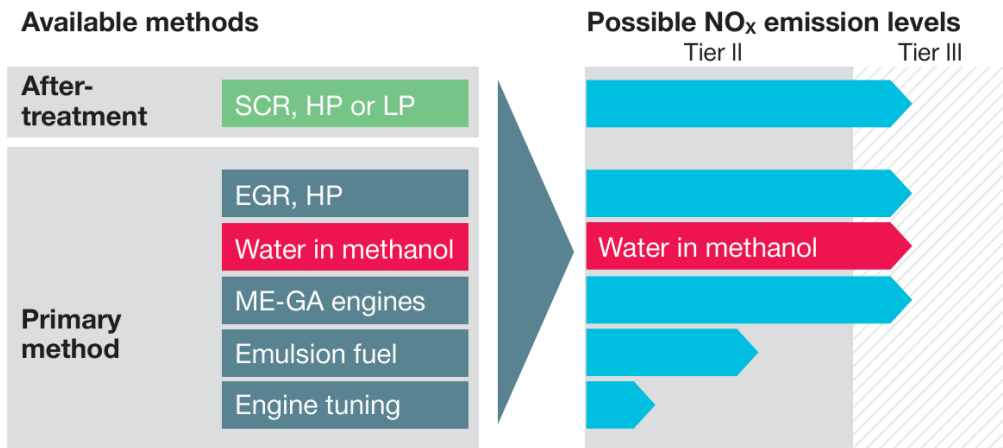
Fuel supply system

ME-LGIMW - water in methanol - Tier III technology

Development of
Tier III compliance by
Water in methanol



Methanol (ME-LGIM)



SCR: selective catalytic reduction; EGR: exhaust gas recirculation
HP: high pressure; LP: low pressure

Fig. 15: NO_x reduction methods

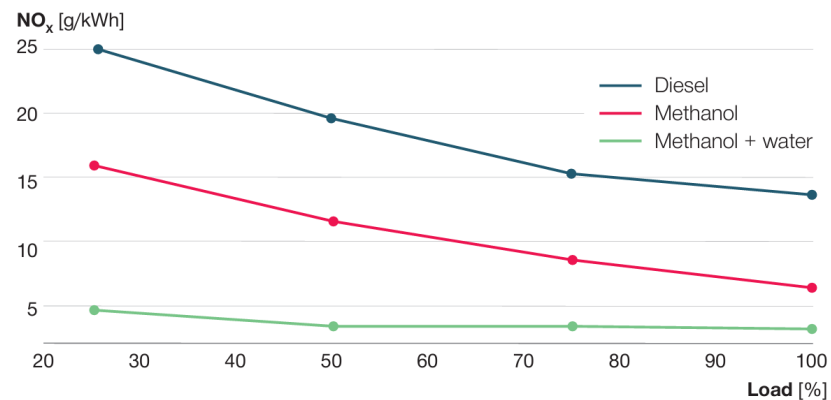
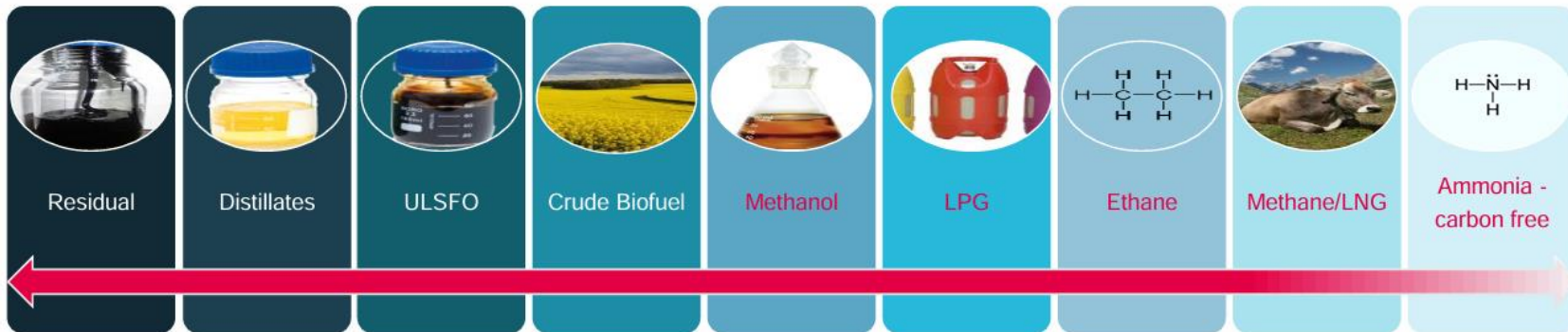


Fig. 16: NO_x reduction as a function of load

MAN B&W

Fuel flexibility

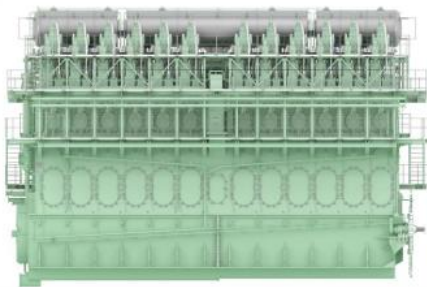
MAN B&W technology available for the combustion of the following fuels



ME-C

ME-LGIM/LGIP

ME-GI/GIE



Ammonia

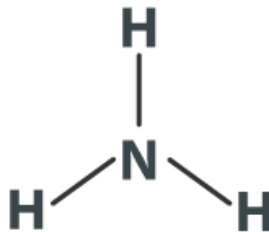
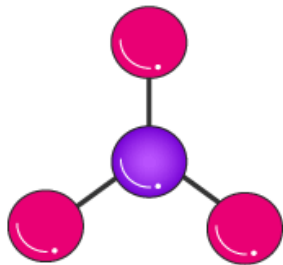
Ammonia (NH_3), colourless, pungent gas composed of nitrogen and hydrogen.

It is the simplest stable compound of these elements and serves as a starting material for the production of many commercially important nitrogen compounds.

The major use of ammonia is as a fertilizer.

In the textile industry, ammonia is used in the manufacture of synthetic fibers, such as nylon and rayon.

AMMONIA STRUCTURE



 **BYJU'S**
The Learning App

Ammonia

- NH_3 is carbon- and sulfur-free and gives a clean combustion with near-zero generation of CO_2 or SO_x .
- The volumetric energy density of NH_3 is higher than for H_2 .
- NH_3 can be cracked to N_2 and H_2 .
- NH_3 is non-explosive unlike H_2 .
- The widespread use of ammonia in industrial processes and as an agricultural fertilizer means that it is already a commercially attractive product.
- It is less expensive and less complex to transport and store than hydrogen and other fuels in need of cryogenic temperatures.
- The low risk of ignition in an ambient atmosphere makes the storage of large quantities of ammonia safer than hydrogen in terms of fire safety.



Ammonia

- The lower heating value (LHV) of approximately 18.6 MJ/kg for ammonia is comparable to methanol.
- The energy density per unit volume of ammonia (12.7 MJ/L) and the other alternative fuels, is lower than that of MGO (35 MJ/L).
- To carry the same energy content of ammonia relative to MGO will require an approximately 2.8 times larger volume if the ammonia tank is cooled.
- Although ammonia has the potential to become the future fuel, it is a toxic substance that, regulation-wise, has not yet been released for use as a marine fuel.

Energy storage type/chemical structure	Energy content, LHV	Energy density	Fuel tank size relative to MGO	Supply pressure	Emission reduction compared to HFO Tier II [%]			
	[MJ/kg]	[MJ/L]		[bar]	SO _x	NO _x	CO ₂	PM
Ammonia (NH ₃) (liquid, -33°C)	18.6	12.7 (-33°C) 10.6 (45°C)	2.8 (-33°C) ¹ 3.4 (45°C) ¹	80	100	Compliant with regulation	~90	~90
Methanol (CH ₃ OH) (65°C)	19.9	14.9	2.4	10	90-97	30-50	11	90
LPG (liquid, -42°C)	46.0	26.7	1.3 ²	50	90-100	10-15	13-18	90
LNG (liquid, -162°C)	50.0	21.2	1.7 ²	300	90-99	20-30	24	90
LEG (liquid, -89°C)	47.5	25.8	1.4 ²	380	90-97	30-50	15	90
MGO	42.7	35.7	1.0	7-8				
Hydrogen (H ₂) (liquid, -253°C)	120	8.5	4.2					

¹⁾ The relative fuel tank size for ammonia has been provided for both cooled (-33°C) and pressurised tanks (45°C)

²⁾ Assuming fully refrigerated media

Table 1: Alternative fuel comparison

Alternative fuels

Properties



Energy storage type	Specific Energy MJ/kg	Energy Density MJ/L	Required Tank Volume m ³ . (1)	Supply pressure bar	Injection pressure bar	Tank-to-wake Emission Reduction Compared To HFO Tier II			
						SO _x	NO _x	CO2	PM
MGO	42,7	35,9	1000	7-8	950				
Liquefied natural gas (LNG -162 °C)	50,0	22,4	1602	300	300	90-99%	20-30%	24%	90%
Liquid ethane gas (LEG -88 °C)	47,5	17,1	2099	380	380	90-97%	30-50%	15%	90%
liquefied petroleum gas (LPG -42,4 °C)	46,0	23,5	1527	53	600-700	90-100%	10-15%	13-18%	90%
Methanol	19.9	15,8	2272	13	500	90-97%	30-50%	11%	90%
Ammonia (liquid -33 °C)	18,6	11,5	3121	83	600-700	100%	will require Tier III equipment	>95%	>90%
Hydrogen (liquid -253 °C)	120	8,5	4223						

- 1: Given a 1000 m³ tank for MGO. Additional space for insulation is not calculated for in above diagram. All pressure values given a high pressure Diesel injection principle.



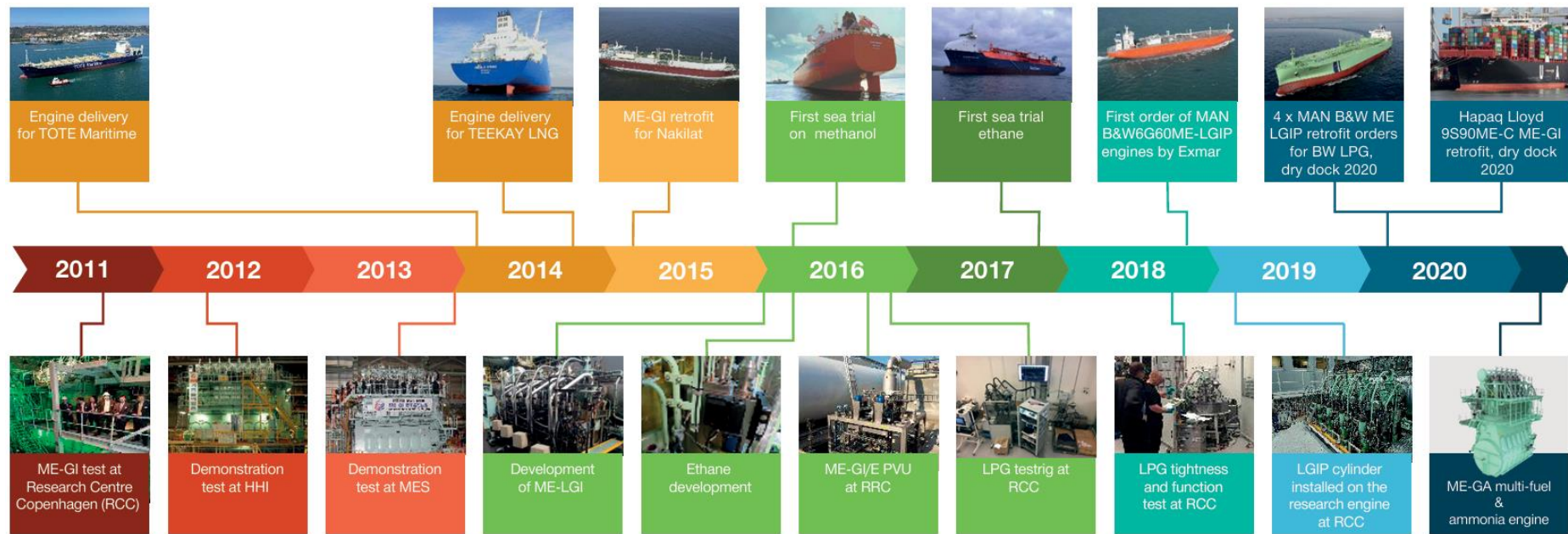


Fig. 3: Fuel diversity and engine types

Ammonia (ME-LGIA)

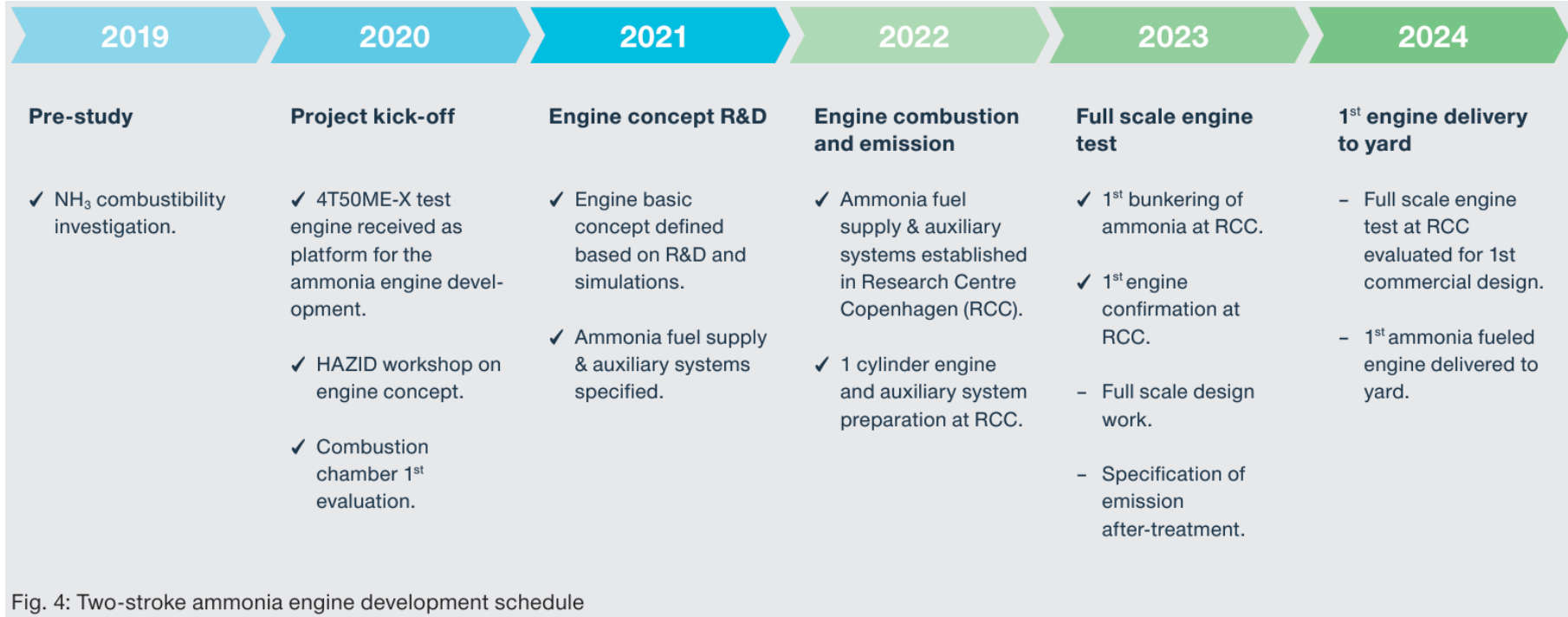


Fig. 4: Two-stroke ammonia engine development schedule

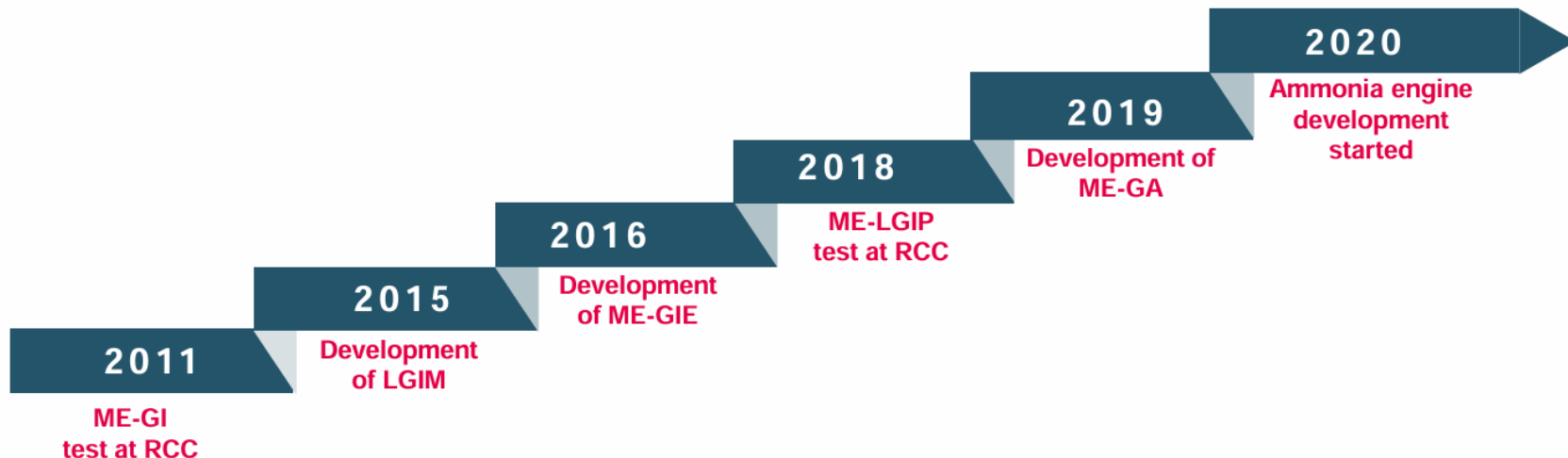
Ammonia (ME-LGIA)

LNG		Ethane	Methanol	LPG	Ammonia
ME-GI	ME-GA	ME-GIE	ME-LGIM	ME-LGIP	→ 2024
					

Fig. 1: MAN B&W dual-fuel two-stroke engine portfolio

Ammonia (ME-LGIA)

MAN B&W two-stroke engines for alternative fuels



Extensive experience in the development of dual-fuel engines



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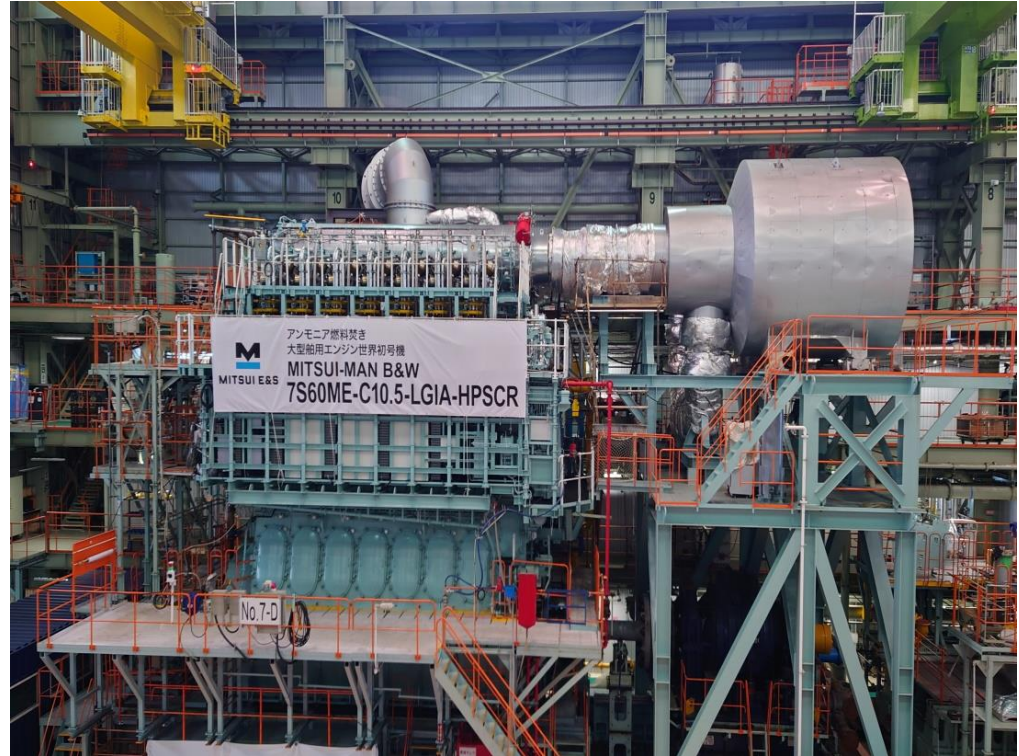
Ammonia (ME-LGIA)

ME → MAN B&W's electronically controlled engine series

LGI → Liquid Gas Injection

A → Ammonia

ME-LGIA = Electronically Controlled Liquid Gas (Ammonia) Injection Engine



2025.02.17

Ammonia (ME-LGIA)

MITSUI E&S has announced the beginning of the test phase for the world's first commercial, two-stroke, dual-fuel ammonia engine. The MAN B&W 7S60ME-LGIA Mk 10.5 prototype engine began its ammonia fuel test-run at MITSUI E&S's Tamano Factory in Japan.

The 60-bore engine will eventually be installed aboard a 200,000-dwt bulk carrier at Imabari Shipbuilding for a joint venture between "K" LINE, NS United and ITOCHU Corp. It also features MAN Energy Solutions' proprietary HPSCR (High Pressure Selective Catalytic Reduction) system for IMO Tier III compliance.



7S60ME-C10.5-LGIA at Mitsui E&S Co., Ltd.



- 7S60ME-C10.5-LGIA at MES has been operated on Diesel. Currently the ammonia auxiliary systems are being finalized.



Ammonia (ME-LGIA)

What is the status of ammonia engine development?

Key milestones

- ✓ **Successful single cylinder test in our Research Engine Copenhagen, including 100% engine load (July '23 to July '24)**
- ✓ **Initiated full scale test in our Research Engine Copenhagen (Nov. '24 to date)**
- ✓ **Reached 100% engine load on ammonia (full scale test)! (Jan. '25)**
- ✓ **Started the first commercial ammonia engine in Mitsui, Tamano, Japan (Feb. '25)**
- ✓ **How can liquid ammonia be stored?**

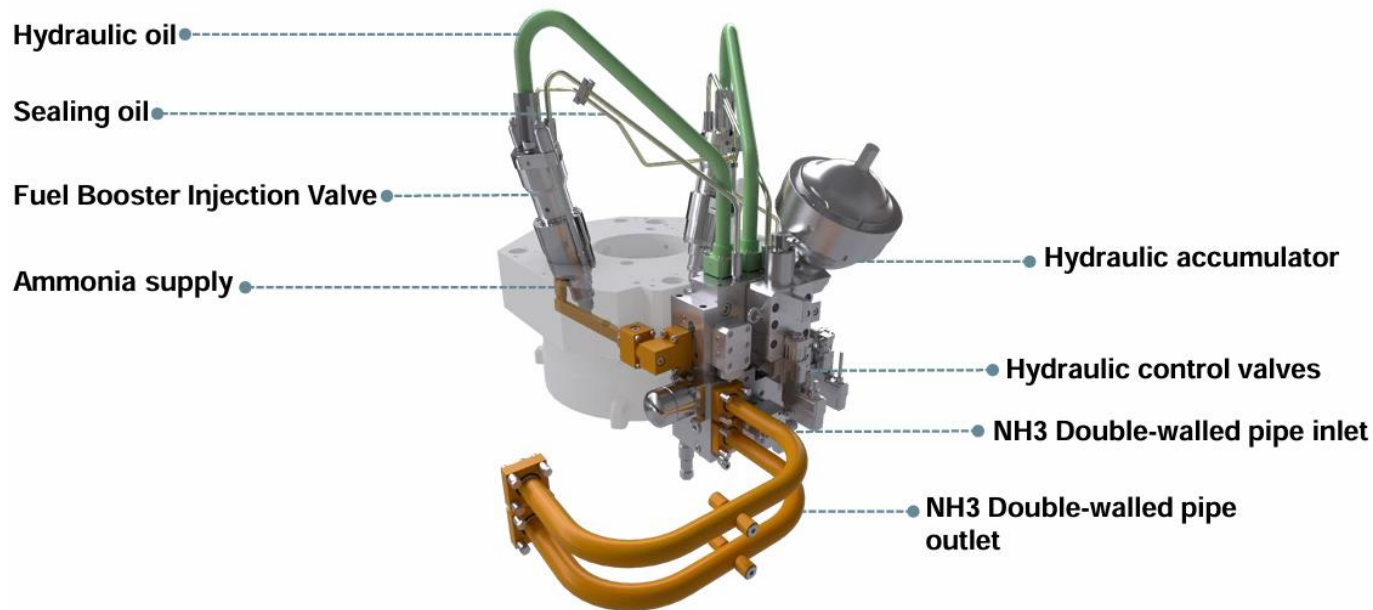


Liquid ammonia can be stored at a pressure above 8.6 bar at ambient temperature (20°C). To keep it liquid when the ambient temperature increases, many ammonia tanks are designed for approximately 18 bar.

Ammonia (ME-LGIA)

Ammonia engine development

The LGI injection system



Ammonia (ME-LGIA)

Ammonia engine development

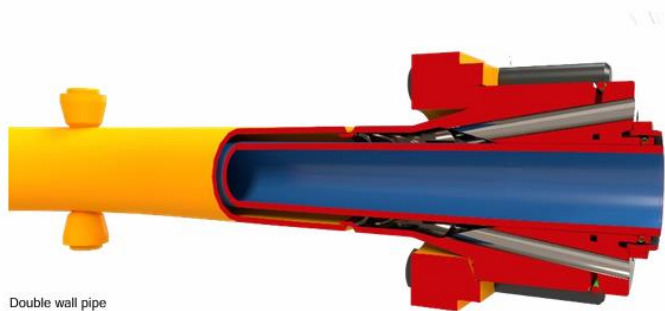
Safety



Safety principle based on well known dual fuel technology

Safe operation ensured by:

- Thorough risk assessment and identification of hazardous zones on the vessel.
- Double walled fuel pipes with supervision of pipe integrity.
- Leakage detection principles from well known dual fuel technology.
- Purging media is N_2 .
- Ammonia catch system for purging.



Double wall pipe



Ammonia (ME-LGIA)

Ammonia engine development

The LGI combustion principle

Ammonia combustibility

- Ammonia is not a hydrocarbon.
- It doesn't burn like hydrocarbons.
- It reacts much slower than hydrocarbons.

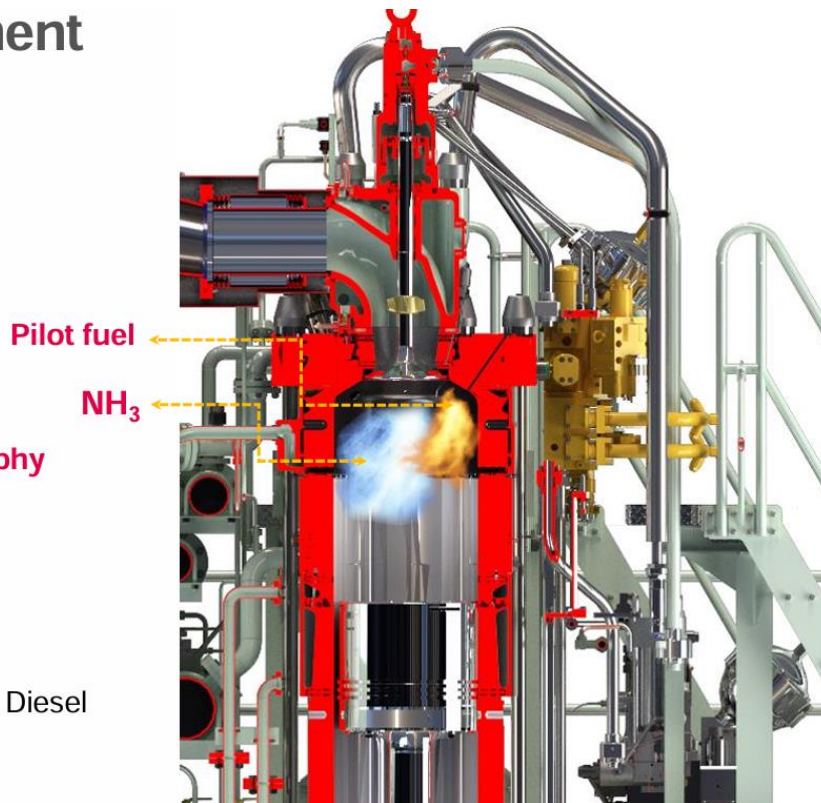
The MAN B&W ammonia engine design philosophy

“Ammonia mode”:

- Small pilot flame.
- Ammonia ignited by the pilot flame.

“Liquid fuel mode”:

- Identical performance as conventional fueled Diesel engine.



Ammonia (ME-LGIA)

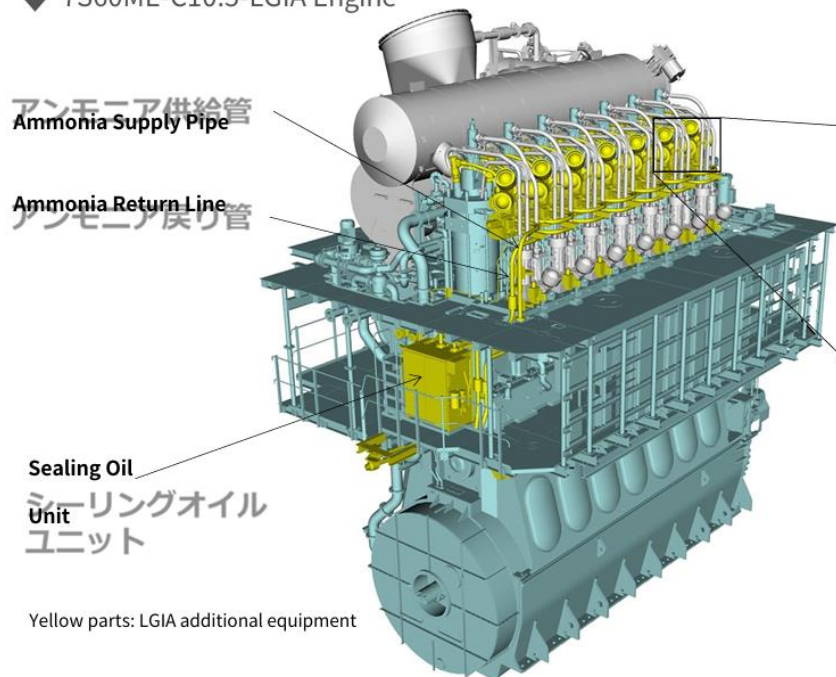


Ammonia (ME-LGIA)

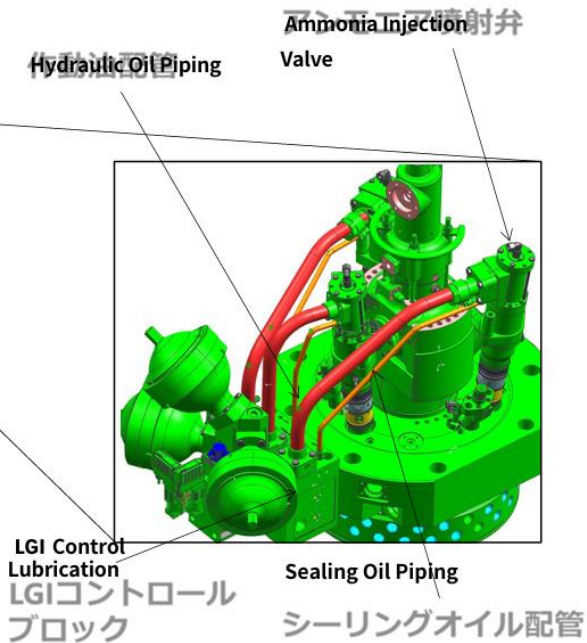
Ammonia-fired engine



◆ 7S60ME-C10.5-LGIA Engine



Yellow parts: LGIA additional equipment



Ammonia (ME-LGIA)

ME-LGIA Design Philosophy



Overview of Ammonia-Fired Engines

- **Dual-Fuel Engine**

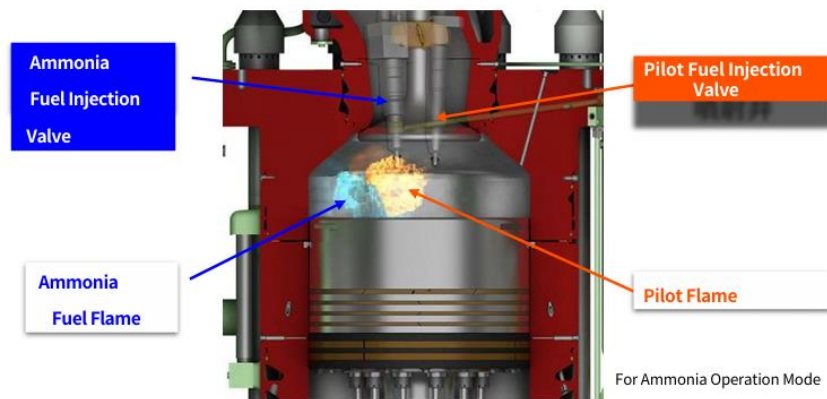
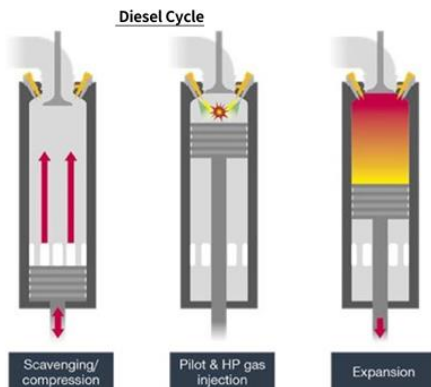
Ammonia operation mode and heavy oil operation mode are available and can be switched.

- **Diesel cycle (diffusion combustion method)**

Fuel is injected at the same timing (top dead center) as in heavy oil-fired engines.

- **Pilot Combustion Method**

In ammonia operation mode, a small amount of **pilot oil (heavy oil fuel valve)** serves as the ignition source, followed by injection of **ammonia** fuel.

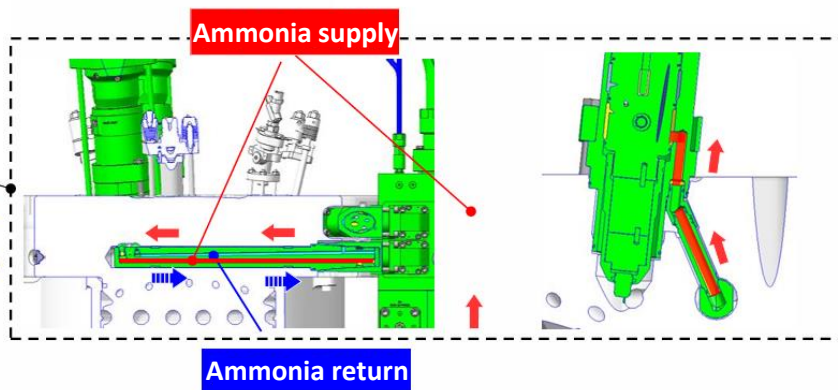
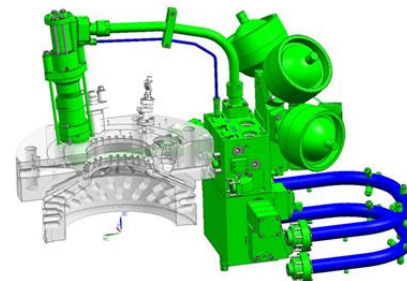
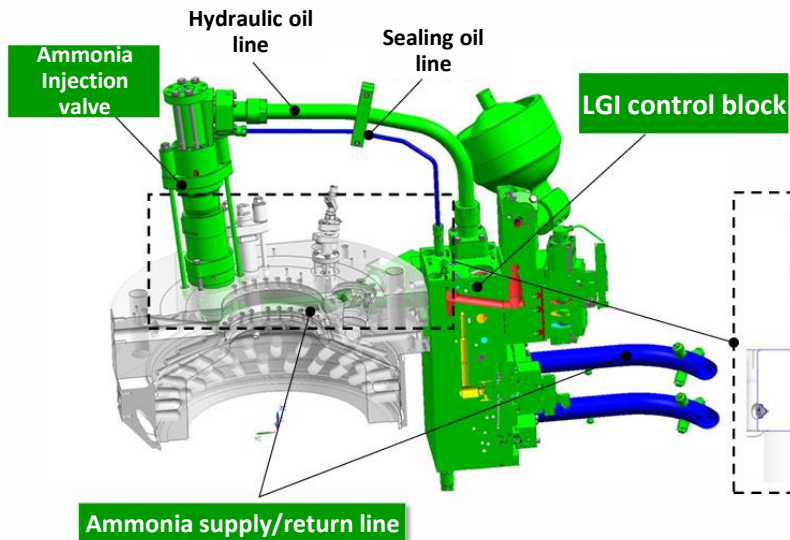


Ammonia (ME-LGIA)

LGI control lubrication



Ammonia supply :



Ammonia (ME-LGIA)

LGI control lubrication

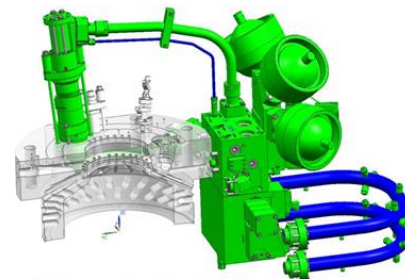
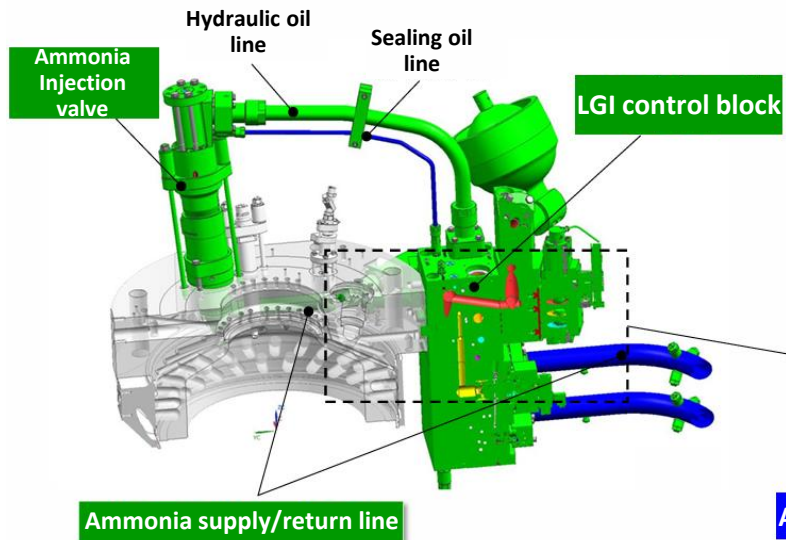


Ammonia supply:

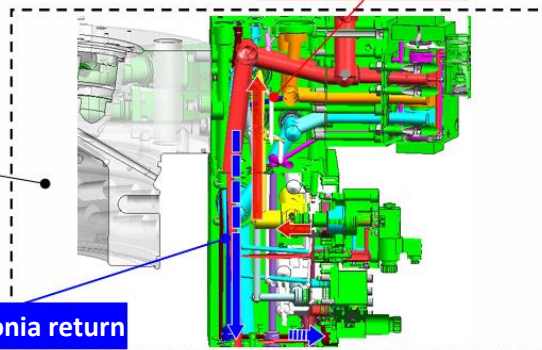
LGI control block

Ammonia supply pipe

Ammonia Injection valve



Ammonia supply



Ammonia return

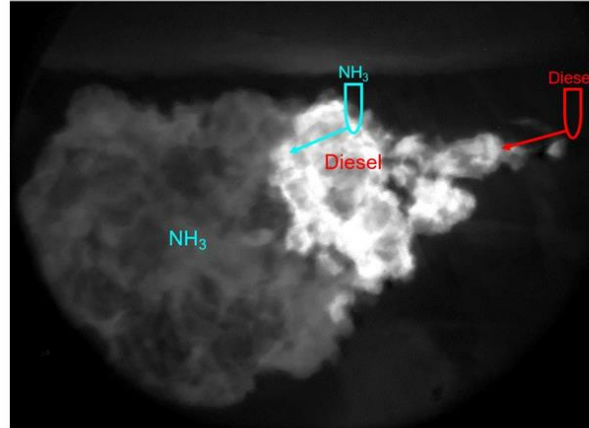


Ammonia (ME-LGIA)

High-speed combustion camera

Part of the testing includes a high speed camera where we can precisely look into the cylinder and identify the combustion dynamics.

- Good visual confirmation of flame speed and ignition properties.



Installation of high-speed camera

Ammonia (ME-LGIA)

Ammonia injection valve

Fuel Booster Injection Valve – Ammonia (FBIV-A) – ongoing development

- Basic design concept known from our methanol and LPG fuelled engines.
- High pressure hydraulic oil acting on top of a piston to increase ammonia pressure from 83 bars to around 650 bar injection pressure.
- Ammonia supplied via lance in cylinder cover and sleeve to FBIV-A.



Ammonia (ME-LGIA)

Fuel Booster Injection Valve



- A device combining a fuel pressurization mechanism with a fuel injection valve.

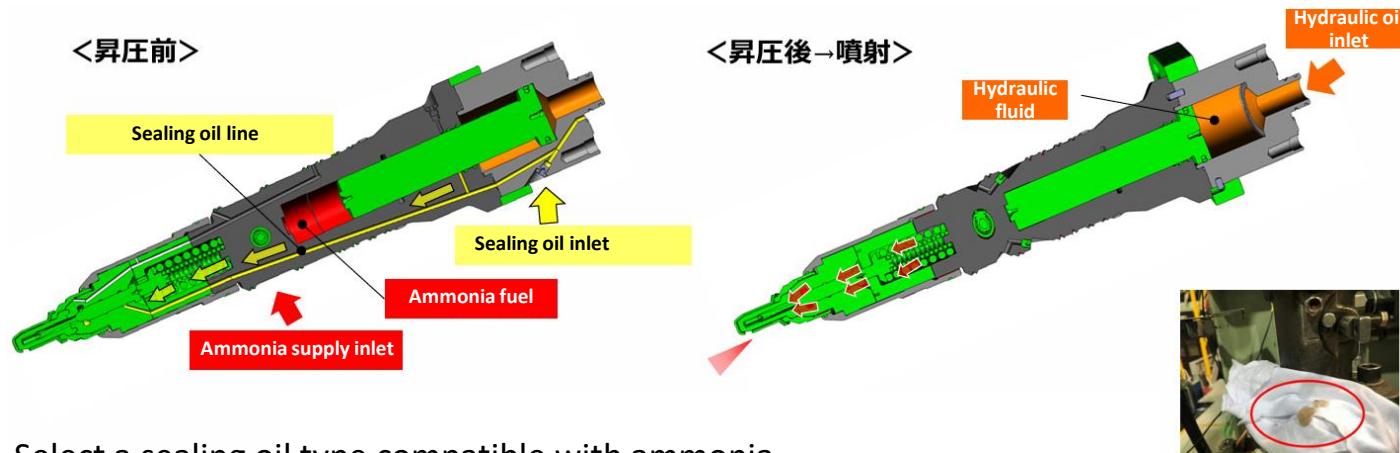
The supplied ammonia fuel is pressurized by the ammonia injection valve and injected into the cylinder.

- Sealing oil* is supplied to prevent ammonia fuel from mixing with the hydraulic fluid.

- The ammonia injection valve is an in-house product.

※Supplied from the sealing oil unit installed in the middle section of the main engine.

Ventilation line + Equipped with **gas detector**.



Select a sealing oil type compatible with ammonia.

System oil and ammonia are incompatible, causing slime-like viscous deposits to form.



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Ammonia (ME-LGIA)

Retrofit Considerations (Major Changes from Conventional Engines)

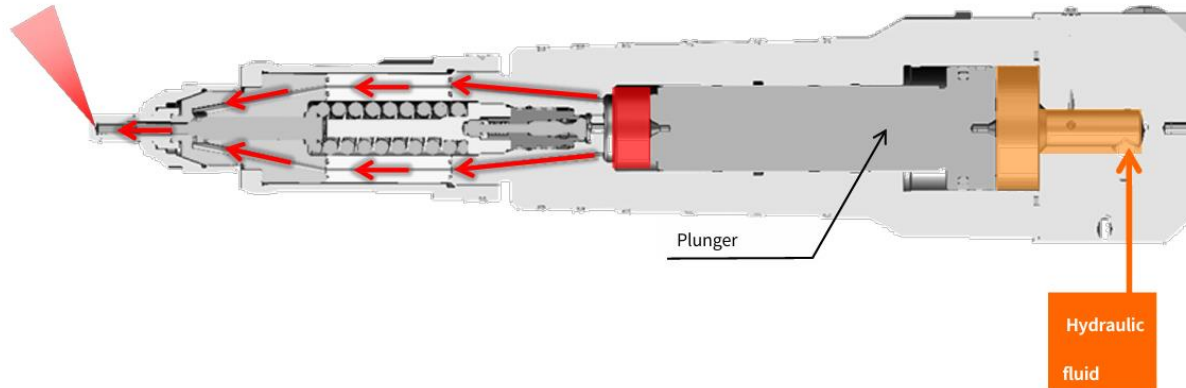


Increased hydraulic fluid consumption due to higher fuel injection rate per unit time

Lower heat output and energy density per unit volume of ammonia fuel



Ammonia injection per cycle increases by approximately **3.4 times**, leading to increased hydraulic oil consumption



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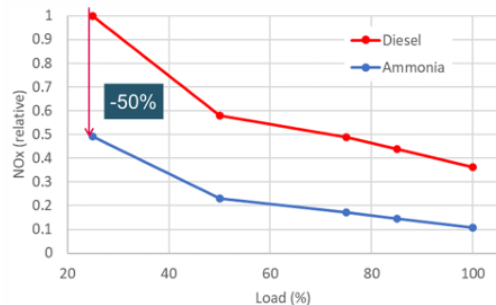
Ammonia (ME-LGIA)

Engine combustion characteristics

Test results from full engine testing at Research Centre Copenhagen

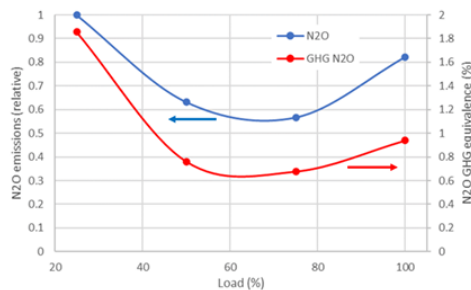
NO_x emissions

Ammonia operation leads to smaller engine-out NO_x emissions compared to diesel operation



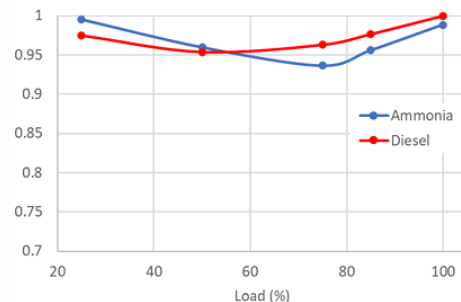
N₂O removed by engine tuning

Very low N₂O emissions (typically <5 ppm)



Combustion stability and repeatability

Operation on ammonia is within same thermal efficiency range as diesel



Absolute values will be determined by final engine tuning

Ammonia (ME-LGIA)

Ammonia engine auxiliary systems

Research Center Copenhagen

Ammonia service tank



Nitrogen purging



Ammonia supply and recirculation system



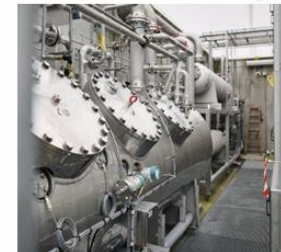
Double wall ventilation and absorber



Fuel valve and return train



Ammonia catch system
KO Drum and recovery tank



Ammonia (ME-LGIA)

Ammonia Catch System

Basic principles

- Designed to avoid direct ammonia vapours to atmosphere when purging.
- Is designed to only release vapours of ammonia up to 5 ppm.
- We have been able to test the system with blow down equivalent to a full engine shutdown due to a test rig set up.
- Approx. skid size: Length: 7m - Width: 2,5m - Height: 3,5m.
- Accumulated water/ammonia mixture to be discharged as chemical waste based on existing regulations.

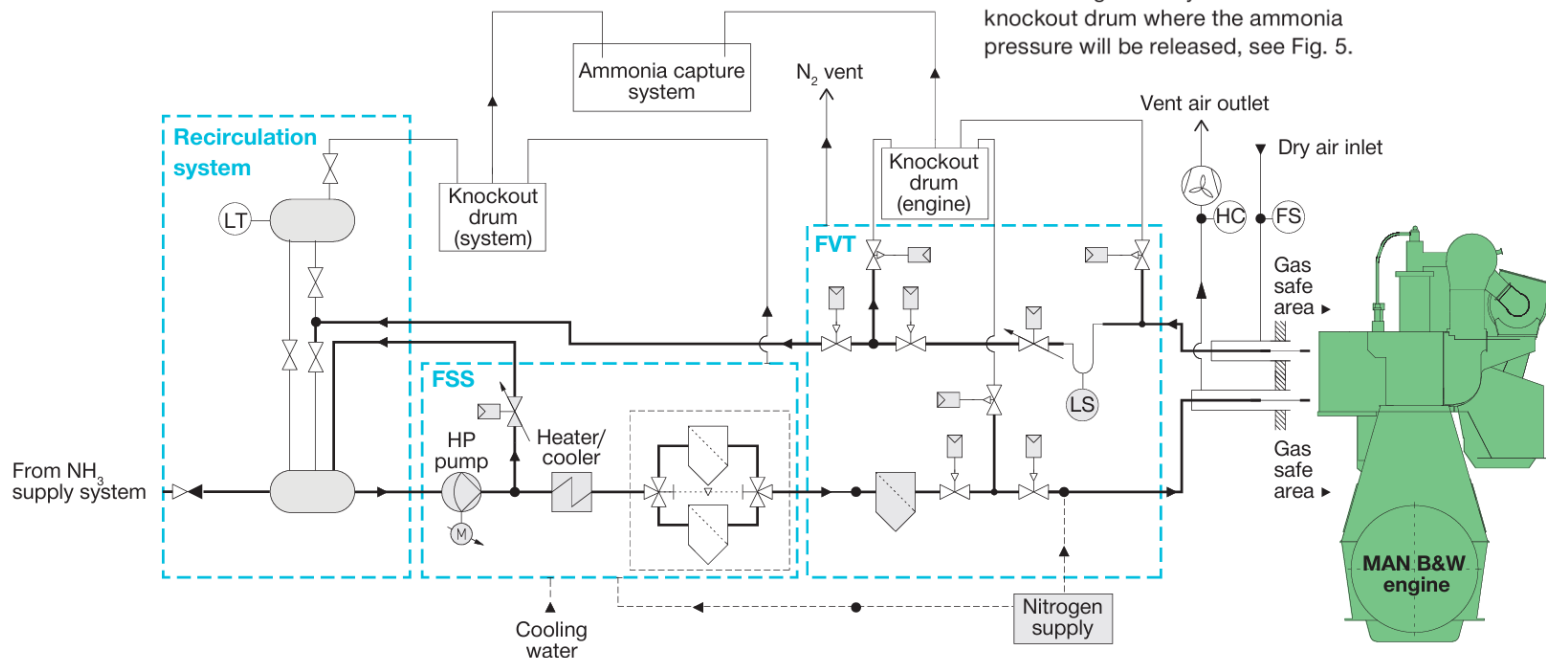
When switching from ammonia to fuel oil, there will be residual ammonia in the system. Purge that with nitrogen and use the ammonia catch system to prevent ammonia release to atmosphere.



Ammonia (ME-LGIA)

4.2.7 Ammonia capture system

Ammonia systems must be designed with an ammonia capture system (ACS) to prevent release of ammonia to the surroundings. The system consists of a knockout drum where the ammonia pressure will be released, see Fig. 5.

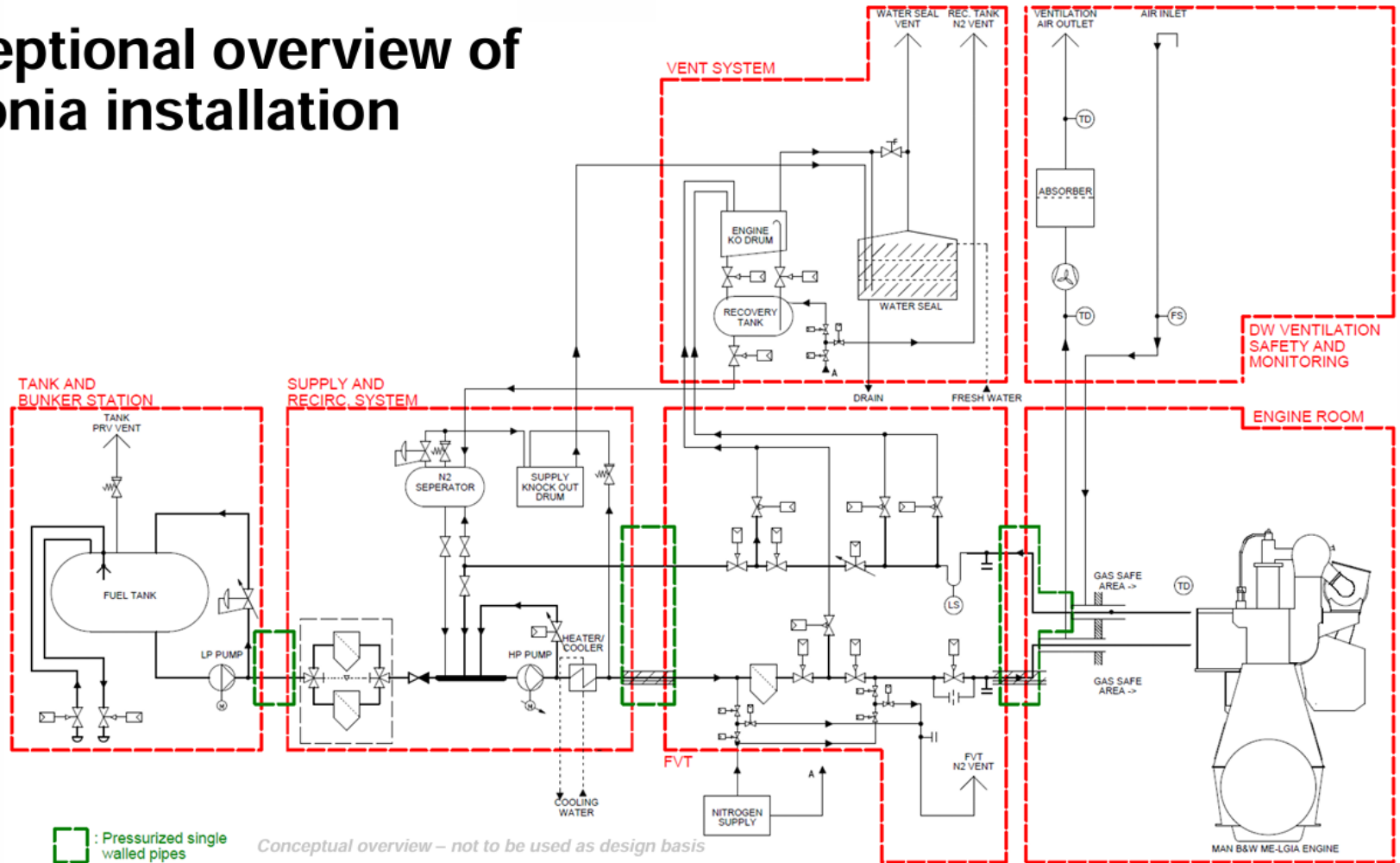


4.2.5 Nitrogen system

Nitrogen must be available for purging the engine after dual-fuel operation, for gas freeing prior to maintenance and for tightness testing after maintenance.

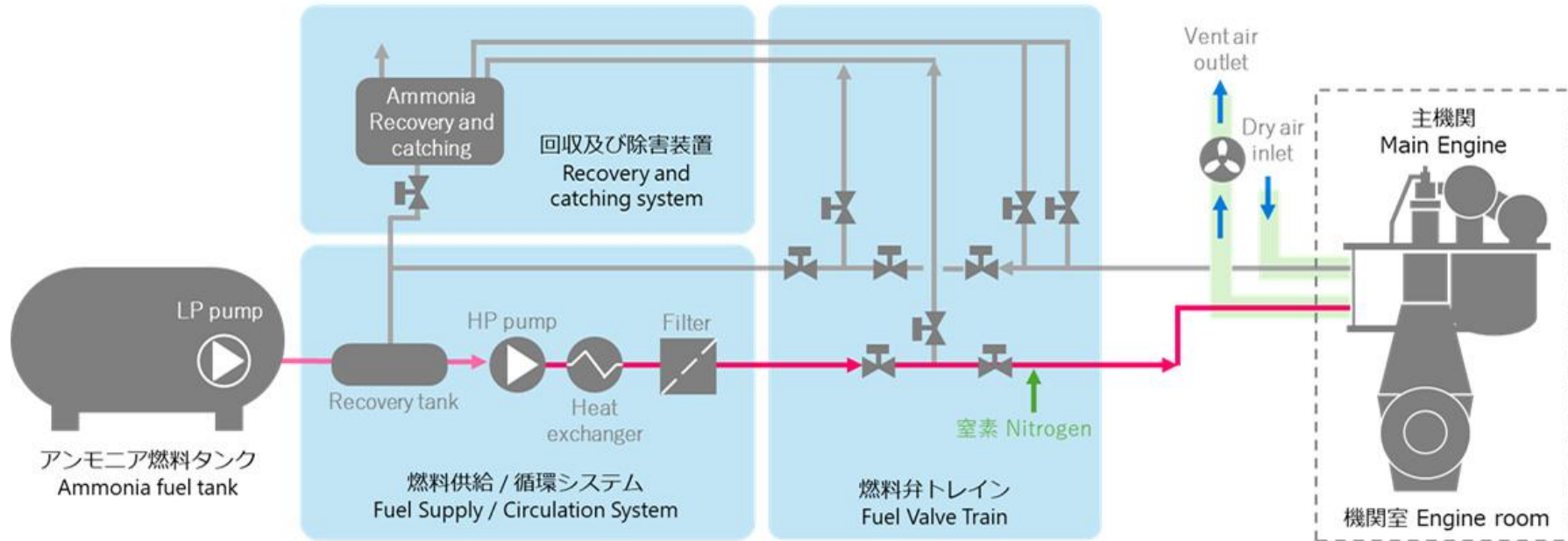
Fig. 5 Principles of the ammonia supply system showing main components

Conceptional overview of ammonia installation



Ammonia (ME-LGIA)

アンモニア燃料供給システム
Ammonia fuel supply system



Ammonia (ME-LGIA)

Safety mechanisms for ME-LGI engines



➤ Monitoring of cylinder pressure

- Compression Pressure Abnormally Low
- Maximum Explosion Pressure Abnormally High
- Ignition Failure (Expansion Pressure Abnormally Low)
- $dp/d\theta$ abnormally high (abnormally high pressure rise rate)

➤ Secondary Fuel (SF) Leak Monitoring

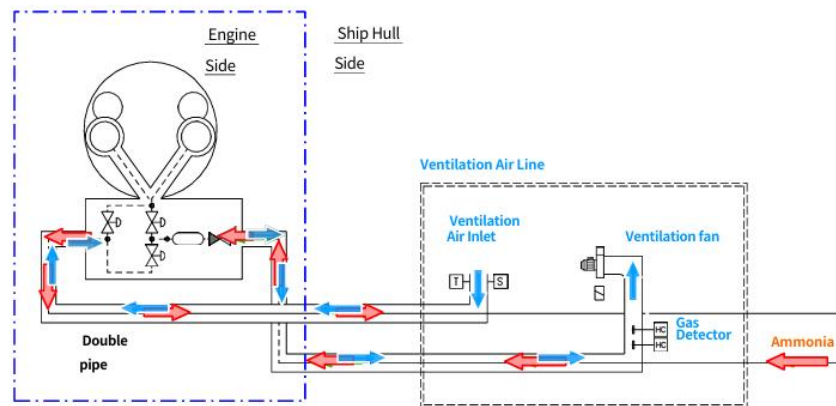
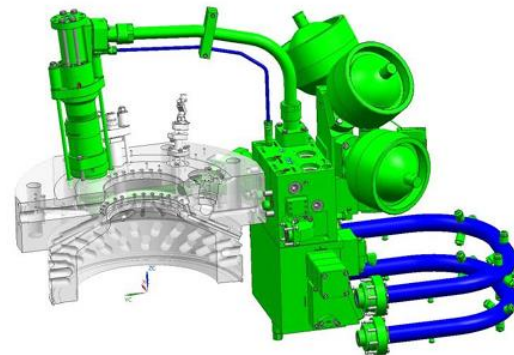
- Gas detector SF Leak Detection



SF Shutdown (Dual Fuel Operation Halt)

※Switch to Heavy Oil Operation

➤ Execution of post-operation purge sequence



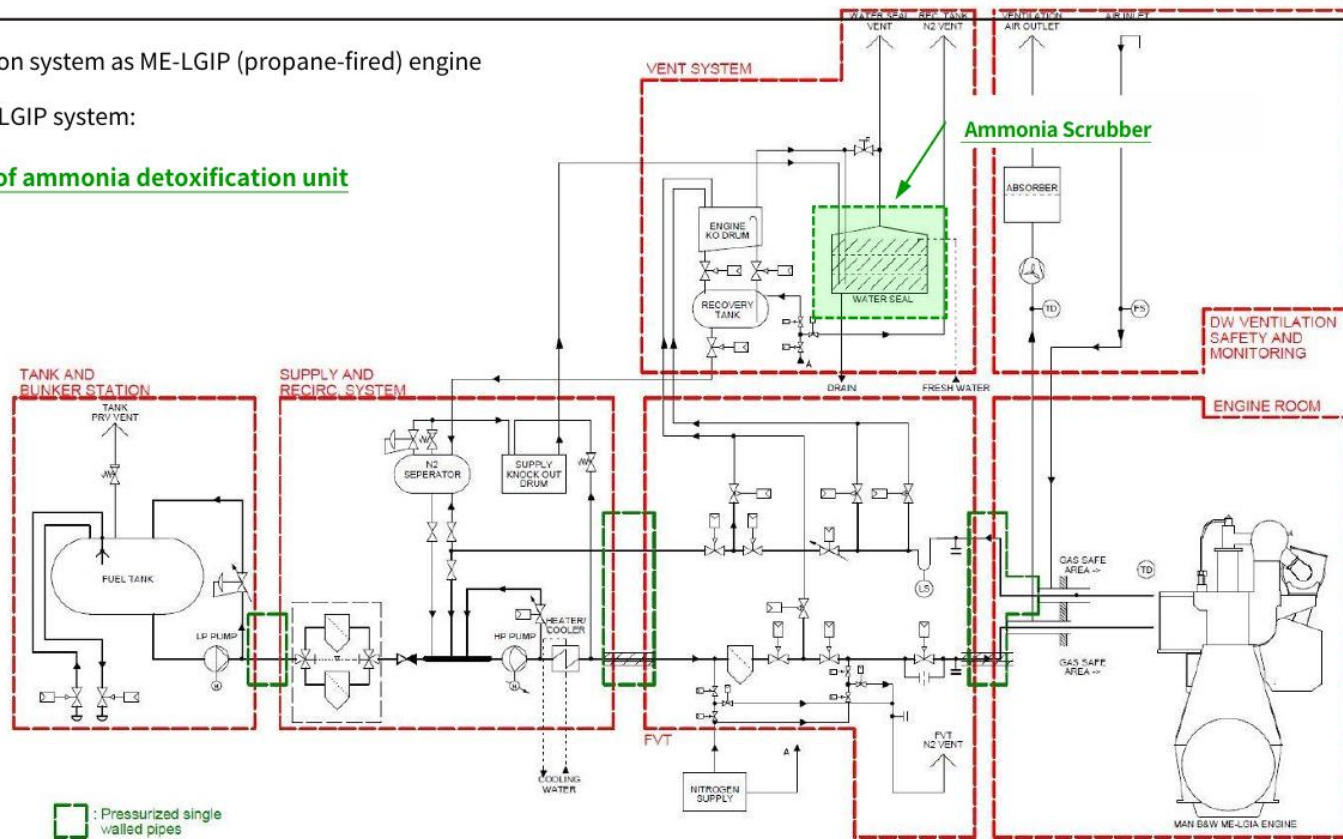
Ammonia (ME-LGIA)

Ammonia Supply System Circuit



- Same circulation system as ME-LGIP (propane-fired) engine
- Changes from LGIP system:

Addition of ammonia detoxification unit



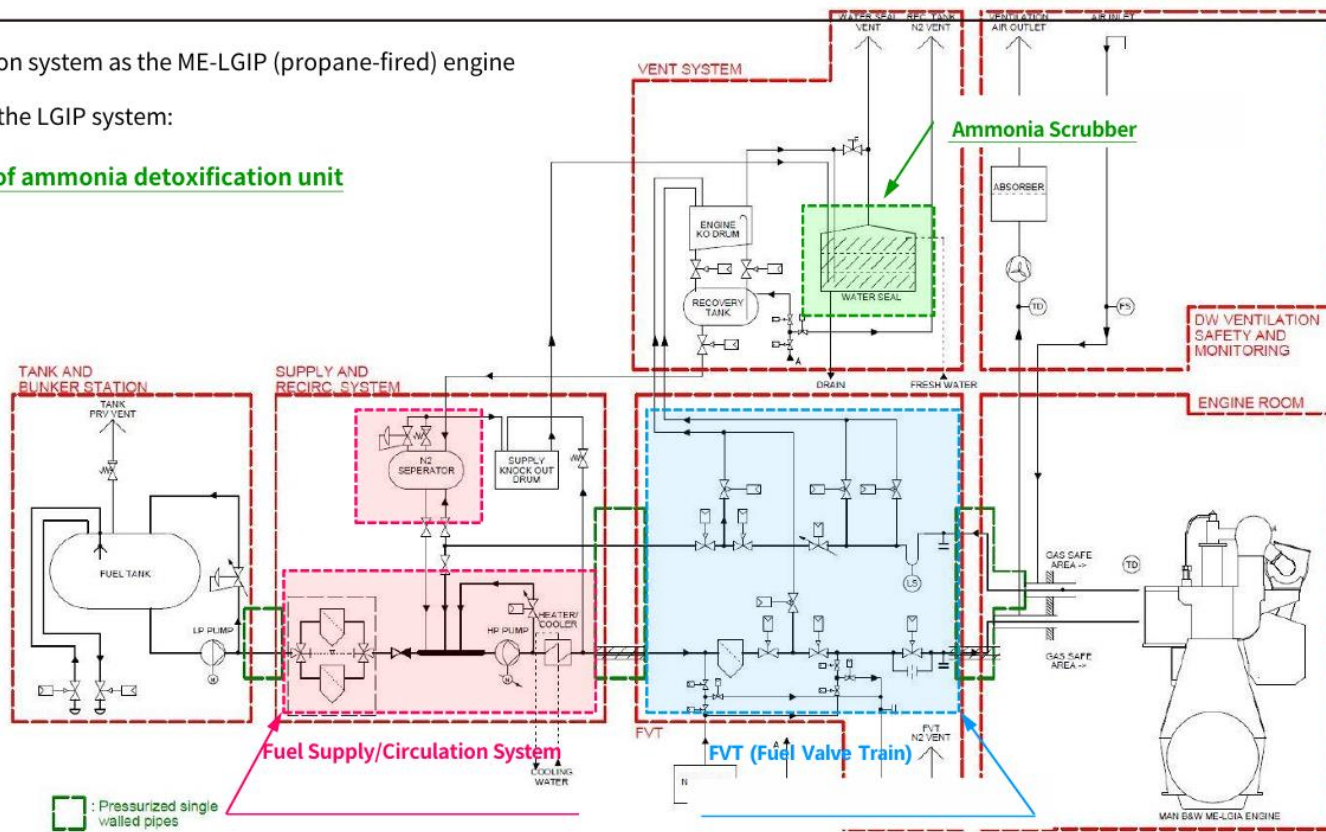
Ammonia (ME-LGIA)

Ammonia supply system circuit



- Same circulation system as the ME-LGIP (propane-fired) engine
- Changes from the LGIP system:

Addition of ammonia detoxification unit



Ammonia (ME-LGIA)

Modular design enables **extensive retrofit** options

By ensuring full fuel flexibility and extensive retrofit capabilities with a proven record, MAN Energy Solutions future proof your investment

Fuel types	ME-C	ME-GI	ME-GA	ME-GIE	ME-LGIM	ME-LGIP
Fuel oil	✓	✓	✓	✓	✓	✓
LNG	Retrofit	✓	✓	Retrofit	Retrofit	Retrofit
LEG (Ethane)	Retrofit	Retrofit	-	✓	Retrofit	Retrofit
Methanol	Retrofit	Retrofit	-	Retrofit	✓	Retrofit
LPG	Retrofit	Retrofit	-	Retrofit	Retrofit	✓
Ammonia	Retrofit	Retrofit	-	Retrofit	Retrofit	Retrofit

Ammonia (ME-LGIA)

Modular design enables extensive retrofit options



By ensuring **full fuel flexibility and extensive retrofit capabilities with a proven record**, MAN Energy Solutions **future proof** your investment

Fuel types	MC	ME-B	ME-C	ME-GI	ME-GA	ME-GIE	ME-LGIM	ME-LGIP
0-0.50% S VLSFO	Design	Design	Design	Design	Design	Design	Design	Design
HFO	Design	Design	Design	Design	Design	Design	Design	Design
Biofuels	Design	Design	Design	Design	Design	Design	Design	Design
LNG	-	-	Retrofit	Design	Design	Retrofit	Retrofit	Retrofit
LEG (Ethane)	-	-	Retrofit	Retrofit	-	Design	Retrofit	Retrofit
Methanol / Ethanol	-	-	Retrofit	Retrofit	-	Retrofit	Design	Retrofit
LPG	-	-	Retrofit	Retrofit	-	Retrofit	Retrofit	Design
Ammonia	-	-	Retrofit	Retrofit	-	Retrofit	Retrofit	Retrofit



Ammonia (ME-LGIA)

Characteristics of ammonia as a marine fuel



	Fuel Oil	Methane	Methanol	Ammonia
Engine configuration	ME-C	ME-GI	ME-LGIM	ME-LGIA
LCV [kJ/kg]	42,700	50,000	19,900	18,600
Energy density [MJ/L]	35.7	21.2	14.9	10.6
Fuel Tank Size Ratio	1.0	1.7	2.4	3.4
MAN-ES Performance	✓	✓	✓	Under Development
CO ₂ Reduction Rate (TtW)	Ref.	24%	5%	95%
Combustion Cycle	Diesel cycle			
Supply Pressure [MPa]	-	Max. 30.0	1.3	8.3
Boiling point [°C] @ atmospheric pressure	-	-162	65	-33

a) Fire-retardant fuel

- Combustible air-fuel ratio: Narrow range of 16%-25%
- Laminar combustion velocity: Slow, about 20% that of methane
- Ignition temperature: High at 650°C

b) Storage Properties

- Low calorific value: 18,600 [kJ/kg]
- Tank capacity: Approximately twice that of methane

c) Toxicity

- Highly irritating to mucous membranes
Causes severe damage to the respiratory tract and lungs in a short time
- Time-weighted average exposure limit: Approximately 25 ppm

d) Corrosive

- Corrosive to copper, copper alloys, alloys with nickel concentrations exceeding 6%, and plastics
- Seal ring material: Teflon® material recommended



difference between ME-B & ME-C engine

ME-B engine

The ME-B engine is a cost-effective version of the earlier **ME-C engine**, designed to meet the market demand for electronically controlled engines with low fuel consumption and emissions. The ME-B engine uses electronic control only for fuel injection (ELFI – Electronic Fuel Injection), while retaining a camshaft-driven exhaust valve and a mechanical cylinder oil lubricator. The fuel injection system consists of a dual hydraulic control unit (HCU) that controls two fuel injection valves per cylinder, allowing for flexible injection timing and rate shaping. The HCU also controls a solenoid valve for starting air distribution, eliminating the need for a mechanical distributor.

The main advantages of the ME-B engine are:

- Lower initial cost compared to ME-C engine
- Reduced maintenance cost due to fewer electronic components
- Improved fuel efficiency and emission performance due to electronic fuel injection
- Compatibility with various fuel types, including low-sulphur fuels and biofuels
- Simple installation and operation

ME-C engine

The ME-C engine is the most advanced electronically controlled two-stroke engine from MAN Diesel & Turbo, featuring electronic control for both fuel injection (FIVA – Fuel Injection Valve Activation) and exhaust valve (ELVA – Electronic Exhaust Valve Activation). The ME-C engine also has an electronic cylinder oil lubricator (ALS – Alpha Lubrication System) that adjusts the lubrication rate according to load and speed. The fuel injection system consists of a single HCU that controls one fuel injection valve per cylinder, allowing for precise injection timing and rate shaping. The exhaust valve system consists of an actuator that controls the opening and closing of the exhaust valve according to a pre-programmed profile.

The main advantages of the ME-C engine are:

- Higher performance and reliability due to full electronic control
- Optimal combustion efficiency and emission performance due to flexible injection timing and rate shaping
- Reduced cylinder oil consumption due to electronic lubrication system
- Enhanced operational flexibility due to variable exhaust valve timing
- Ability to handle high-pressure gas injection for dual-fuel operation

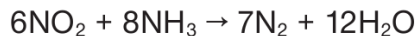


Ammonia (SCR for ME-LGIA)

4.3.1 Selective catalytic reduction technology

To reduce emissions of nitrogen oxides (NO and NO₂, commonly referred to as NO_x) and to fulfil the regionally different emission regulations, engines from MAN Energy Solutions have been equipped with, for example, advanced selective catalytic reduction (SCR)

In the catalytic reaction, NH₃ and NO_x are converted to diatomic nitrogen (N₂) and water (H₂O):



By ensuring a complete combustion, the emission of unburned NH₃ (ammonia slip) and the formation of nitrous oxide (N₂O) will be minimised.

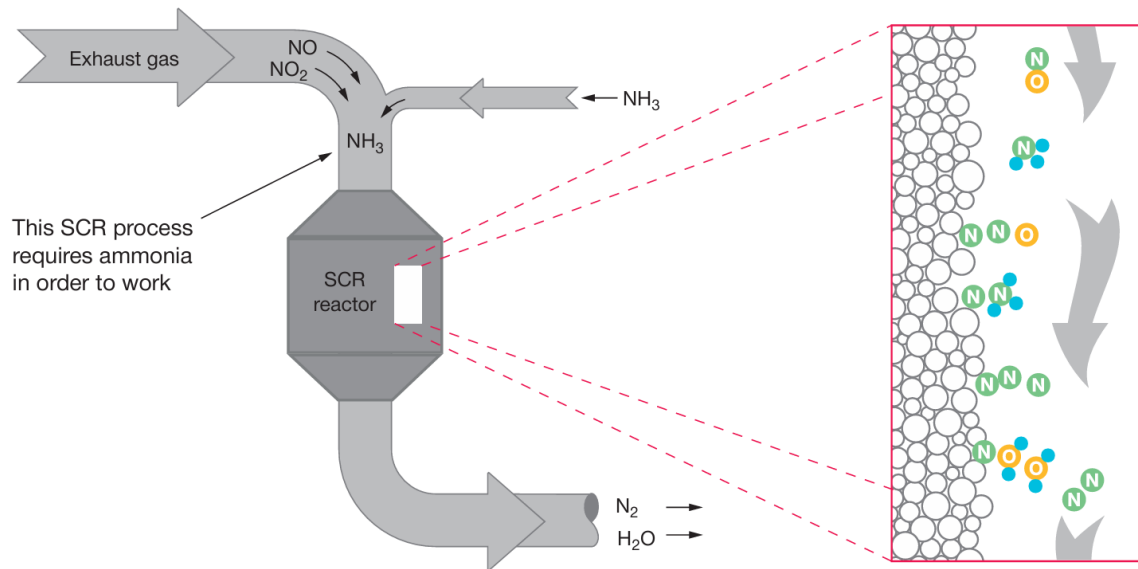


Fig. 6: The selective catalytic reduction process

Ammonia (SCR for ME-LGIA)

Engine emissions

How do we handle potential Nitrous Oxide (N_2O) emissions?

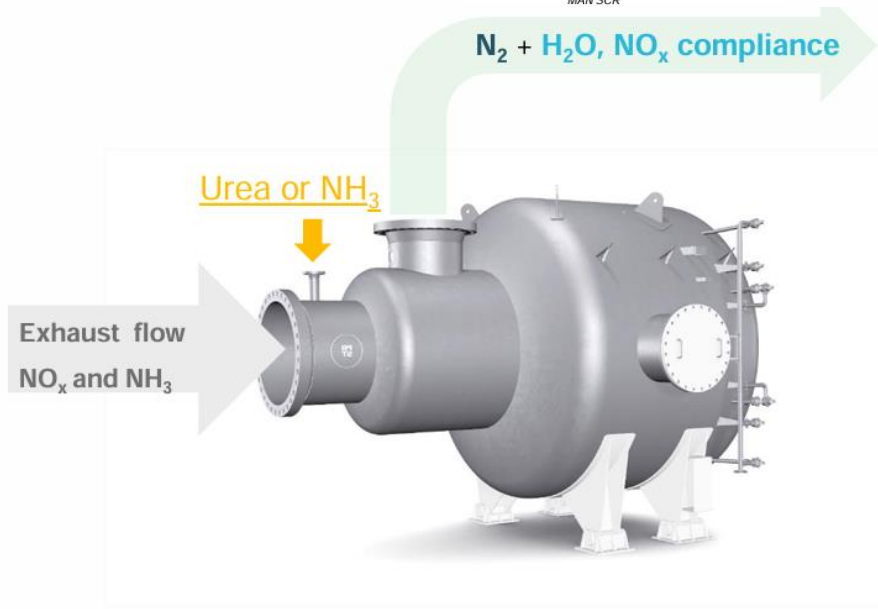
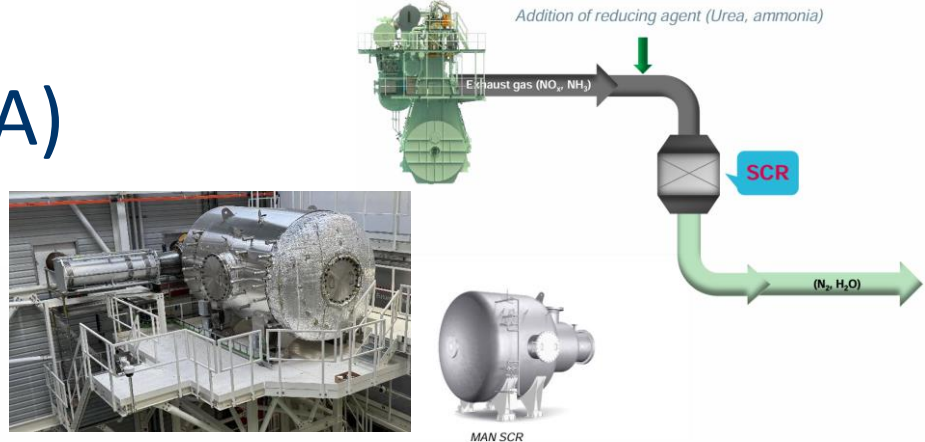
N_2O is a very potent GHG with GWP of 298.

It will also be accounted in on-going adopted regulations

- Nitrous oxide (N_2O) will be removed by engine tuning.

Ammonia slip and NO_x emissions

- Unburned NH_3 and NO_x is removed in the SCR reactor.
- Dosing of additional ammonia to SCR reaction if needed.
- Known SCR technology is suitable and MAN HP-SCR reactor can be applied.



Ammonia (ME-LGIA)

Two-stroke ammonia engine development update

The MAN B&W two-stroke ammonia engine is designated ME-LGIA

MAN B&W ME-LGIA status

- 12 months of ammonia combustion testing at **R**esearch **C**entre **C**openhagen completed on a two-stroke test engine.
- Combustion stability is very good.
- N_2O emissions are extremely low.
- NO_x emissions are around 40% lower than fuel oil.
- Similar pilot oil amount as for methanol and LPG.

4 cylinder 50-bore test engine at RCC



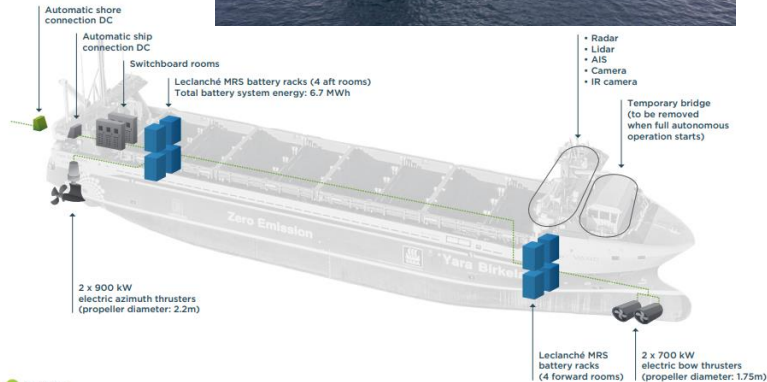
7S60ME-C10.5-LGIA at Mitsui E&S Co., Ltd.



Electricity

Electrical energy is not covered in this class because it is not directly related to the internal combustion engine.

Some examples of battery-powered vessels are shown in below.



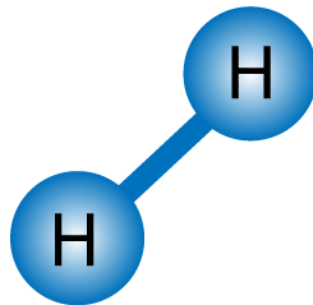
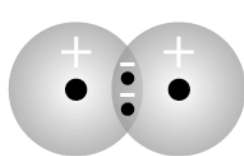
Hydrogen

Hydrogen is a chemical element; it has symbol H and atomic number 1.

It is the lightest and most abundant chemical element in the universe, constituting about 75% of all normal matter.

Under standard conditions, hydrogen is a gas of diatomic molecules with the formula H_2 , called dihydrogen, or sometimes hydrogen gas, molecular hydrogen, or simply hydrogen.

Dihydrogen is colorless, odorless, non-toxic, and highly combustible.



Hydrogen (H_2)

BYJU'S
The Learning App



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Fuel Type	Calorific Value (MJ/kg)	Storage Requirement Onboard	Utilisation Efficiency		Utilisation by-Products	Availability of Utilisation Technologies in Commercial Use
			FC (Electric Efficiency)	ICE		
Natural Gas (liquefied)	50–55	–163 °C (Pressurised or unpressurised)	45–60% (SOFC) 35–45%(PEMFC)	35–40%	CO ₂ , H ₂ O, NO _x (ICE)	ICE: Yes, available for wide commercial use FC: Underdevelopment, no commercial case in marine application
Hydrogen	120–140	350–700 bar (gases), –252.8 °C (liquefied), Material-based storage	50–60% (PEMFC/SOFC)	30–35%	H ₂ O, NO _x (ICE)	ICE: Yes, available for commercial use FC: Yes, available for commercial use
Ammonia	22.5	21 °C under 8.8 bar or –33 °C under atmospheric pressure	30–60% (SOFC) 15–30%(AFC)	35–40%	H ₂ O, N ₂ , NO _x (ICE)	ICE: underdevelopment FC: underdevelopment, no commercial case in marine application
Methanol	22.7	liquid at ambient temperature	30–50%(SOFC) 20–30%(DMFC)	Up to 40%	CO ₂ , H ₂ O, CO(ICE), NO _x (ICE)	ICE: Yes, available for wide commercial use FC: Underdevelopment, no commercial case in marine application
Bio-diesel	38–46	liquid at ambient temperature	10–30% (SOFC)	Varies	CO ₂ , H ₂ O, CO(ICE), NO _x (ICE)	ICE: Yes, available for wide commercial use FC: Underdevelopment, no commercial case in marine application
Electricity	–	High energy density battery systems such as: NMC, LFP, NCA and solid state battery	–	–	–	Yes, currently only available for small size shore-based ship

Fuel	Energy Density LHV (MJ/kg)	Volumetric Energy Density (GJ/m ³)	Renewable Synthetic Production Cost (MJ/MJ)	Storage Pressure (Bar)	Liquified Storage Temperature (°C)
Compressed hydrogen	120	4.7	1.7	700	20
Liquid hydrogen	120	8.5	1.8	1	-253
Ethanol	26.7	21.1	3.6	1	20
Methanol	19.9	15.8	2.6	1	20
Liquid methane	50	23.4	2.3	1	-162
Liquid ammonia	18.6	12.7	1.8	1 or 10	-34 or 20



Hydrogen

Color	GREY HYDROGEN	BLUE HYDROGEN	TURQUOISE HYDROGEN*	GREEN HYDROGEN
Process	SMR or gasification	SMR or gasification with carbon capture (85-95%)	Pyrolysis	Electrolysis
Source	Methane or coal 	Methane or coal 	Methane 	Renewable electricity 

Note: SMR = steam methane reforming.

* Turquoise hydrogen is an emerging decarbonisation option.

0

CO₂ emissions during the combustion of
green hydrogen

1 kg

of green hydrogen production requires
~53 kWh of electricity

40%

lower emissions possible in steel
production with green hydrogen

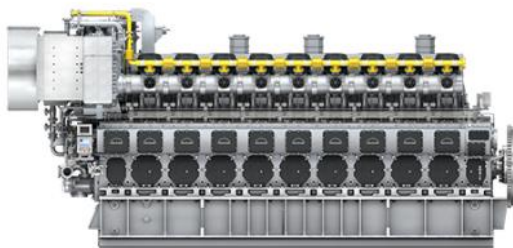


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Hydrogen

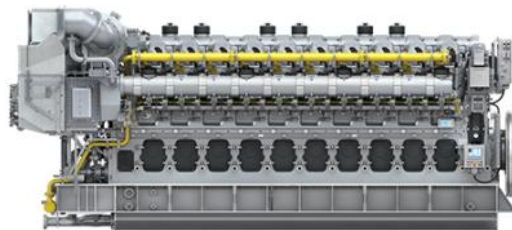
MAN hydrogen solutions

MAN gas-powered engines are H₂-ready and operable in stationary mode with a hydrogen content of up to 25% by volume in a gas-fuel mix. This includes the MAN 35/44G TS, 51/60G and 51/60G TS gas engines. 100% H₂ combustion is in development.



MAN 51/60G

The MAN 51/60G offers superior performance with its two-stage turbocharging. Offering best-in-class performance and low emissions, it sets new standards for gas engines.



MAN 35/44G

The MAN 35/44G offers ultimate performance and maximum efficiency while keeping emissions low. Also available with two-stage turbocharging.



Hydrogen

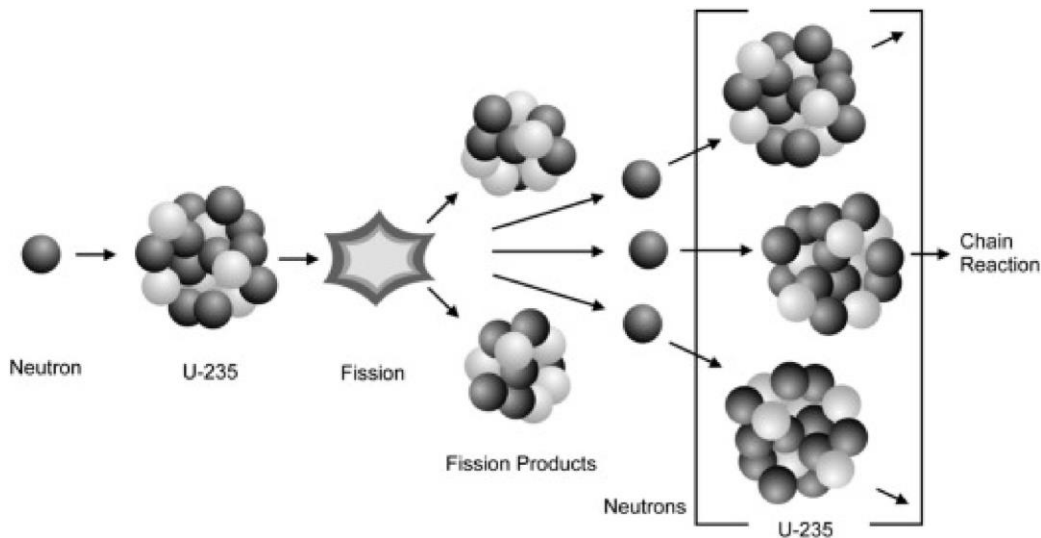


Nuclear

All nuclear power generation, currently in commercial use, is based on nuclear fission.

The main fuel used in nuclear fission is uranium-235 (^{235}U), which accounts for 0.711 % of natural uranium by weight.

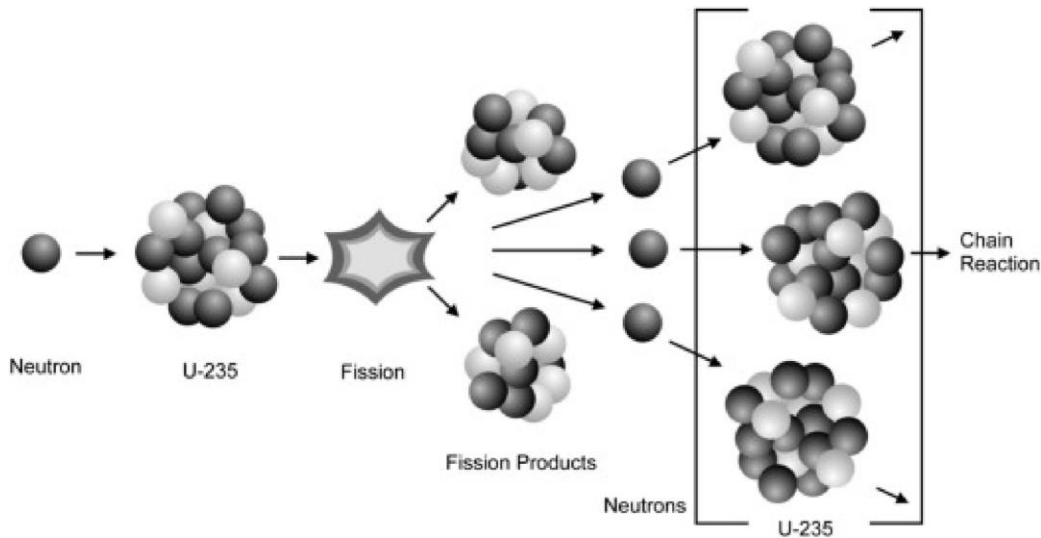
The remainder of natural uranium is composed of ^{238}U (99.284 %) and ^{234}U (0.005 %).



Nuclear

When ^{235}U absorbs an extra neutron, it splits into fission products and two or three neutrons, releasing a large amount of energy in the form of heat and radiation.

The split neutrons collide with other ^{235}U atoms, and this process continues as a chain reaction.

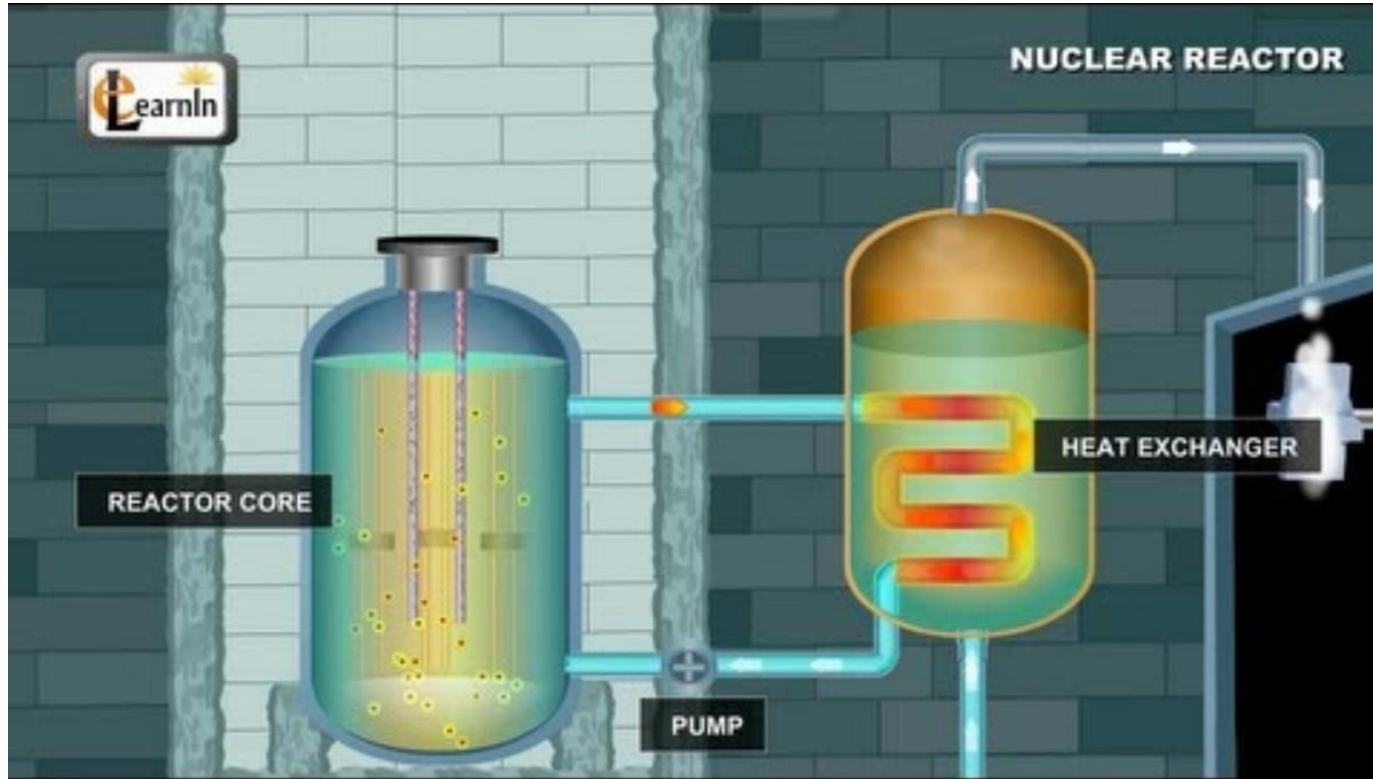


Nuclear

**How does a
nuclear power
plant work?**



Nuclear



Nuclear



Nuclear



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